RaceTracker — VK4WIP UI validation pack

Every new screen, every event (click or entry), with what the plan says and what the built app showed.
Branch feature/vk4wip-ui · generated 21 Oct 2026, 11:15 am · 109 events · 12 screens

How to validate

Each event block has: **Plan** (section of IMPLEMENTATION_SPEC.md), **Event** (exactly what was clicked or entered), **Plan says** (wording quoted from the plan — not the developer's own description), **App showed** (on-screen messages captured as text from the running app) and a screenshot taken straight after the event. The full plan is reproduced in the appendix.

Suggested prompt for Claude.ai — attach this PDF and say:
"This PDF documents a built UI against the plan in its appendix (IMPLEMENTATION_SPEC.md). For every event, compare 'App showed' and the screenshot with 'Plan says' and the referenced plan section. List every mismatch, missing behaviour, styling difference from §3 (colours, buttons, runner states) and accessibility concern, with the event ID and the plan section. Also list plan items (§4, §7, §8) that no event exercises."

Notes for the reviewer. The race is generated for the run (started two hours before the screenshots), so times differ from the plan's 07:47 examples. Base screens were seeded with earlier calls (an hour and half an hour before) including deliberate data problems, so the overdue alert and every data check have something to show. Known, deliberate differences from the plan are listed in DECISIONS.md (e.g. no Start checkpoint, the setup wizard has 5–6 steps, the race list stays on Home). All events were completed by the automation.

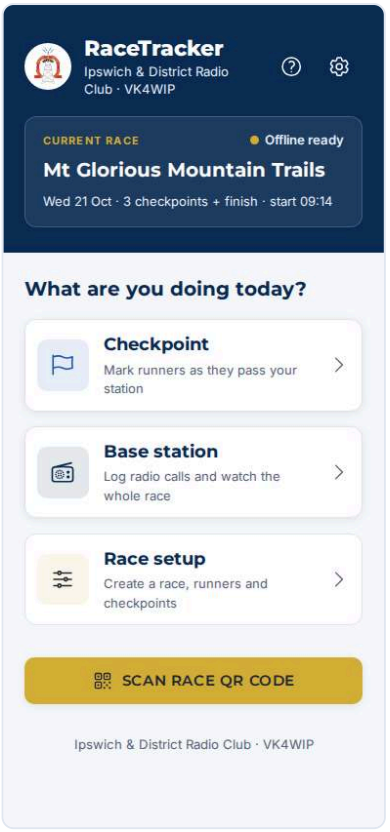
Screens

- Home — HOME-01 … HOME-03 (3)
- Settings — SET-01 … SET-06 (6)
- Share race config — SHARE-01 … SHARE-03 (3)
- Race setup — SETUP-01 … SETUP-04 (4)
- Checkpoint — Mark off — CP-01 … CP-16 (16)
- Checkpoint — Callout sheet — CO-01 … CO-08 (8)
- Checkpoint — Read-out — RO-01 … RO-08 (8)
- Checkpoint — Overview — OV-01 … OV-11 (11)
- Base — Receive — RX-01 … RX-19 (19)
- Base — DNF / DNS — ST-01 … ST-12 (12)
- Base — Overview — BO-01 … BO-03 (3)
- Base — Reports & data checks — RP-01 … RP-16 (16)

Home

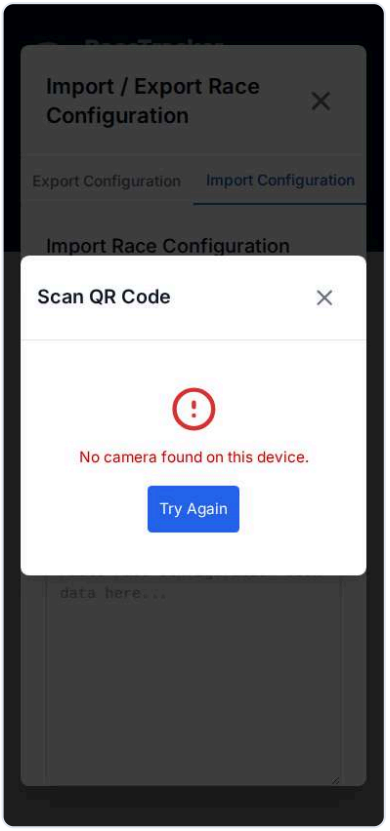
HOME-01 390×844

Plan	§4.1
Event	**Open the app at /**
Plan says	*Navy header: badge, "RaceTracker", "Ipswich & District Radio Club · VK4WIP", "Offline ready" status. Current race card (name, date, checkpoint count, start time). Three role cards (≥76px): Checkpoint, Base station, Race setup — icon tile, title, one-line description, chevron. Primary button at bottom: Scan race QR code (existing import flow). Footer lockup: "Ipswich & District Radio Club · VK4WIP".*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



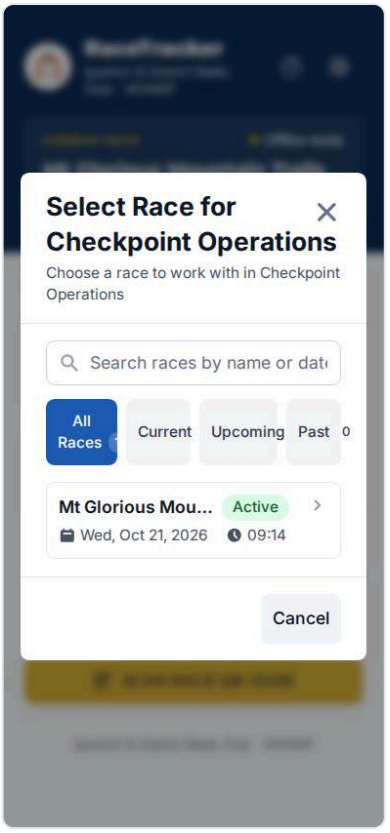
HOME-02 390×844

Plan	§4.1
Event	**Click "Scan race QR code"**
Plan says	*Opens the existing import flow (QR scanner).*
App showed	• Import / Export Race Configuration
Result	☐ Matches plan ☐ Issue: _____



HOME-03 390×844

Plan	§4.1
Event	**Close the scanner, then click the "Checkpoint" role card**
Plan says	*Role cards keep the existing navigation (choose race, then checkpoint). Navigation protection unchanged.*
App showed	• Select Race for Checkpoint Operations
Result	☐ Matches plan ☐ Issue: _____



Settings

SET-01 390×844

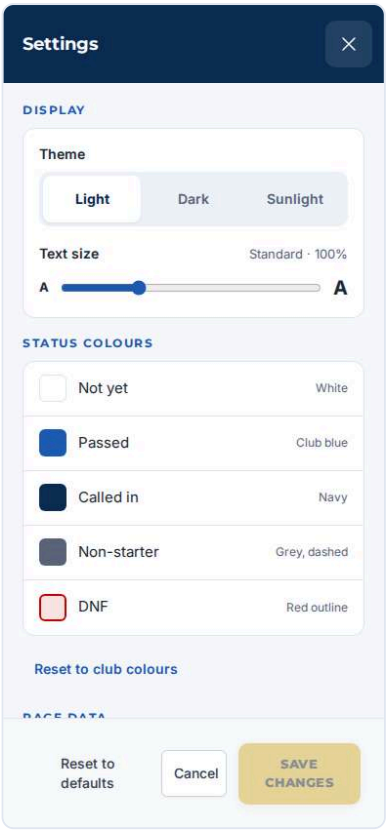
Plan §4.4

Event **Home → click the Settings (cog) button**

Plan says *Sections: Display (Theme segmented: Light / Dark / Sunlight; text-size slider), Status colours (rows with swatch, name, description; "Reset to club colours"), Race data (Share or import race config).*

App showed
- Settings
- Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database

Result ☐ Matches plan ☐ Issue: _____



SET-02 390×844

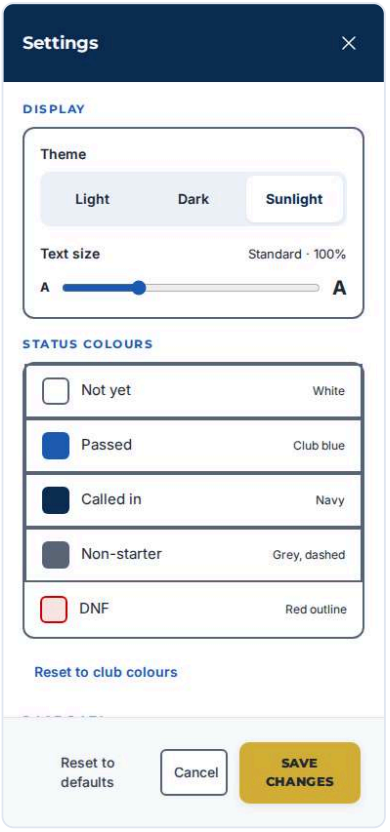
Plan §4.4

Event **Click Theme → "Sunlight"**

Plan says *Sunlight = new high-contrast theme option (pure white bg, ink text, thicker borders).*

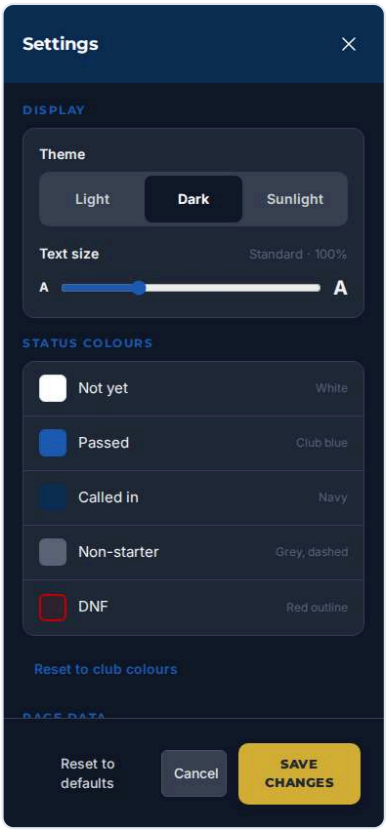
App showed
- html classes: touch-optimized sunlight-mode
- Settings
- Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database

Result ☐ Matches plan ☐ Issue: _____



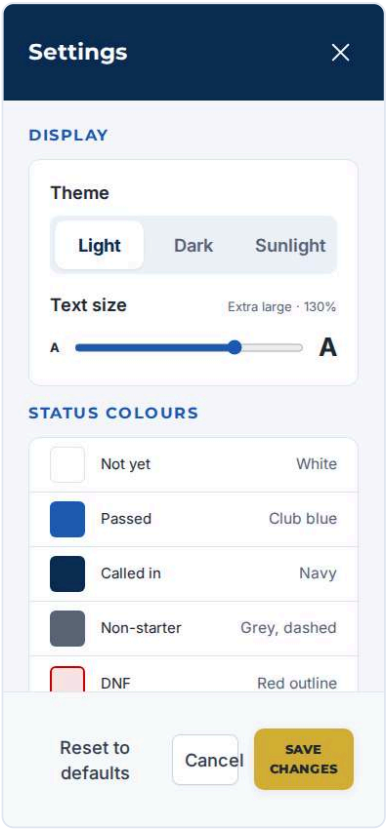
SET-03 390×844

Plan	§4.4 / G7
Event	**Click Theme → "Dark"**
Plan says	*Theme segmented: Light / Dark / Sunlight.*
App showed	• html classes: touch-optimized dark • Settings • Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database
Result	☐ Matches plan ☐ Issue: _____



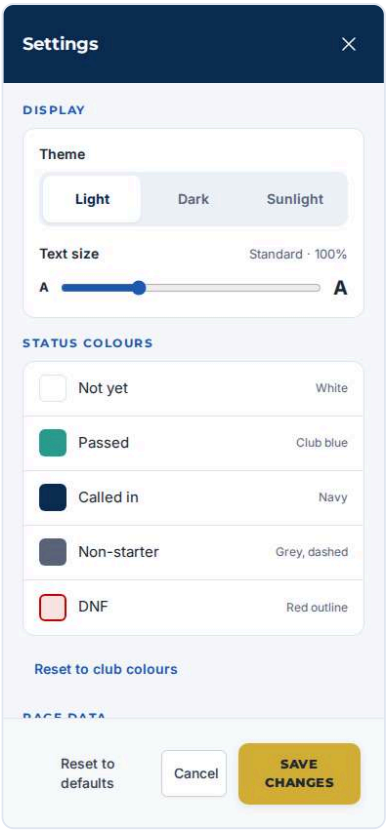
SET-04 390×844

Plan	§3.2 / §4.4
Event	**Click "Light", then drag Text size to 130%**
Plan says	*Keep the existing --font-scale setting; all sizes scale with it.*
App showed	• --font-scale = 1.3 • Settings • Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database
Result	☐ Matches plan ☐ Issue: _____



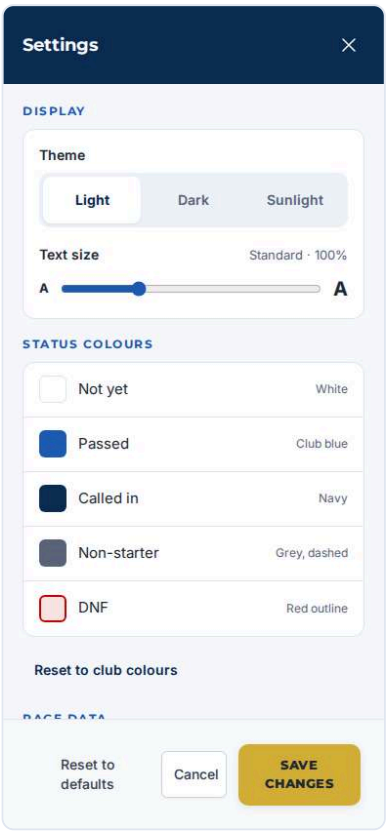
SET-05 390×844

Plan	§4.4 / G7
Event	**Change the "Passed" colour to #2a9d8f with its swatch**
Plan says	*Users may still recolour in Settings.*
App showed	• --color-status-passed = 42 157 143 • Settings • Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database
Result	☐ Matches plan ☐ Issue: _____



SET-06 390×844

Plan	§4.4
Event	**Click "Reset to club colours"**
Plan says	*Status colours … "Reset to club colours".*
App showed	• --color-status-passed = 30 92 179 (club blue = 30 92 179) • Settings • Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database
Result	☐ Matches plan ☐ Issue: _____



Share race config

SHARE-01 390×844

Plan §4.3

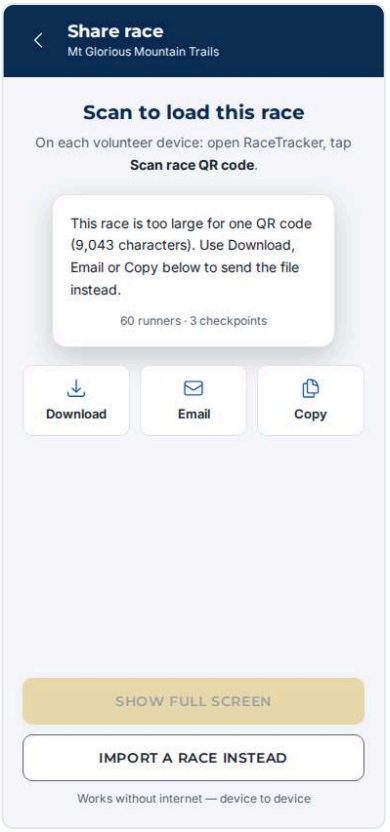
Event **Settings → Race data → "Share or import race config"**

Plan says *Existing QR export: large QR in a white card with shadow, runner/checkpoint count; Download / Email / Copy tiles; primary "Show full screen" (QRDisplayFullscreen). Note "Works without internet — device to device".*

App showed

- Settings
- Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database
- Share race
- Scan to load this race

Result ☐ Matches plan ☐ Issue: _____



SHARE-02 390×844

Plan §4.3

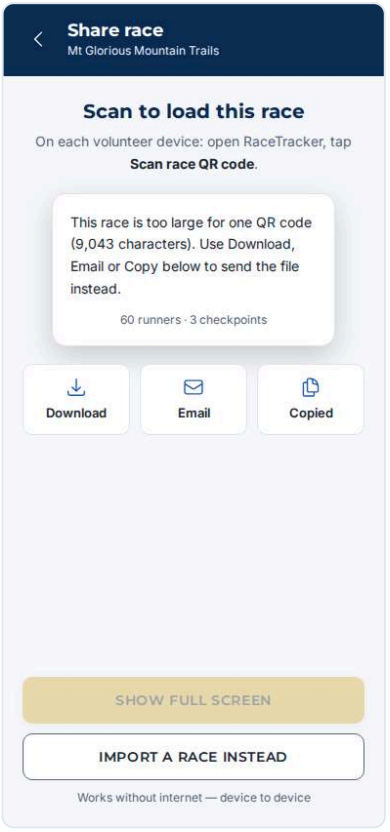
Event **Click the "Copy" tile**

Plan says *Download / Email / Copy tiles.*

App showed

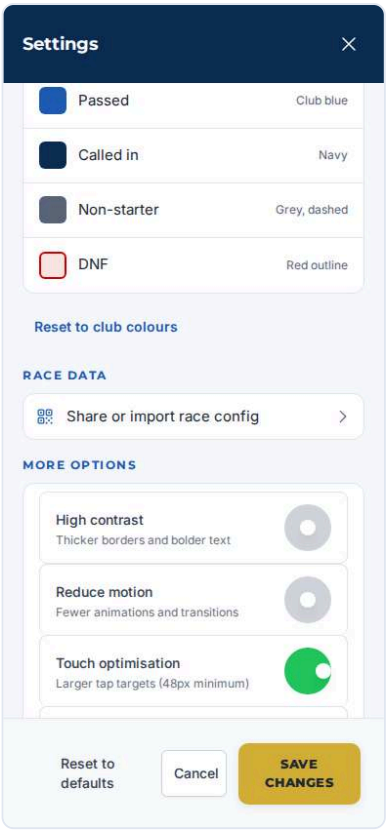
- Settings
- Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database
- Share race
- Scan to load this race

Result ☐ Matches plan ☐ Issue: _____



SHARE-03 390×844

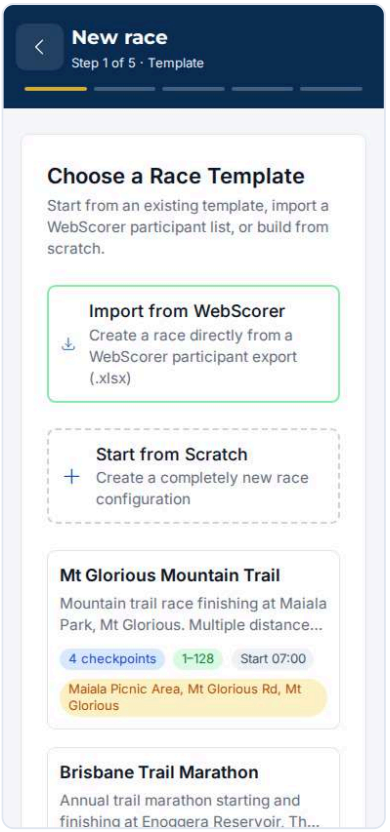
Plan	§4.3
Event	**Click back ("Close share race")**
Plan says	*Returns to Settings.*
App showed	• Settings • Clear database Permanently deletes all races, runners and settings on this device. This cannot be undone. Clear database
Result	☐ Matches plan ☐ Issue: _____



Race setup

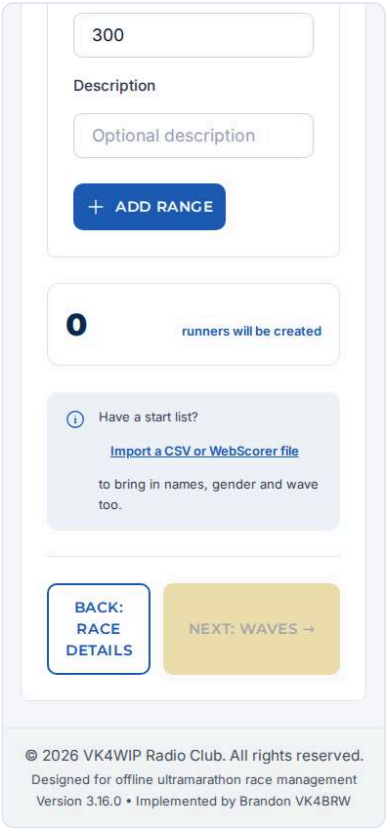
SETUP-01 390×844

Plan	§4.2
Event	**Open /race-maintenance/setup**
Plan says	*Navy header with step "Step 3 of 4 · Runners" and a 4-segment gold progress bar. Restyle only; keep the wizard.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



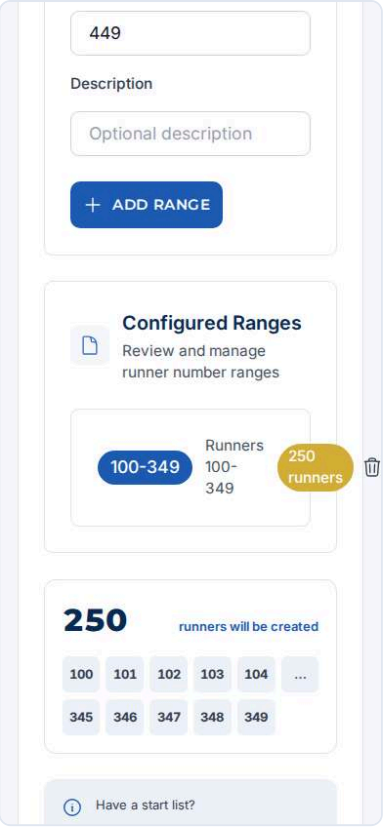
SETUP-02 390×844

Plan	§4.2
Event	**Click "Start from Scratch", enter name / date / start time, click Next**
Plan says	*Keep the wizard (Back secondary, Next primary).*
App showed	• 0 runners will be created
Result	☐ Matches plan ☐ Issue: _____



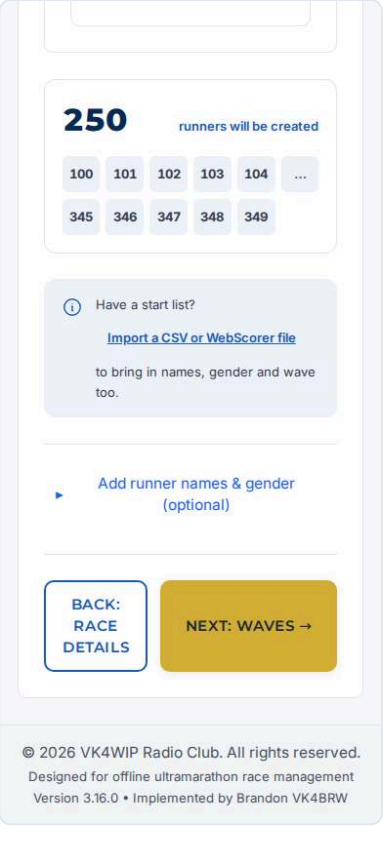
SETUP-03 390×844

Plan	§4.2 / G1
Event	**Enter From 100, To 349 and click "Add Range"**
Plan says	*Runner ranges input (ranges allowed here only), live count ("250 runners will be created") and first/last preview chips. Link to CSV / WebScorer import. Back (secondary) + Next (primary).*
App showed	• 250 runners will be created 100 101 102 103 104 … 345 346 347 348 349
Result	☐ Matches plan ☐ Issue: _____



SETUP-04 390×844

Plan	§4.2
Event	**Scroll to the bottom of the runners step**
Plan says	*Link to CSV / WebScorer import. Back (secondary) + Next (primary).*
App showed	• 250 runners will be created 100 101 102 103 104 … 345 346 347 348 349
Result	☐ Matches plan ☐ Issue: _____



Checkpoint — Mark off

CP-01 390×844

Plan	§4.5 Layout
Event	**Open /checkpoint/3**
Plan says	*AppBar (checkpoint number, location + operator callsign) with 3 StatTiles (Passed, Still to come, To call in gold) → quick-entry input + gold Mark → status line (aria-live) → group chips (20 per group) → 4-column tile grid (64px rows) → CheckpointTabBar.*
App showed	• Tap a number as the runner passes. Tap again to fix its time.
Result	☐ Matches plan ☐ Issue: _____



CP-02 390×844

Plan	§4.5 Behaviour 1
Event	**Tap tile 101 (not passed)**
Plan says	*Tap a not-passed tile → mark passed at now (actualTime = now, commonTime = block).*
App showed	• 101 marked at 11:14
Result	☐ Matches plan ☐ Issue: _____



CP-03 390×844

Plan §4.5 Behaviour 6

Event **Type "103, 104 105" in quick entry and press Enter**

Plan says *Quick entry accepts multiple numbers separated by commas/spaces (individual numbers only), marks each at now, jumps the grid to the group of the first number, message "3 runners marked at 07:47".*

App showed
- 3 runners marked at 11:14
- 3 runners marked UNDO

Result ☐ Matches plan ☐ Issue: _____



CP-04 390×844

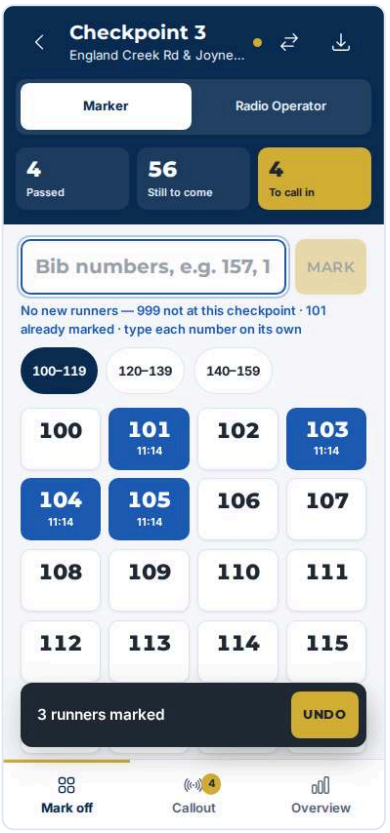
Plan G1 / §4.5 Behaviour 6

Event **Type "110-115 999 101" and press Enter**

Plan says *Individual numbers only (no ranges outside Race setup). Unknown / already-marked numbers are not marked.*

App showed
- No new runners — 999 not at this checkpoint · 101 already marked · type each number on its own
- 3 runners marked UNDO

Result ☐ Matches plan ☐ Issue: _____



CP-05 390×844

Plan	§4.5 Behaviour 6
Event	**Type "125" and press Enter**
Plan says	*Jumps the grid to the group of the first number.*
App showed	• 125 marked at 11:14 • 125 marked UNDO
Result	☐ Matches plan ☐ Issue: _____



CP-06 390×844

Plan	§4.5
Event	**Tap group chip "100–119"**
Plan says	*Group chips (20 per group: 100–119 …, horizontally scrolling).*
App showed	• 125 marked at 11:14 • 125 marked UNDO
Result	☐ Matches plan ☐ Issue: _____



CP-07 390×844

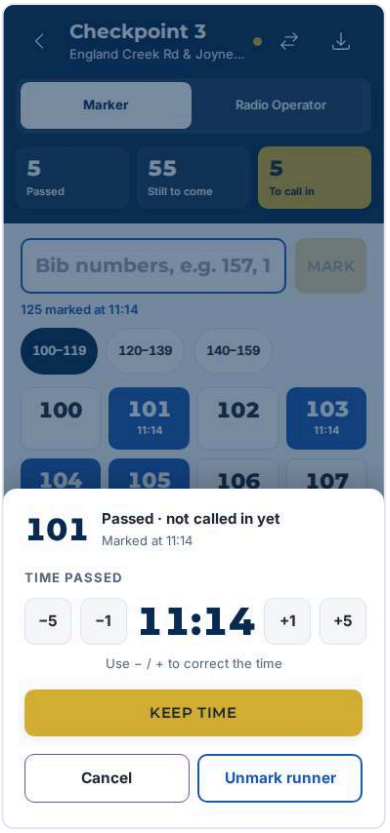
Plan §4.5 Behaviour 2–3

Event **Tap tile 101 again (passed)**

Plan says *Tap a passed or called-in tile → opens the Edit sheet (never unmarks directly). Edit sheet: bib, status line ("Passed · not called in yet" / "Called in to base"), "Marked at HH:MM". TimeStepper (clamped). Primary Save HH:MM (or "Keep time" if unchanged). Secondary Unmark runner. Cancel.*

App showed
- 125 marked at 11:14
- 125 marked UNDO
- 101
- 11:14

Result ☐ Matches plan ☐ Issue: _____



CP-08 390×844

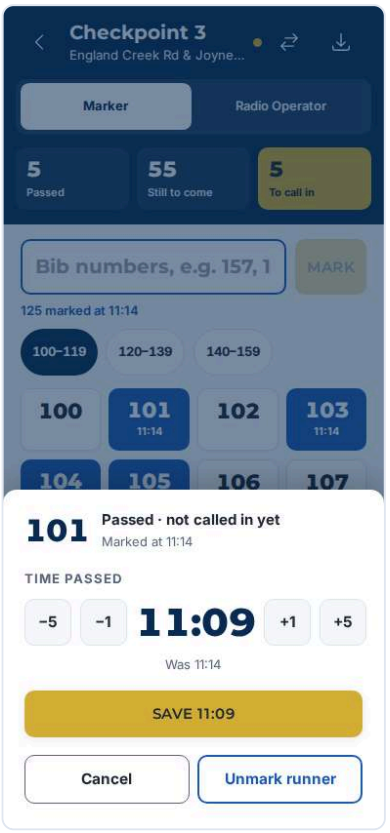
Plan §3.5 TimeStepper

Event **In the sheet tap "−5"**

Plan says *TimeStepper: −5 / −1 / big time / +1 / +5. Clamp to [race start, now]. Primary changes to "Save HH:MM".*

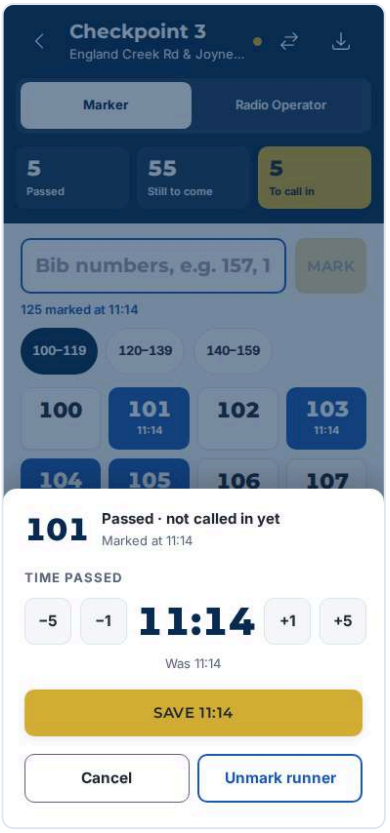
App showed
- 125 marked at 11:14
- 125 marked UNDO
- 101
- 11:09

Result ☐ Matches plan ☐ Issue: _____



CP-09 390×844

Plan	§3.5 TimeStepper
Event	**Tap "+5" three times**
Plan says	*Clamp to [race start, now] — the time cannot go past now.*
App showed	• 125 marked at 11:14 • 125 marked UNDO • 101 • 11:14
Result	☐ Matches plan ☐ Issue: _____



CP-10 390×844

Plan	§4.5 Behaviour 3
Event	**Tap "−5" twice, then "Save HH:MM"**
Plan says	*Save changes the time; status line confirms.*
App showed	• 101 time changed to 11:04 • 101 time changed UNDO
Result	☐ Matches plan ☐ Issue: _____



CP-11 390×844

Plan	§4.5 Behaviour 4
Event	**Tap tile 103 → "Unmark runner"**
Plan says	*Unmark → tile becomes "unmarked earlier" (dashed blue, was HH:MM) and an UndoBar "142 unmarked · Undo" appears.*
App showed	• 103 unmarked UNDO
Result	☐ Matches plan ☐ Issue: _____



CP-12 390×844

Plan	§4.5 Behaviour 4 / G5
Event	**Tap "Undo" on the undo bar**
Plan says	*Undo restores the original record exactly.*
App showed	• 103 restored at 11:14
Result	☐ Matches plan ☐ Issue: _____



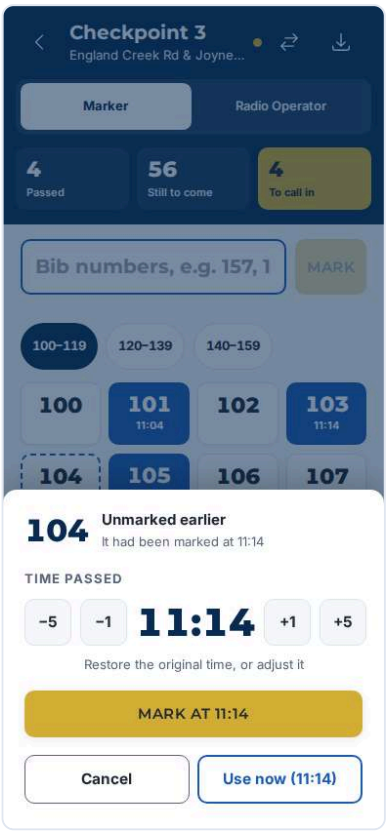
CP-13 390×844

Plan	§4.5 Behaviour 4
Event	**Tap tile 104 → "Unmark runner", wait for the undo bar to clear**
Plan says	*Tile becomes "unmarked earlier" (dashed blue, was HH:MM).*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



CP-14 390×844

Plan	§4.5 Behaviour 5
Event	**Tap tile 104 (unmarked earlier)**
Plan says	*Sheet in restore mode: "Unmarked earlier · It had been marked at 07:37", TimeStepper preset to the original, primary Mark at 07:37, secondary Use now (HH:MM).*
App showed	• 104 • 11:14
Result	☐ Matches plan ☐ Issue: _____



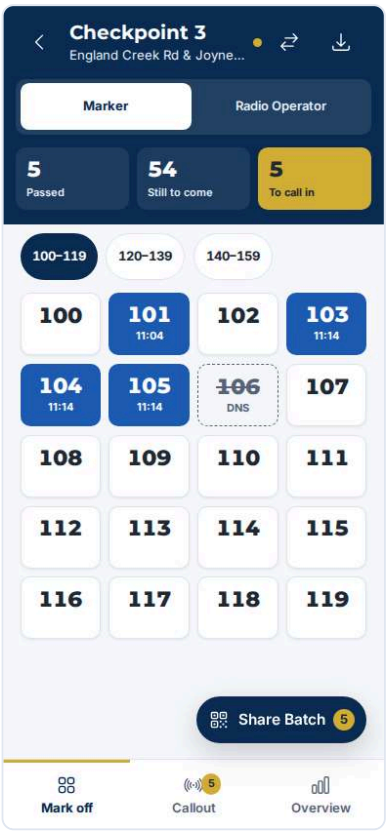
CP-15 390×844

Plan	§4.5 Behaviour 5
Event	**Tap "Mark at HH:MM"**
Plan says	*Restores the original time.*
App showed	• 104 marked at 11:14 • 104 marked UNDO
Result	☐ Matches plan ☐ Issue: _____



CP-16 390×844

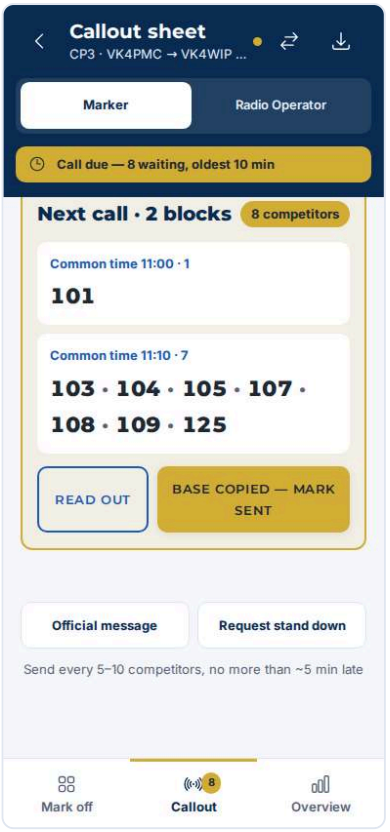
Plan	§4.5 Behaviour 7
Event	**In Overview mark 106 DNS, back to Mark off, tap tile 106**
Plan says	*DNS/DNF tiles are not editable here (use Overview).*
App showed	• 106 is DNS — change it in Overview
Result	☐ Matches plan ☐ Issue: _____



Checkpoint — Callout sheet

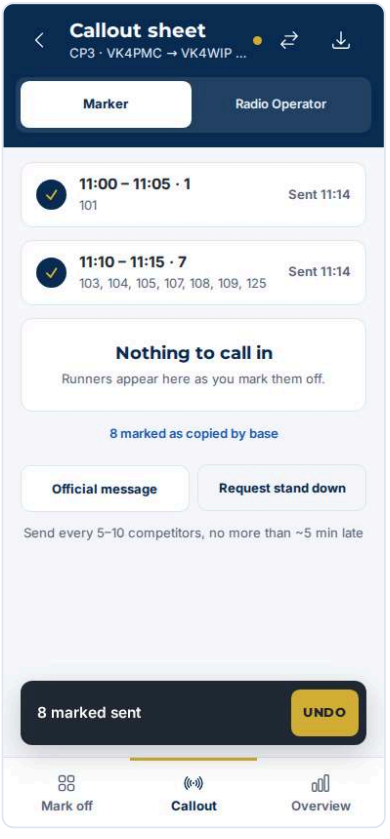
CO-01 390×844

Plan	§4.6
Event	**Mark 107 108 109, then tap the "Callout" tab**
Plan says	*AppBar subtitle: CP3 · VK4PMC → VK4WIP · 147.225. Call-due banner (gold) when ≥ 5 runners waiting or oldest ≥ 5 min: "Call due — 9 waiting, oldest 6 min". Next call card (gold border): heading "Next call · 2 blocks", badge "9 competitors", one white card per pending block: "Common time 07:40 · 5" + numbers in Montserrat 800 22px separated by " · ". Buttons: Read out + Base copied — mark sent. Below: Official message and Request stand down. No radio phrases on screen.*
App showed	• Call due — 8 waiting, oldest 10 min
Result	☐ Matches plan ☐ Issue: _____



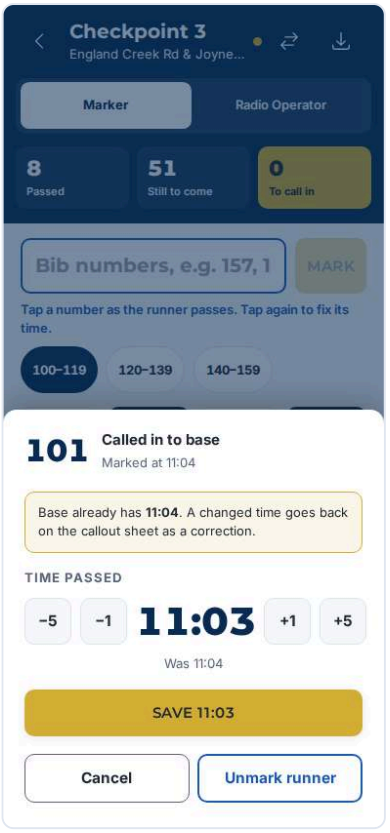
CO-02 390×844

Plan	§4.6
Event	**Tap "Base copied — mark sent"**
Plan says	*Marks every runner in the pending blocks calledIn = true, callInTime = now. Called-in blocks collapse to one compact row (navy tick, 07:30 – 07:35 · 4, numbers, "Sent 07:36").*
App showed	• 8 marked as copied by base • 8 marked sent UNDO
Result	☐ Matches plan ☐ Issue: _____



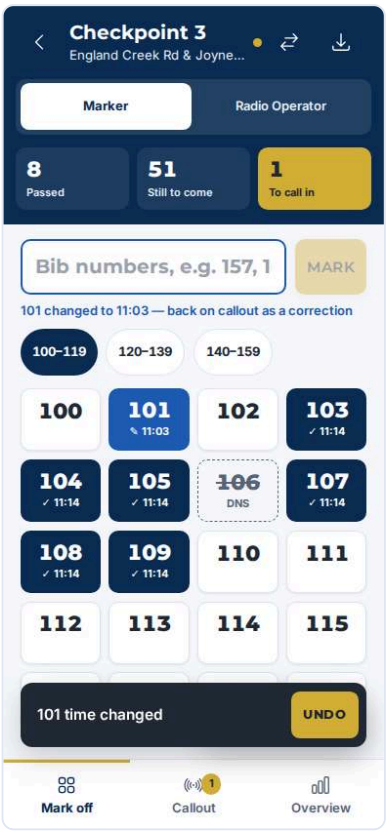
CO-03 390×844

Plan	§4.5 Behaviour 3 / §4.6
Event	**Mark off → tap 101 (called in) → "−1"**
Plan says	*If the runner was called in and the time changes → warning box "Base already has 07:31. A changed time goes back on the callout sheet as a correction."*
App showed	• Tap a number as the runner passes. Tap again to fix its time. • 101 • 11:03
Result	☐ Matches plan ☐ Issue: _____



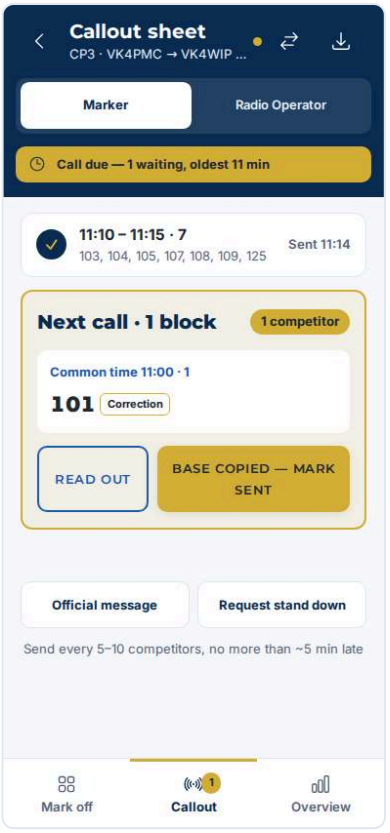
CO-04 390×844

Plan	§4.5 Behaviour 3
Event	**Tap "Save HH:MM"**
Plan says	*Saving sets status back to passed with correction = true so it reappears on the callout sheet; tile shows ✎.*
App showed	• 101 changed to 11:03 — back on callout as a correction • 101 time changed UNDO
Result	☐ Matches plan ☐ Issue: _____



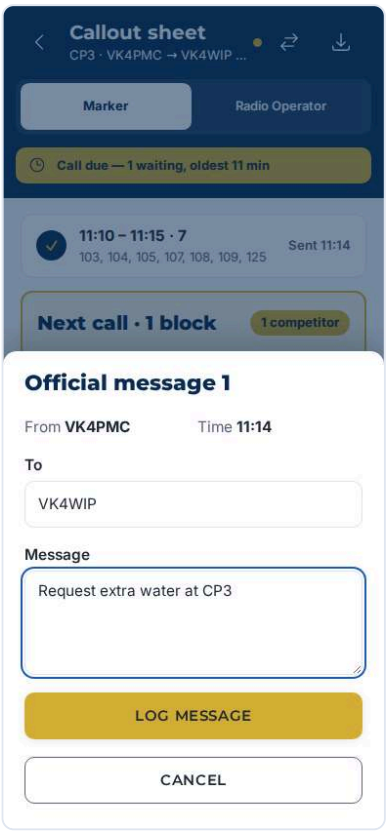
CO-05 390×844

Plan	§4.6
Event	**Tap the "Callout" tab**
Plan says	*Corrections (runners with correction = true) appear in the pending list tagged "Correction".*
App showed	• Call due — 1 waiting, oldest 11 min
Result	☐ Matches plan ☐ Issue: _____



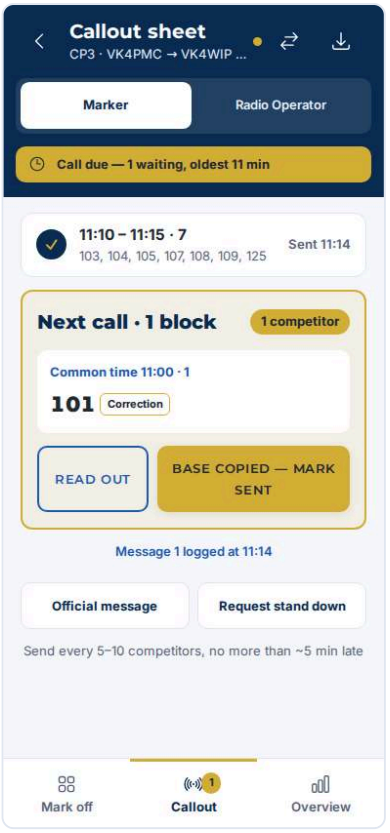
CO-06 390×844

Plan	§4.6
Event	**Tap "Official message", type a message**
Plan says	*Official message ... (plain buttons; v1 can open a simple notes modal / log entry).*
App showed	• Call due — 1 waiting, oldest 11 min • Official message 1
Result	☐ Matches plan ☐ Issue: _____



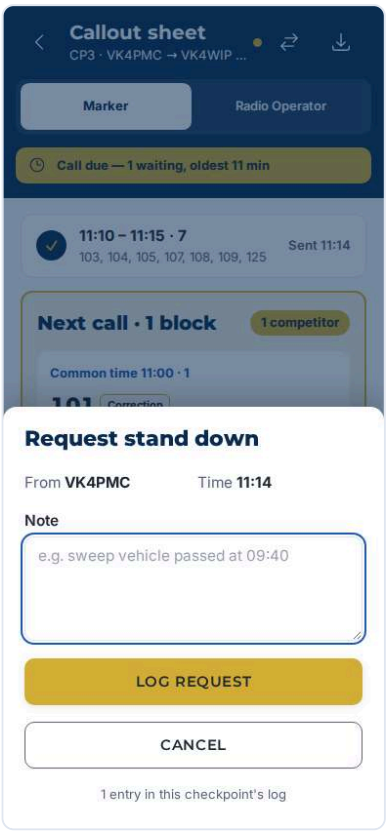
CO-07 390×844

Plan	§4.6
Event	**Tap "Log message"**
Plan says	*v1 can open a simple notes modal / log entry.*
App showed	• Call due — 1 waiting, oldest 11 min • Message 1 logged at 11:14
Result	☐ Matches plan ☐ Issue: _____



CO-08 390×844

Plan	§4.6
Event	**Tap "Request stand down"**
Plan says	*Request stand down (plain button; v1 notes modal / log entry). No radio phrases on screen.*
App showed	• Call due — 1 waiting, oldest 11 min • Message 1 logged at 11:14 • Request stand down
Result	☐ Matches plan ☐ Issue: _____



Checkpoint — Read-out

RO-01 390×844

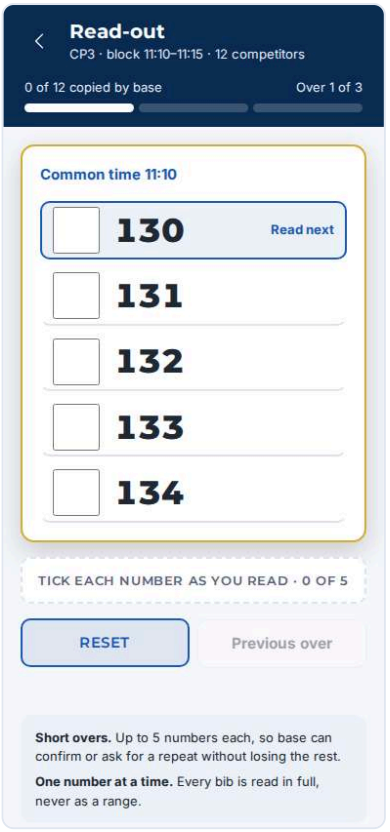
Plan §4.7

Event **Cancel; send the pending correction (Base copied — mark sent); mark 12 runners (130–141); tap "Read out"**

Plan says *Header: "Read-out", CP3 · block 07:45–07:50 · 27 competitors, progress "X of N copied by base", segmented bar (one segment per over). Numbers split into overs of up to 5. Each bib is its own row. Current over: lead "Common time 07:45"; one row per number with a checkbox; next unticked row highlighted ("Read next"). Primary Base copied — next locked until every number in the over is ticked ("Tick each number as you read · 2 of 5").*

App showed
- Common time 11:10 130 Read next 131 132 133 134

Result ☐ Matches plan ☐ Issue: _____



RO-02 390×844

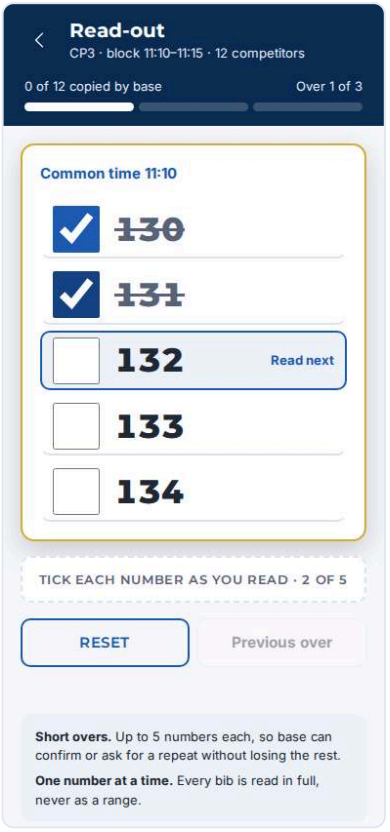
Plan §4.7

Event **Tick the first two numbers**

Plan says *Ticked rows go muted with strike-through; next unticked row highlighted; "Tick each number as you read · 2 of 5".*

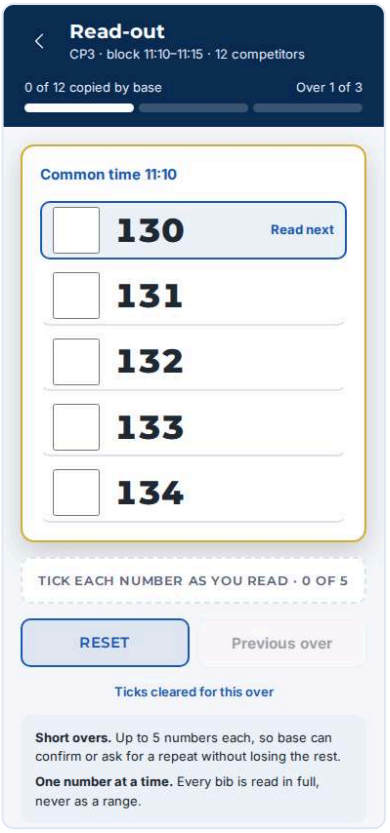
App showed
- Common time 11:10 130 131 132 Read next 133 134

Result ☐ Matches plan ☐ Issue: _____



RO-03 390×844

Plan	§4.7
Event	**Tap "Reset"**
Plan says	*Reset clears ticks on the current over.*
App showed	• Common time 11:10 130 Read next 131 132 133 134 • Ticks cleared for this over
Result	☐ Matches plan ☐ Issue: _____



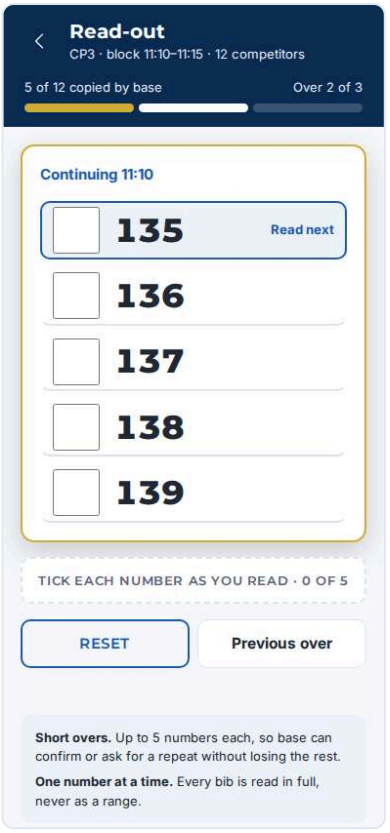
RO-04 390×844

Plan	§4.7
Event	**Tick all numbers in the over**
Plan says	*Primary Base copied — next unlocks when every number is ticked.*
App showed	• Common time 11:10 130 131 132 133 134
Result	☐ Matches plan ☐ Issue: _____



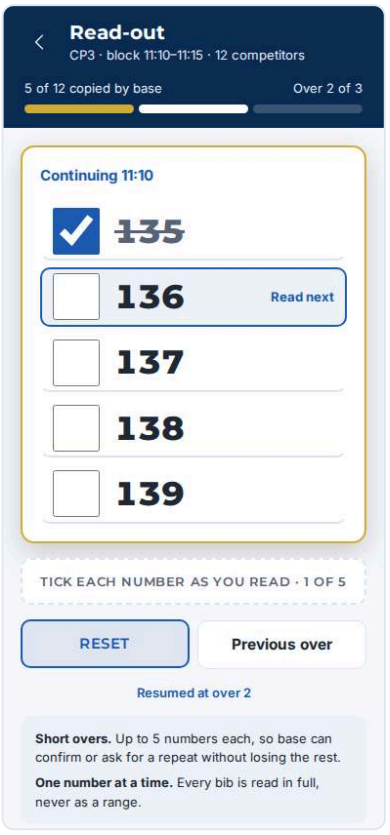
RO-05 390×844

Plan	§4.7
Event	**Tap "Base copied — next"**
Plan says	*Next over: lead "Continuing 07:45"; progress advances.*
App showed	• Continuing 11:10 135 Read next 136 137 138 139
Result	☐ Matches plan ☐ Issue: _____



RO-06 390×844

Plan	§4.7
Event	**Tick one number, tap back, then tap "Read out" again**
Plan says	*Persist read-out progress so an interrupted read resumes at the same over.*
App showed	• Continuing 11:10 135 136 Read next 137 138 139 • Resumed at over 2
Result	☐ Matches plan ☐ Issue: _____



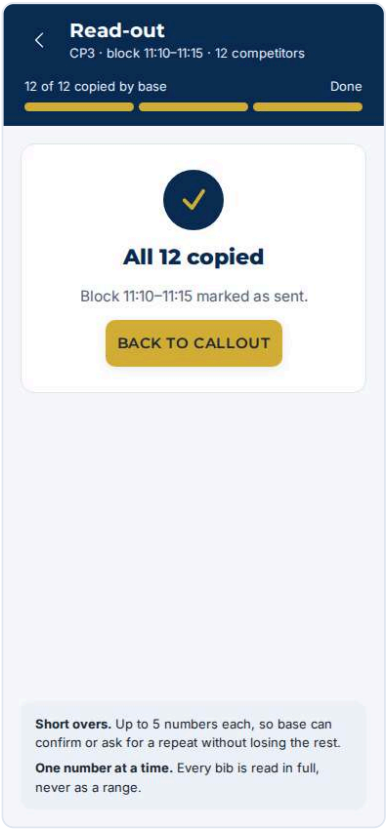
RO-07 390×844

Plan	§4.7
Event	**Tap "Previous over"**
Plan says	*Previous over steps back.*
App showed	• Common time 11:10 130 131 132 133 134
Result	☐ Matches plan ☐ Issue: _____



RO-08 390×844

Plan	§4.7
Event	**Complete every over (tick all → next) to the end**
Plan says	*Finishing the last over → done state "All 27 copied", which marks the block called in.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



Checkpoint — Overview

OV-01 390×844

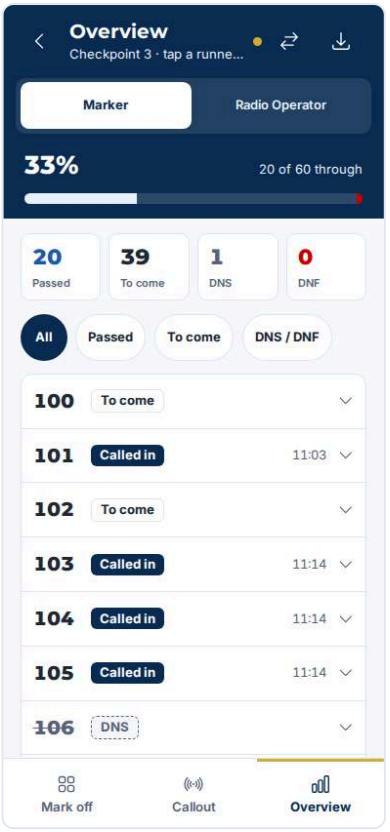
Plan §4.8

Event **Back to callout, tap the "Overview" tab**

Plan says *Header: % through, "142 of 250 through", stacked bar (called in / passed not called / to come / out), role="img" with aria-label. 4 StatTiles: Passed, To come, DNS, DNF. Filter pills: All · Passed · To come · DNS / DNF. Scrollable runner list; tap a row to expand (one open at a time, aria-expanded).*

App showed
- progress bar aria-label: 20 called in, 0 passed not called in, 39 to come, 1 withdrawn

Result ☐ Matches plan ☐ Issue: _____



OV-02 390×844

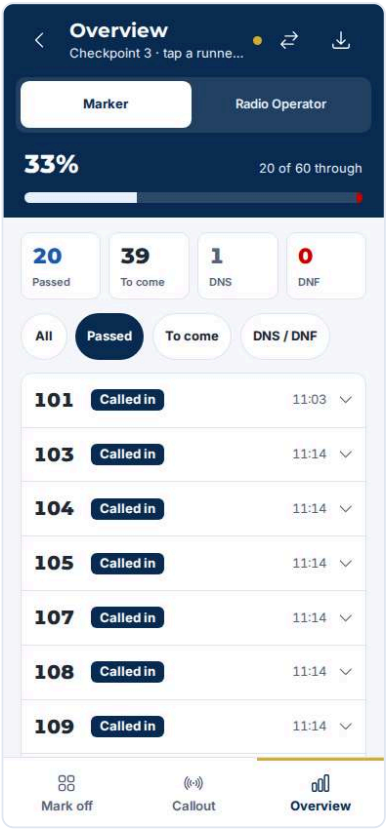
Plan §4.8

Event **Tap filter "Passed"**

Plan says *Filter pills: All · Passed · To come · DNS / DNF.*

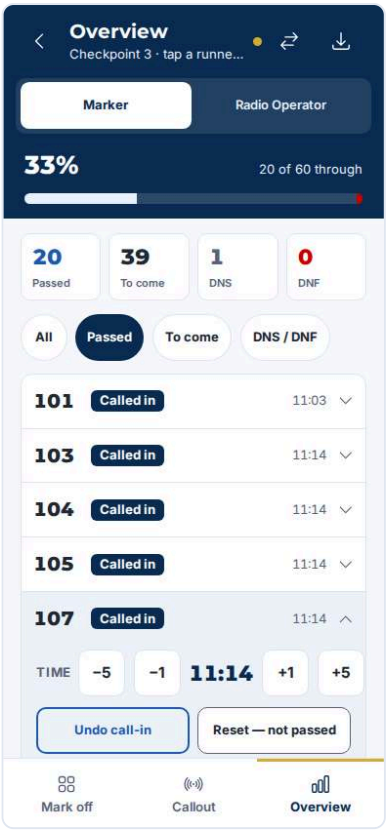
App showed *(no status messages — see screenshot)*

Result ☐ Matches plan ☐ Issue: _____



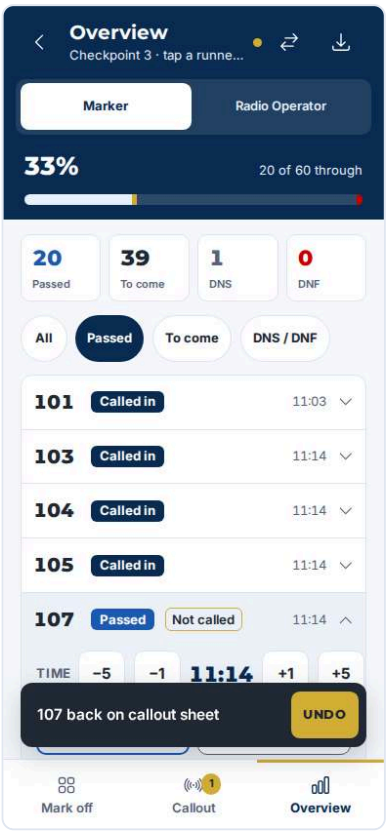
OV-03 390×844

Plan	§4.8 table (Called in)
Event	**Tap row 107 (called in)**
Plan says	*Called in: TimeStepper, Undo call-in, Reset — not passed, Mark DNF (+ note "Base already has HH:MM. Changing the time sends a correction.")*
App showed	• 11:14
Result	☐ Matches plan ☐ Issue: _____



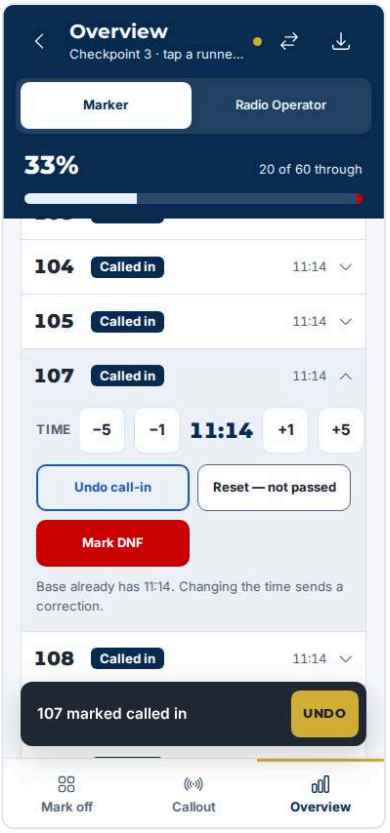
OV-04 390×844

Plan	§4.8
Event	**Tap "Undo call-in"**
Plan says	*Every action shows UndoBar; totals, bar and the Callout badge update live.*
App showed	• 11:14 • 107 back on callout sheet UNDO
Result	☐ Matches plan ☐ Issue: _____



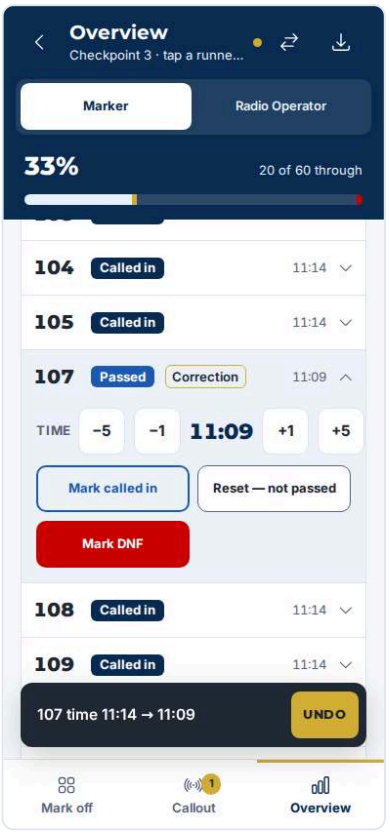
OV-05 390×844

Plan	§4.8 table (Passed)
Event	**Row 107 now Passed: tap "Mark called in"**
Plan says	*Passed: TimeStepper, Mark called in, Reset — not passed, Mark DNF.*
App showed	• 11:14 • 107 marked called in UNDO
Result	☐ Matches plan ☐ Issue: _____



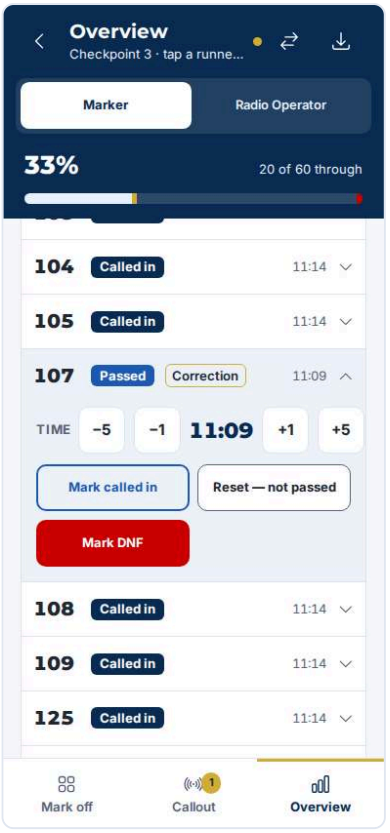
OV-06 390×844

Plan	§4.8
Event	**Tap the time "−5" on row 107 (called in)**
Plan says	*Changing the time of a called-in runner → status passed + correction = true.*
App showed	• 11:09 • 107 time 11:14 → 11:09 UNDO
Result	☐ Matches plan ☐ Issue: _____



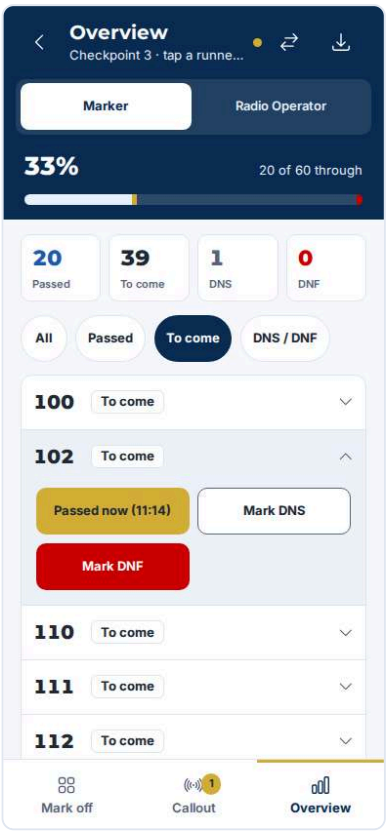
OV-07 390×844

Plan	§4.8
Event	**Tap "Reset — not passed", then "Undo"**
Plan says	*Every action shows UndoBar.*
App showed	• 11:09
Result	☐ Matches plan ☐ Issue: _____



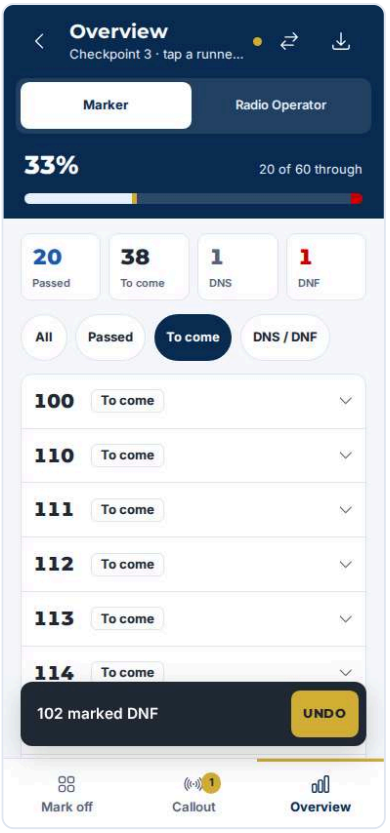
OV-08 390×844

Plan	§4.8 table (To come)
Event	**Filter "To come", tap row 102**
Plan says	*To come: Passed now (HH:MM) (gold), Mark DNS, Mark DNF.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



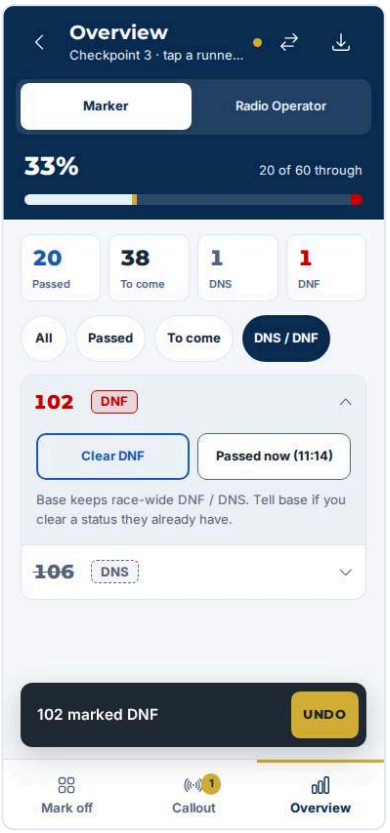
OV-09 390×844

Plan	§4.8
Event	**Tap "Mark DNF"**
Plan says	*DNF shown with red style (§3.3); counts update.*
App showed	• 102 marked DNF UNDO
Result	☐ Matches plan ☐ Issue: _____



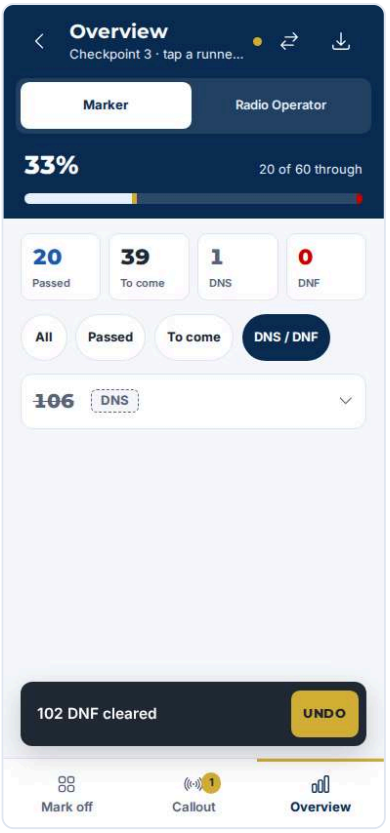
OV-10 390×844

Plan	§4.8 table (DNS / DNF)
Event	**Filter "DNS / DNF", tap row 102**
Plan says	*DNS / DNF: Clear DNS/DNF, Passed now (+ note that base is flagged).*
App showed	• 102 marked DNF UNDO
Result	☐ Matches plan ☐ Issue: _____



OV-11 390×844

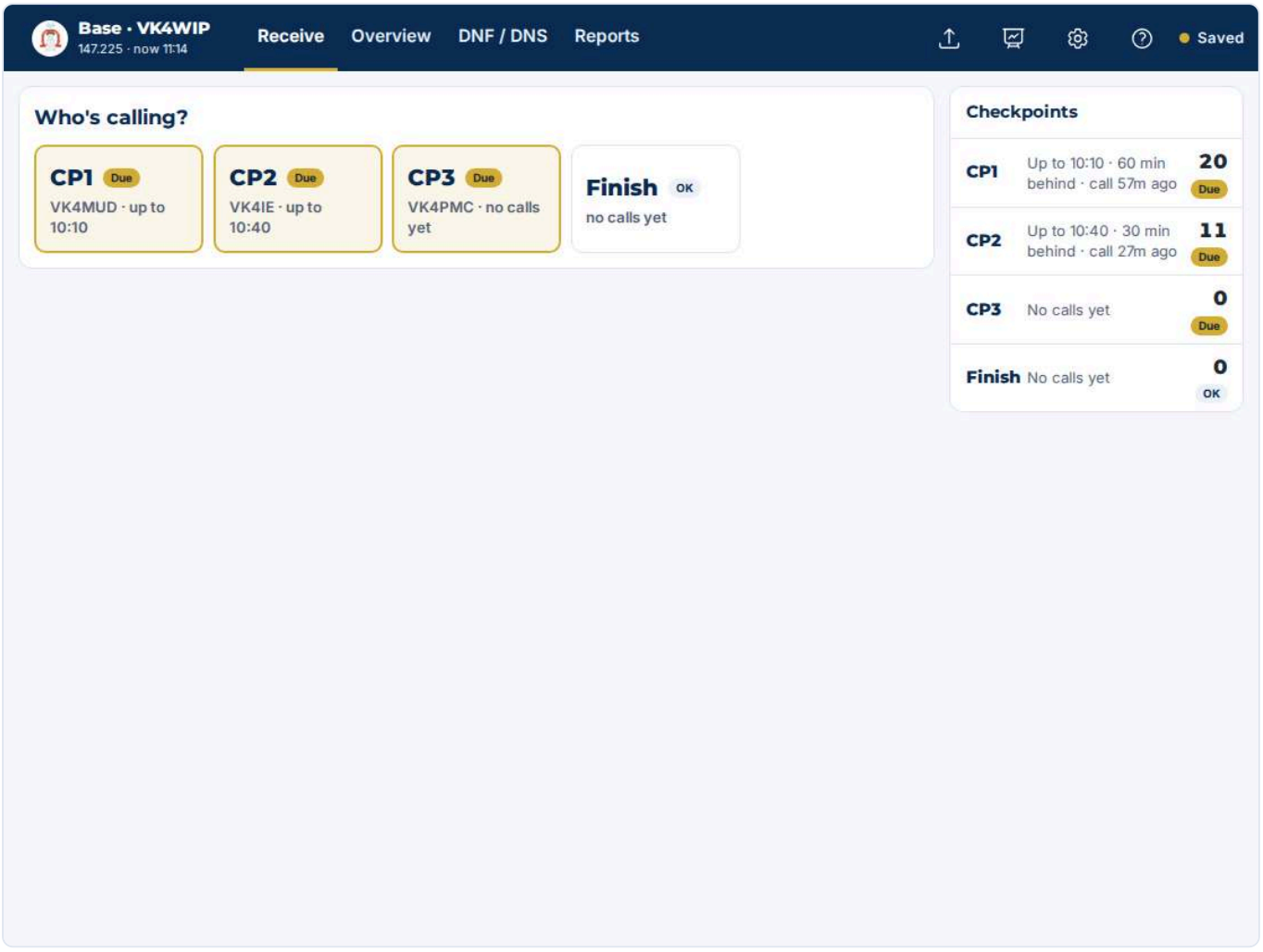
Plan	§4.8
Event	**Tap "Clear DNF"**
Plan says	*Clear DNS/DNF; UndoBar.*
App showed	• 102 DNF cleared UNDO
Result	☐ Matches plan ☐ Issue: _____



Base — Receive

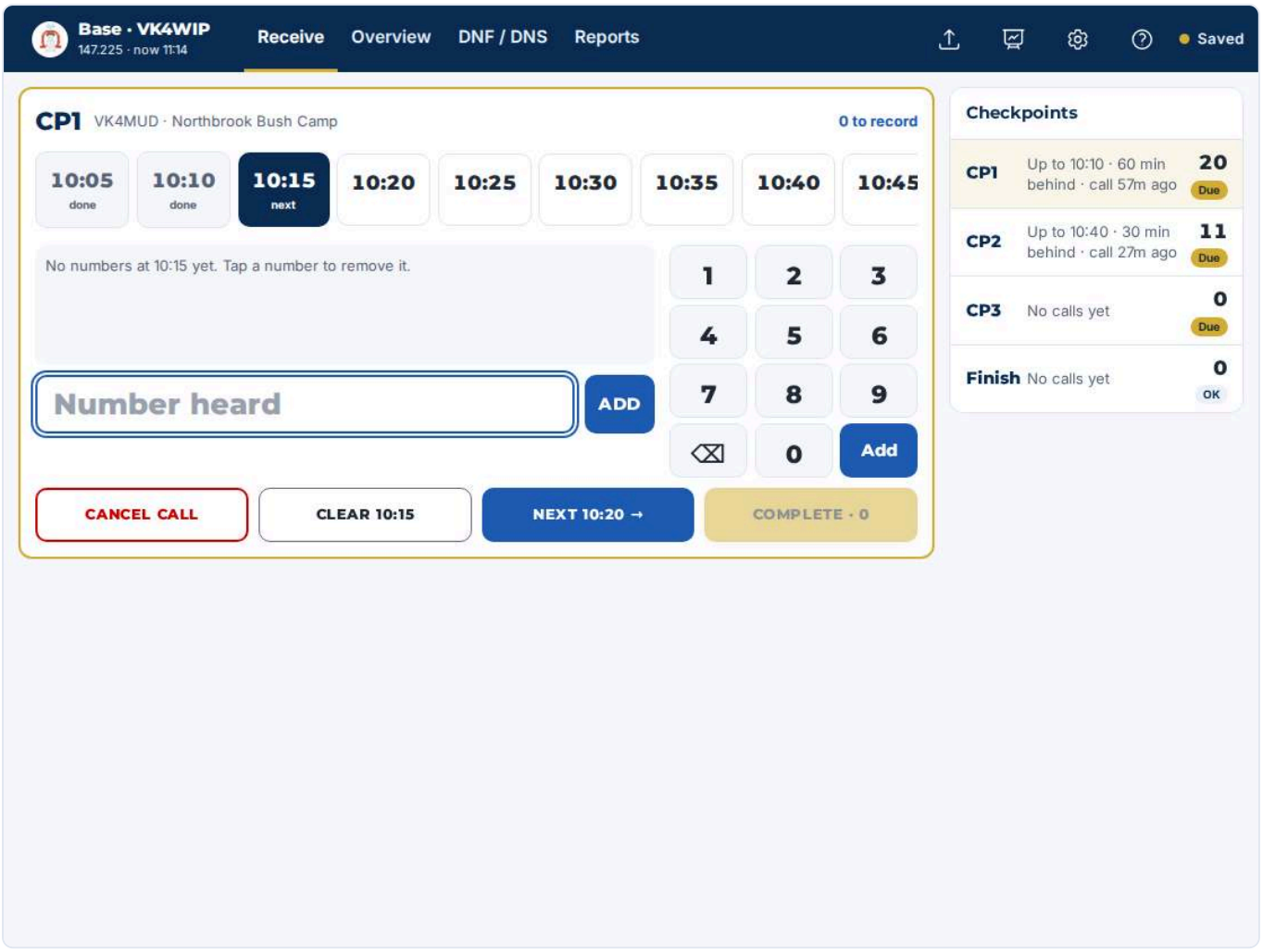
RX-01 1112×834

Plan	§4.9 / §3.5 BaseTabs
Event	**Open /base-station/operations (tablet 1112×834)**
Plan says	*AppBar: badge, "Base · VK4WIP", "147.225 · now HH:MM", inline tabs Receive · Overview · DNF / DNS · Reports, single row on tablet. Idle "Who's calling?": grid of compact buttons (≥64px) for Start, CP1–CP4, Finish: short name, status pill (Due gold / OK / Closed), "VK4PMC · up to 07:15". Checkpoint board: rows with short name, "Up to 07:15 · 30 min behind · call 11m ago", count, Due/OK/Closed pill.*
App showed	• page scroll: 0px
Result	☐ Matches plan ☐ Issue: _____



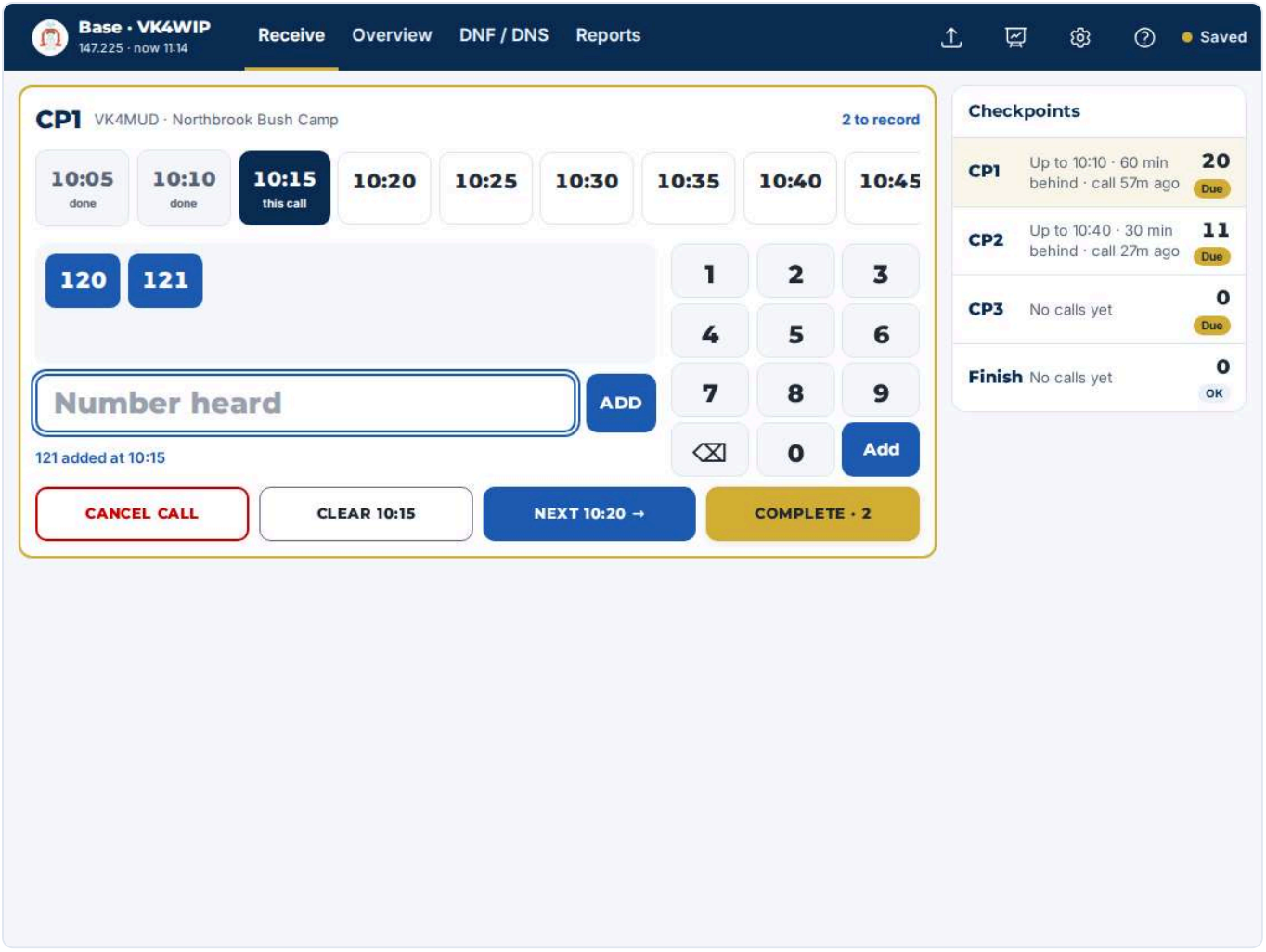
RX-02 1112×834

Plan	§4.9
Event	**Tap "CP1"**
Plan says	*Tap a checkpoint → straight into receiving. Header row: CP short name, operator · location, "N to record". BlockPicker from lastBlock − 5 to the current block. Default selection = that checkpoint's next unreceived block (lastBlock + 5, capped at now). Chips: done (≤ lastBlock, muted), next, this call, selected = navy. "Exact" input.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



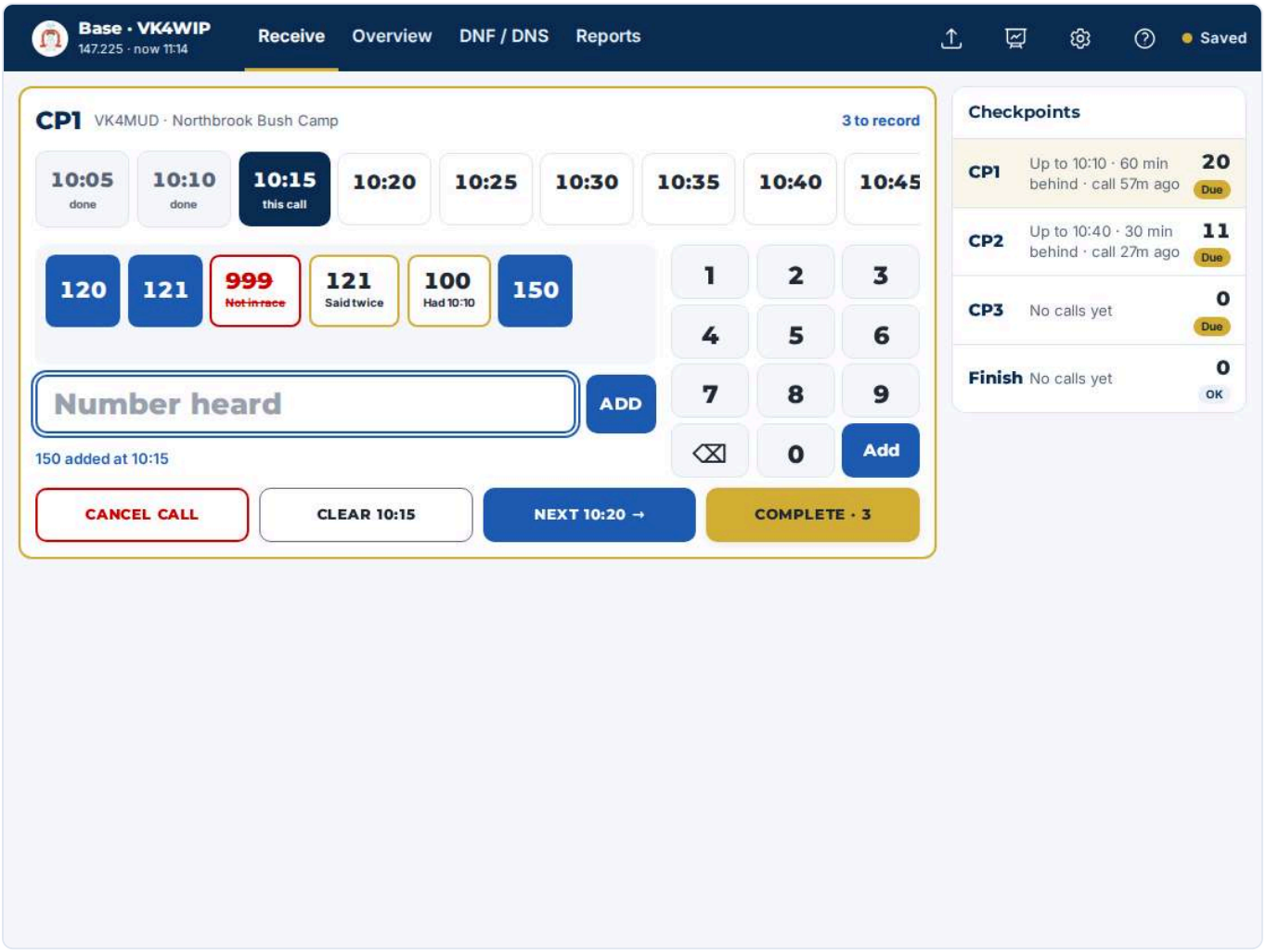
RX-03 1112×834

Plan	§4.9 / §3.5 NumberPad
Event	**Type 120 on the keyboard + Enter; then press keypad 1, 2, 1 and "Add"**
Plan says	*Input + Add + NumberPad side by side. Enter adds. Each added number becomes a BibChip validated live.*
App showed	• 121 added at 10:15
Result	☐ Matches plan ☐ Issue: _____



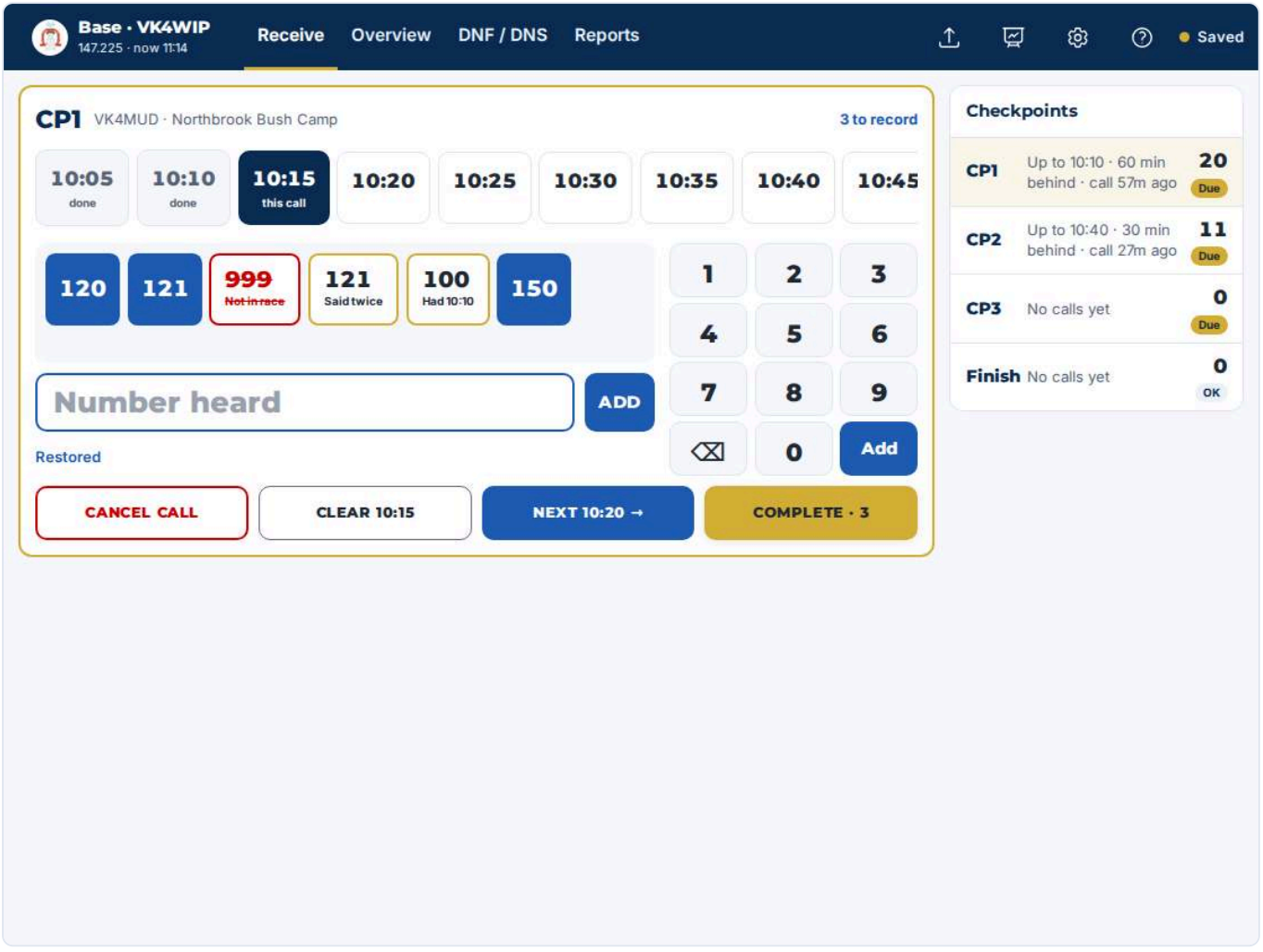
RX-04 1112×834

Plan	§4.9 validation table
Event	**Add 999, 121 again, 100, 150**
Plan says	*Not in race → rejected (red) "Not in race"; same bib earlier in this call → warning (gold) "Said twice"; already logged at this checkpoint → warning "Had 07:40"; otherwise ok (blue). Only ok + info are recorded.*
App showed	• 150 added at 10:15
Result	☐ Matches plan ☐ Issue: _____



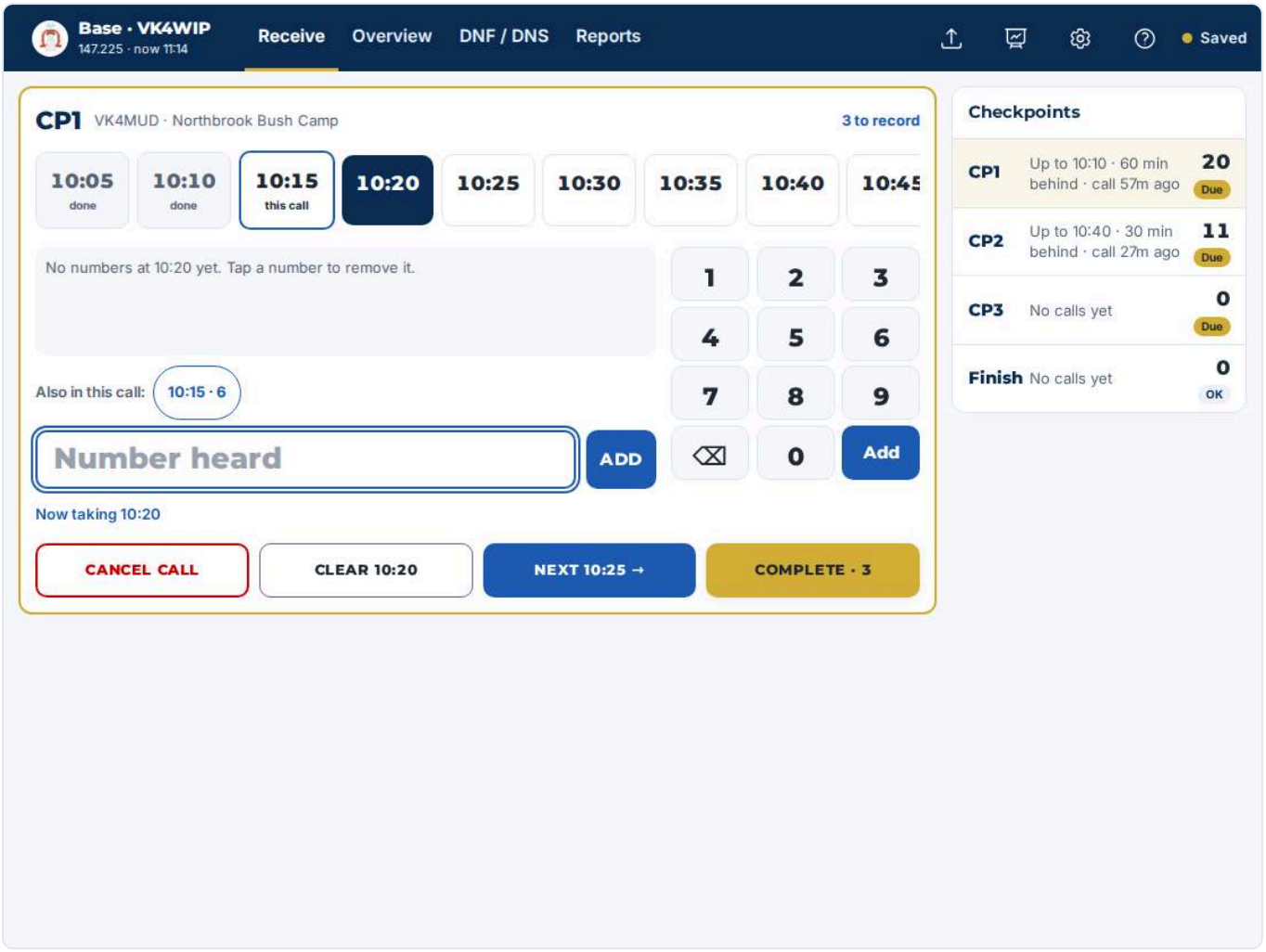
RX-05 1112×834

Plan	§4.9
Event	**Tap chip 999 (remove), then "Undo"**
Plan says	*Tap a chip to remove (Undo). Status line (aria-live) + inline Undo when available.*
App showed	• Restored
Result	☐ Matches plan ☐ Issue: _____



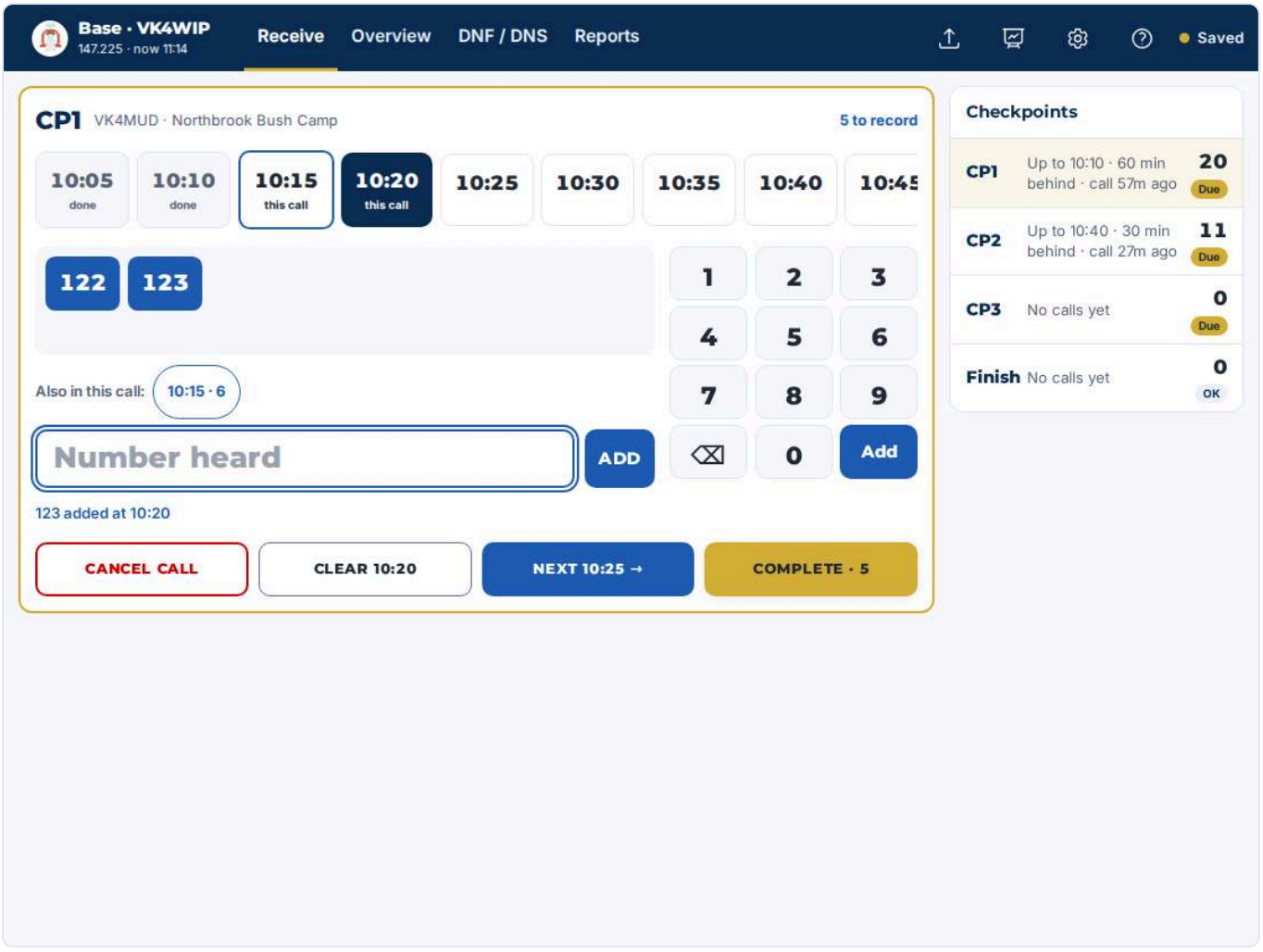
RX-06 1112×834

Plan	§4.9
Event	**Tap "Next HH:MM →"**
Plan says	*Next → move to the next 5-minute block, clear the box; disabled "At now" at the current block. Switching block clears the box and the typed number (nothing is lost).*
App showed	• Now taking 10:20
Result	☐ Matches plan ☐ Issue: _____



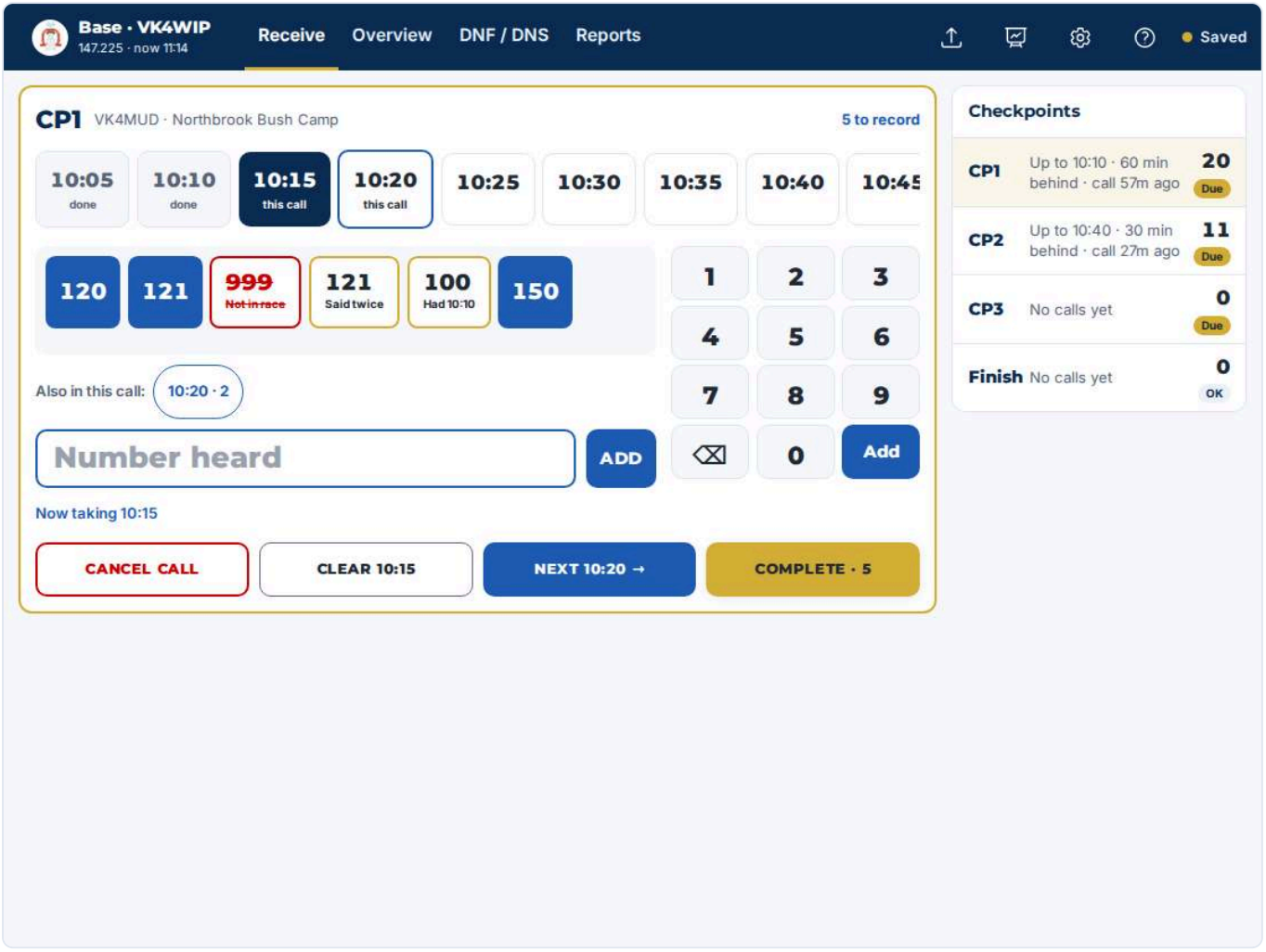
RX-07 1112×834

Plan	§4.9
Event	**Add 122, 123**
Plan says	*Numbers box shows only the current block's numbers. Other blocks in the call show underneath as pills: "Also in this call: 07:40 · 5".*
App showed	• 123 added at 10:20
Result	☐ Matches plan ☐ Issue: _____



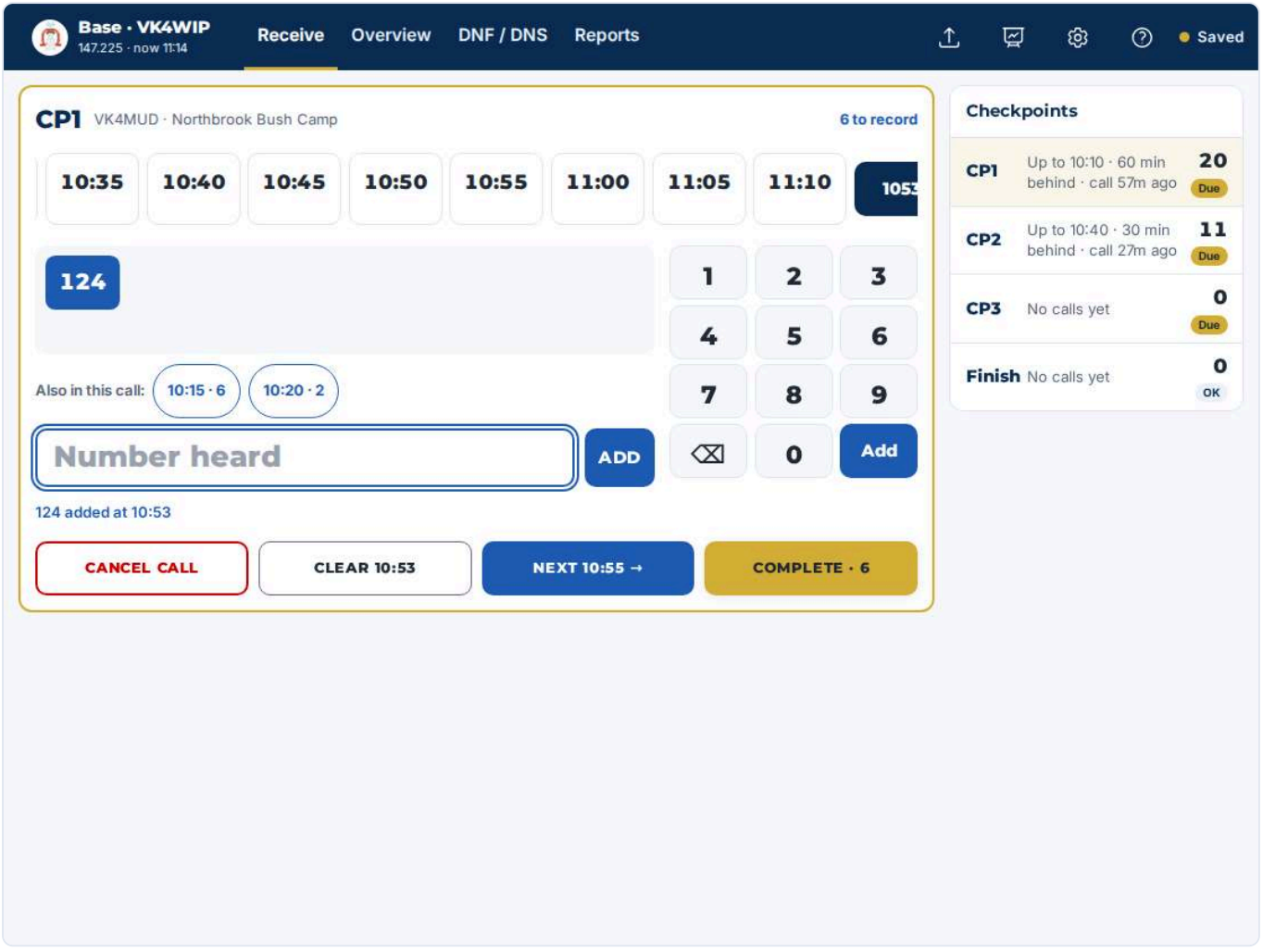
RX-08 1112×834

Plan	§4.9
Event	**Tap the "Also in this call" pill**
Plan says	*Tap to switch back.*
App showed	• Now taking 10:15
Result	☐ Matches plan ☐ Issue: _____



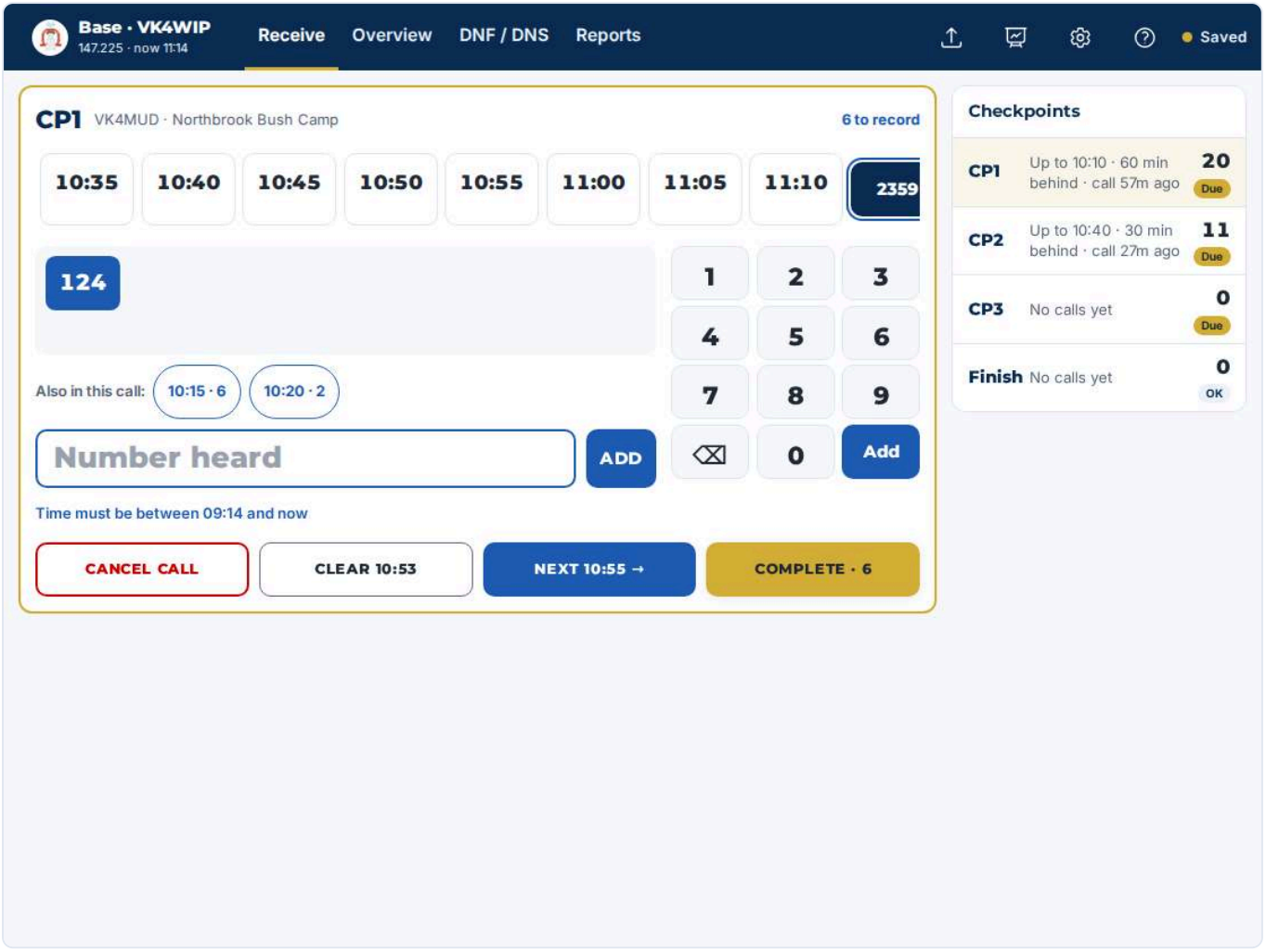
RX-09 1112×834

Plan	§4.9
Event	**Type an exact time in "Exact" (1053)**
Plan says	*"Exact" input accepts HH:MM or HHMM between race start and now; sets a non-block time and shows it selected.*
App showed	• 124 added at 10:53
Result	☐ Matches plan ☐ Issue: _____



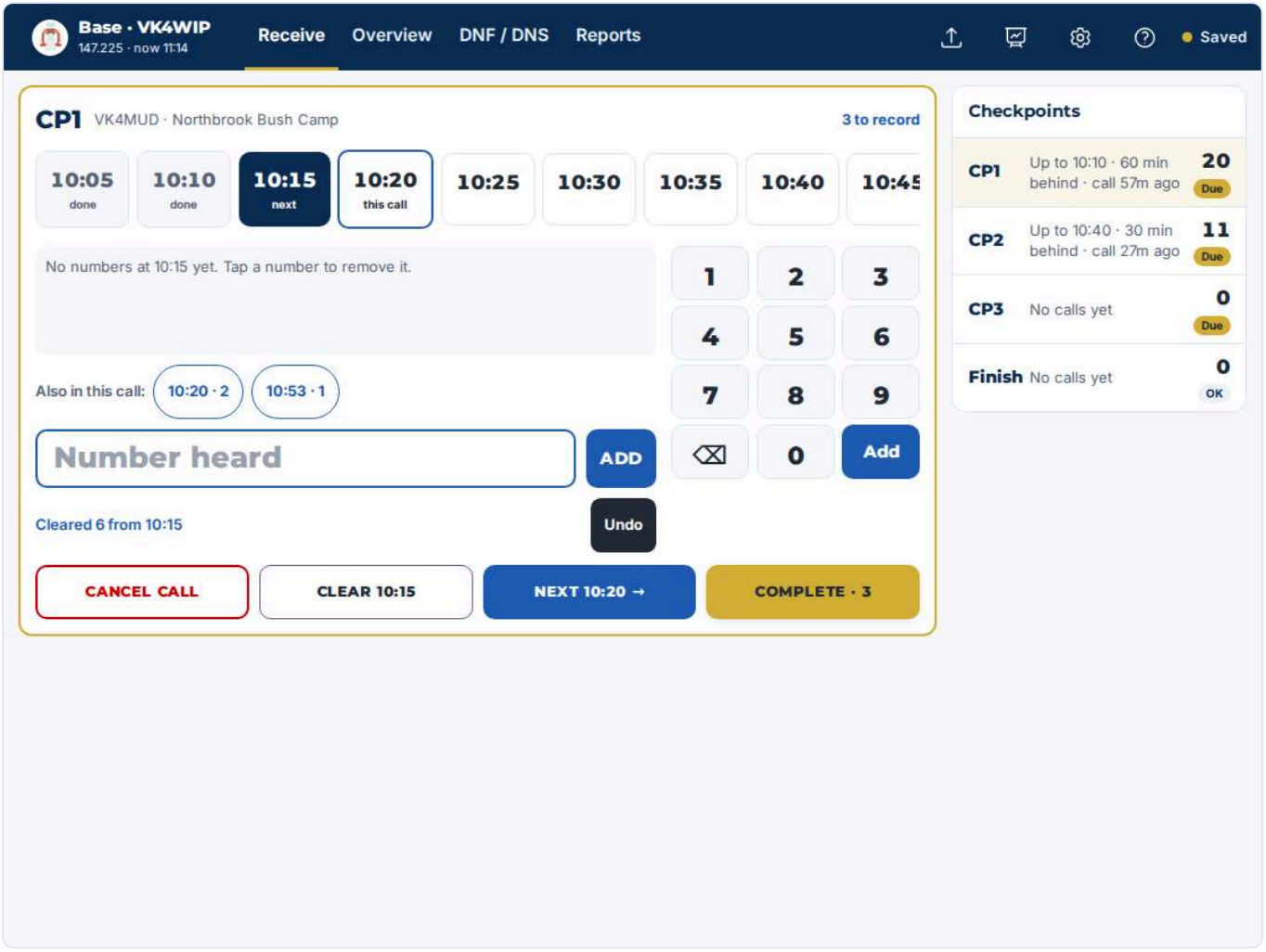
RX-10 1112×834

Plan	§4.9
Event	**Type an exact time in the future (2359)**
Plan says	*Exact time must be between race start and now.*
App showed	• Time must be between 09:14 and now
Result	☐ Matches plan ☐ Issue: _____



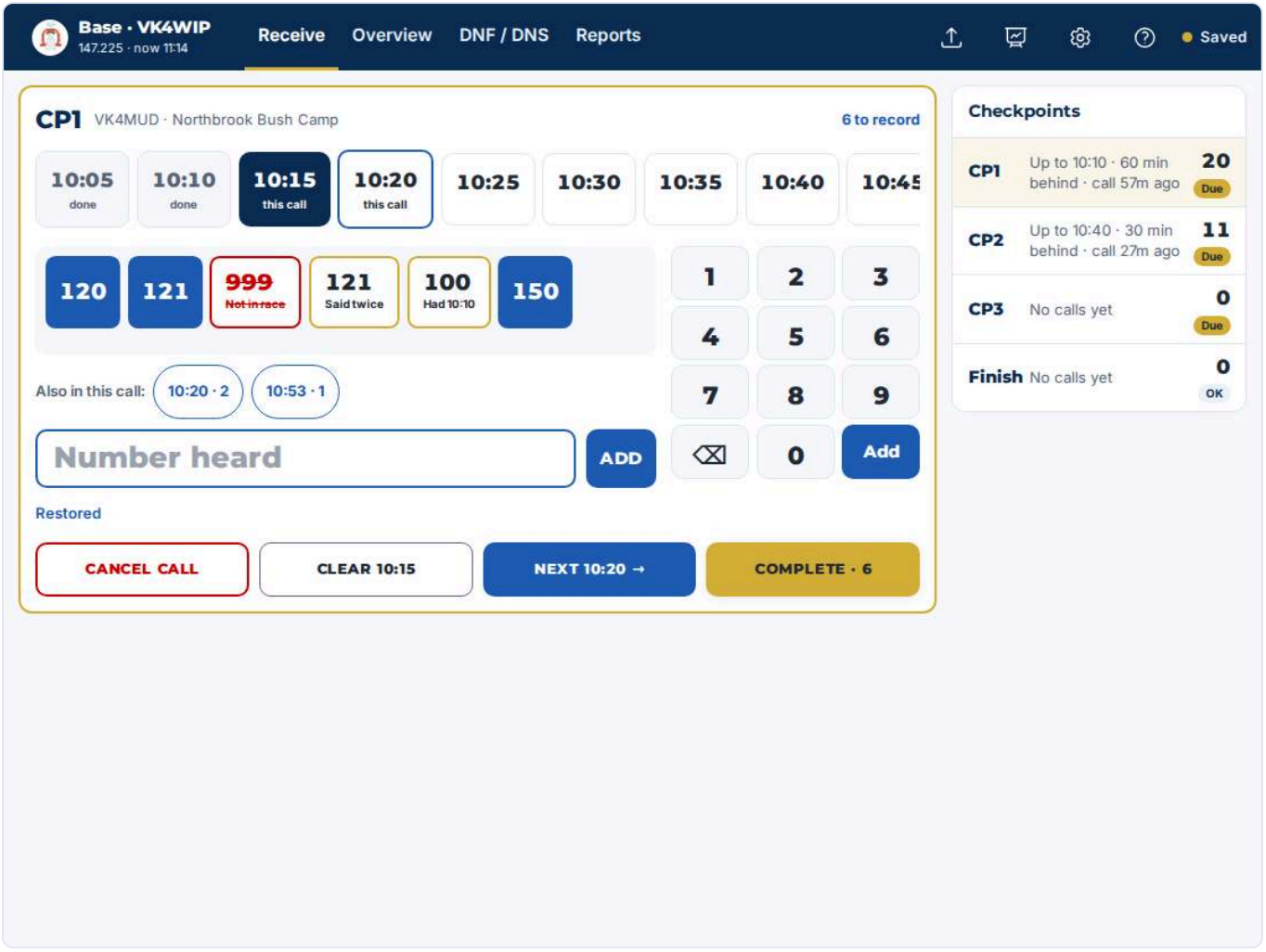
RX-11 1112×834

Plan	§4.9
Event	**Pick the "next" block again, tap "Clear HH:MM"**
Plan says	*Clear 07:40 — clears all numbers in the current block (Undo).*
App showed	• Cleared 6 from 10:15
Result	☐ Matches plan ☐ Issue: _____



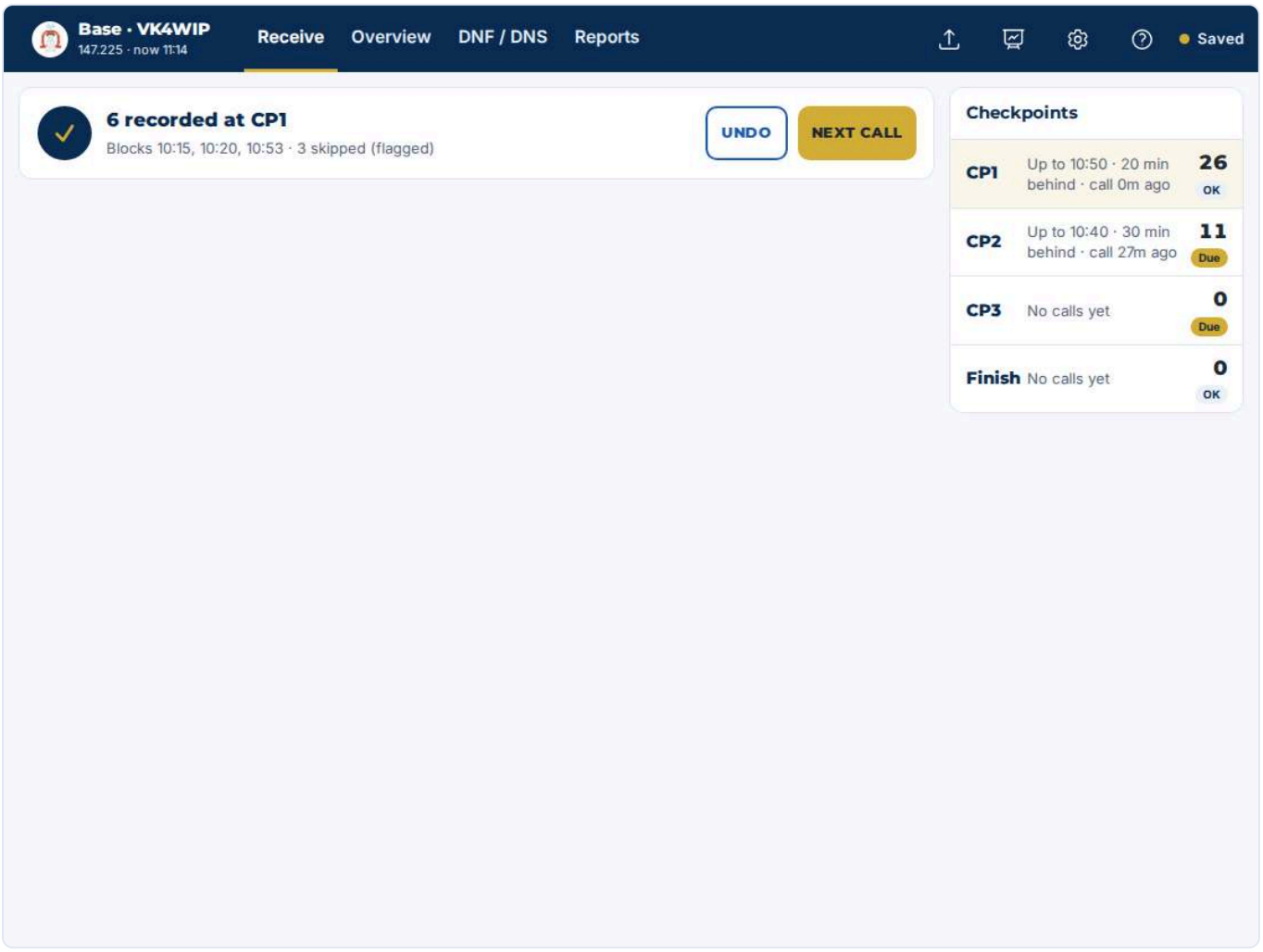
RX-12 1112×834

Plan	§4.9 / G5
Event	**Tap "Undo"**
Plan says	*Undo restores the cleared numbers.*
App showed	• Restored
Result	☐ Matches plan ☐ Issue: _____



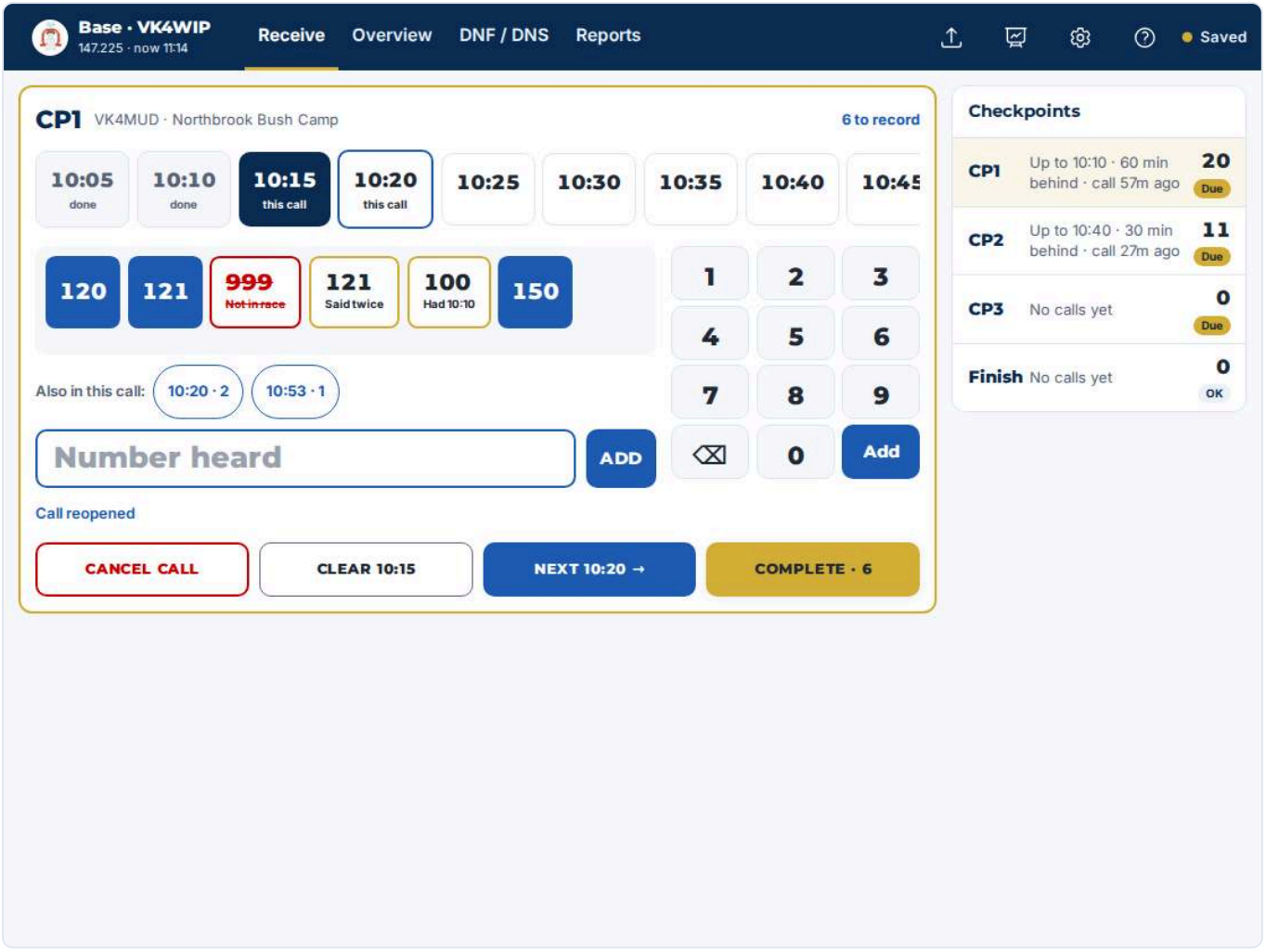
RX-13 1112×834

Plan	§4.9
Event	**Tap "Complete · N"**
Plan says	*Complete → writes base_station_runners, writes a radio_calls record, updates the board (count, last call = now, lastBlock = max block received). Done card: "N recorded at CP3", "Blocks 07:40, 07:45 · 2 skipped (flagged)", Undo (reopens the call, reverts writes) and Next call.*
App showed	• 6 recorded at CP1 Blocks 10:15, 10:20, 10:53 · 3 skipped (flagged) UNDO NEXT CALL
Result	☐ Matches plan ☐ Issue: _____



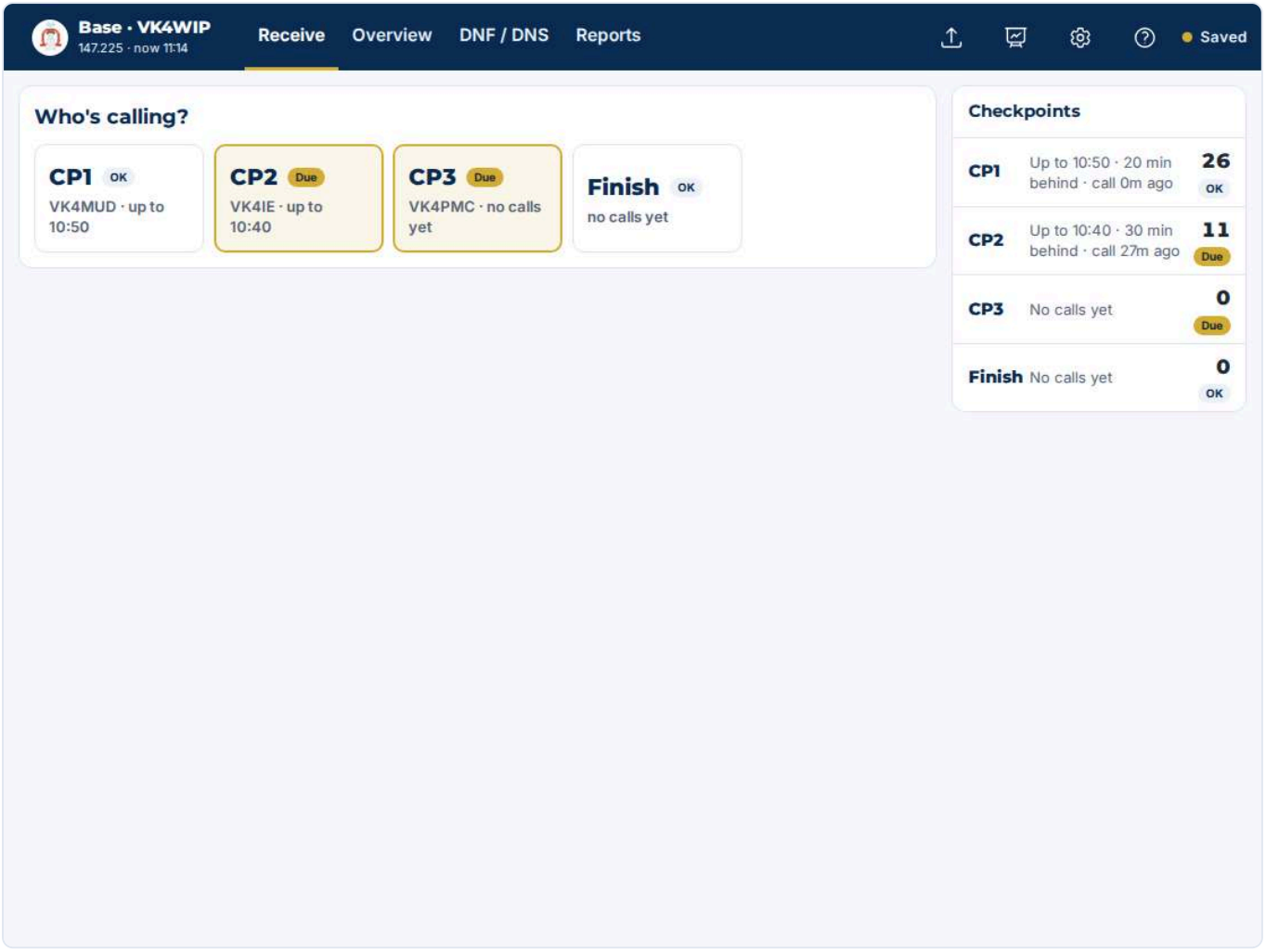
RX-14 1112×834

Plan	§4.9
Event	**Tap "Undo" on the done card**
Plan says	*Undo (reopens the call, reverts writes).*
App showed	• Call reopened
Result	☐ Matches plan ☐ Issue: _____



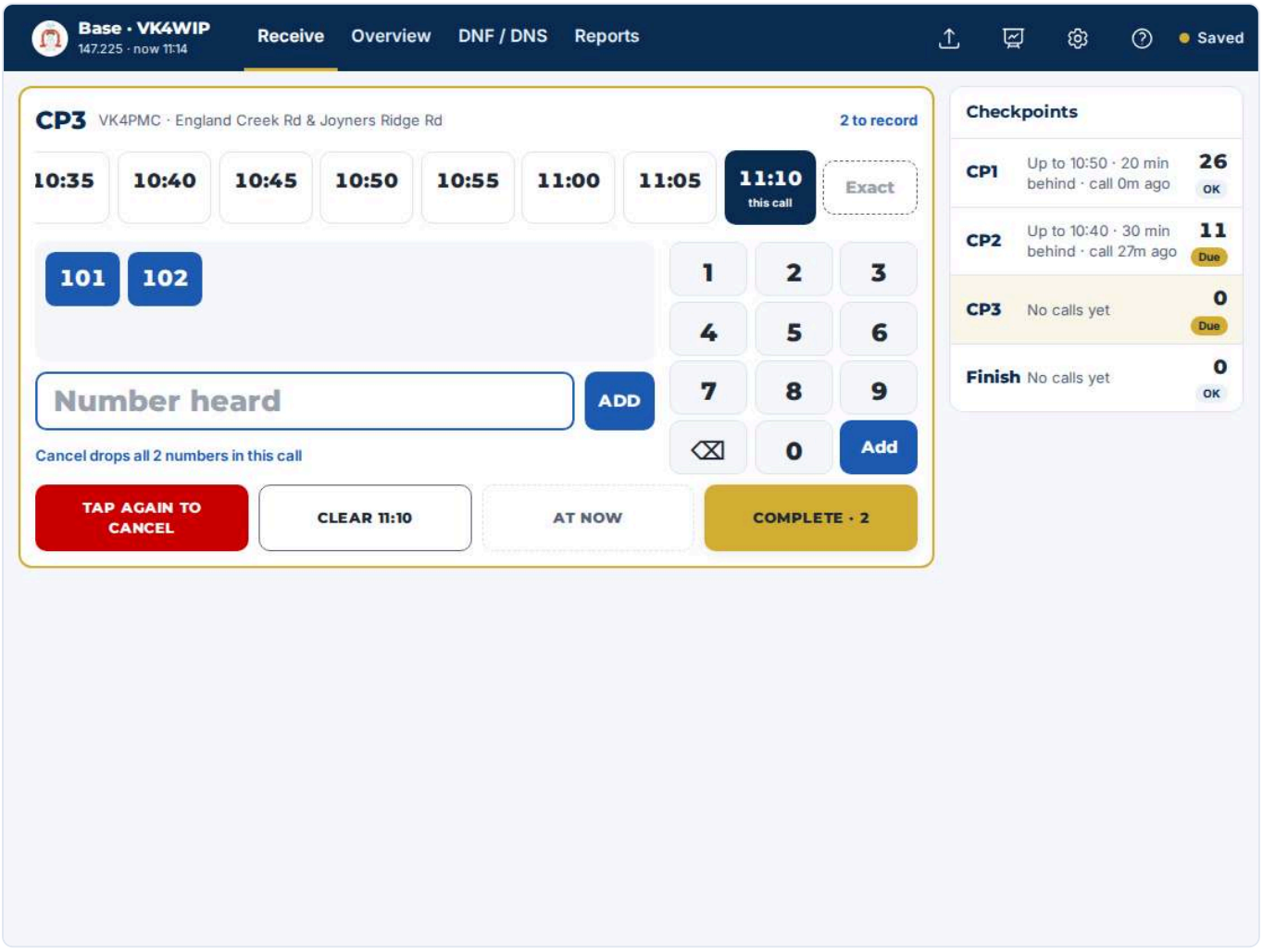
RX-15 1112×834

Plan	§4.9
Event	**Tap "Complete", then "Next call"**
Plan says	*Next call returns to "Who's calling?"; board updated.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



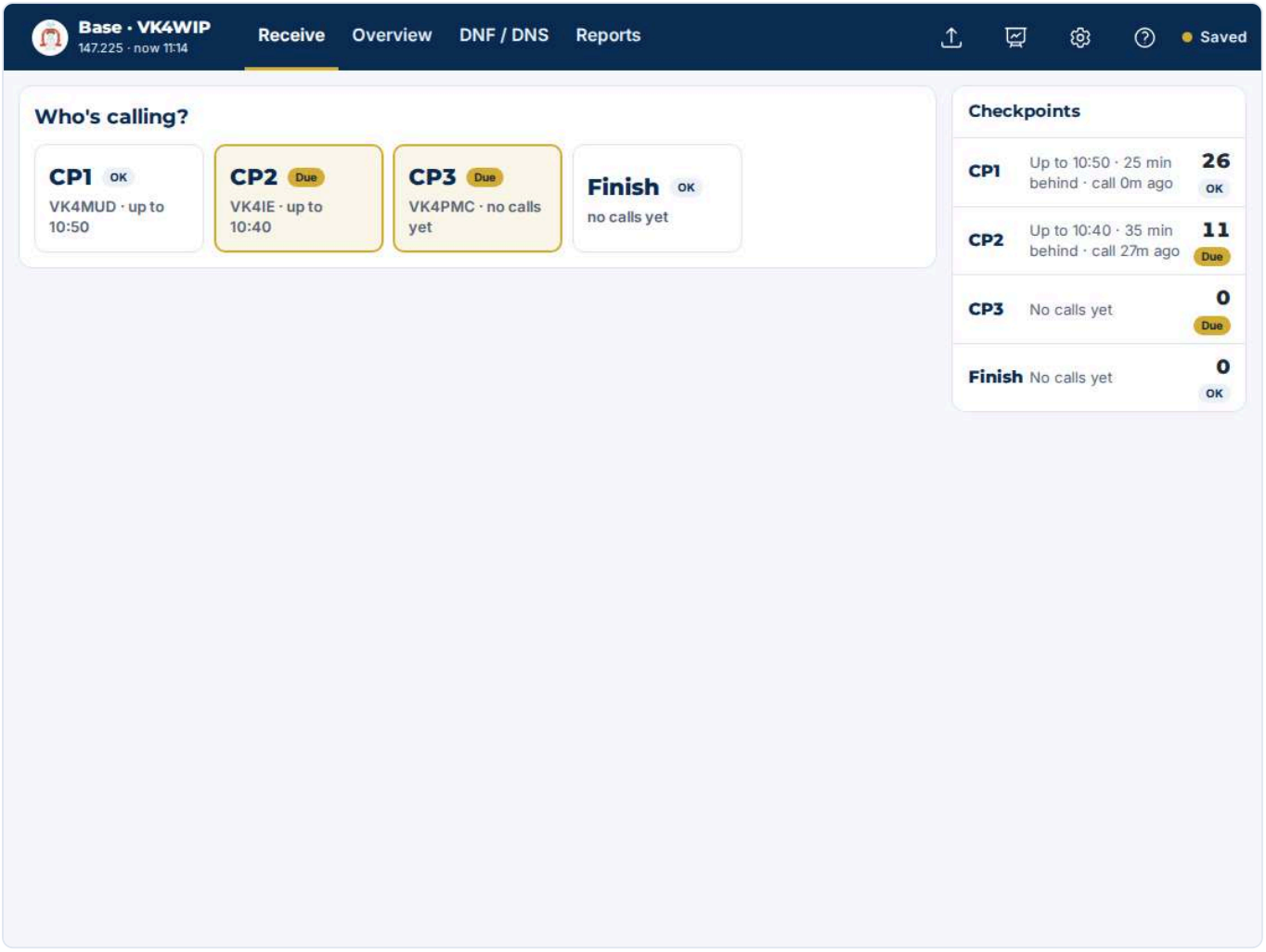
RX-16 1112×834

Plan	§4.9 / G5
Event	**Tap "CP3", add 101 102, tap "Cancel call" once**
Plan says	*Cancel call — red outline; if numbers exist, first tap turns solid red "Tap again to cancel" and says how many will be dropped.*
App showed	• Cancel drops all 2 numbers in this call
Result	☐ Matches plan ☐ Issue: _____



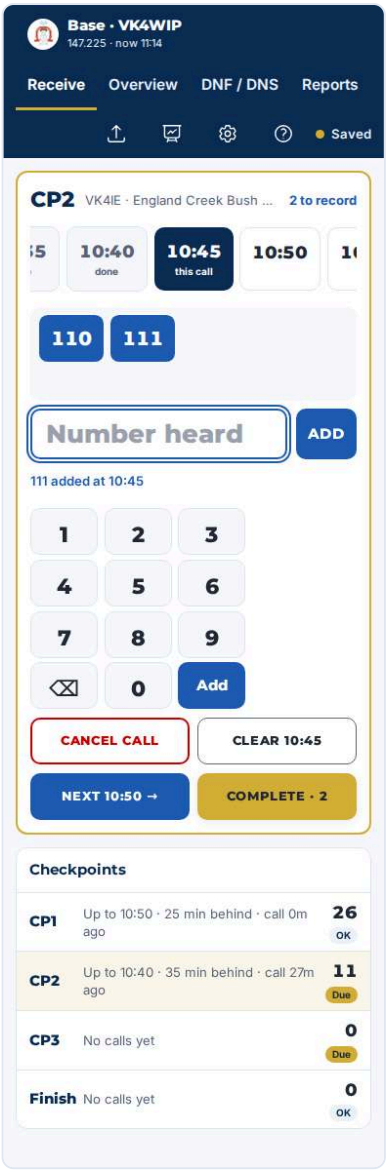
RX-17 1112×834

Plan	§4.9
Event	**Tap "Tap again to cancel"**
Plan says	*Second tap drops the call.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



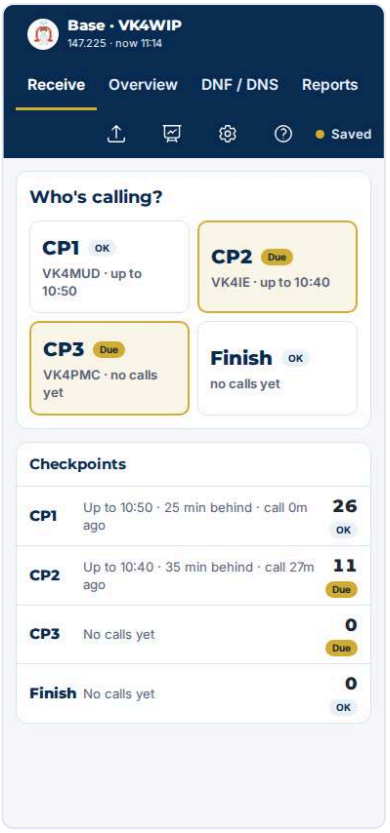
RX-18 390×844

Plan	§4.9 Layout
Event	**Phone 390×844: tap "CP2", add 110 111**
Plan says	*Phone: stacks, receive panel first. Tabs horizontally scrollable on phone.*
App showed	• 111 added at 10:45
Result	☐ Matches plan ☐ Issue: _____



RX-19 390×844

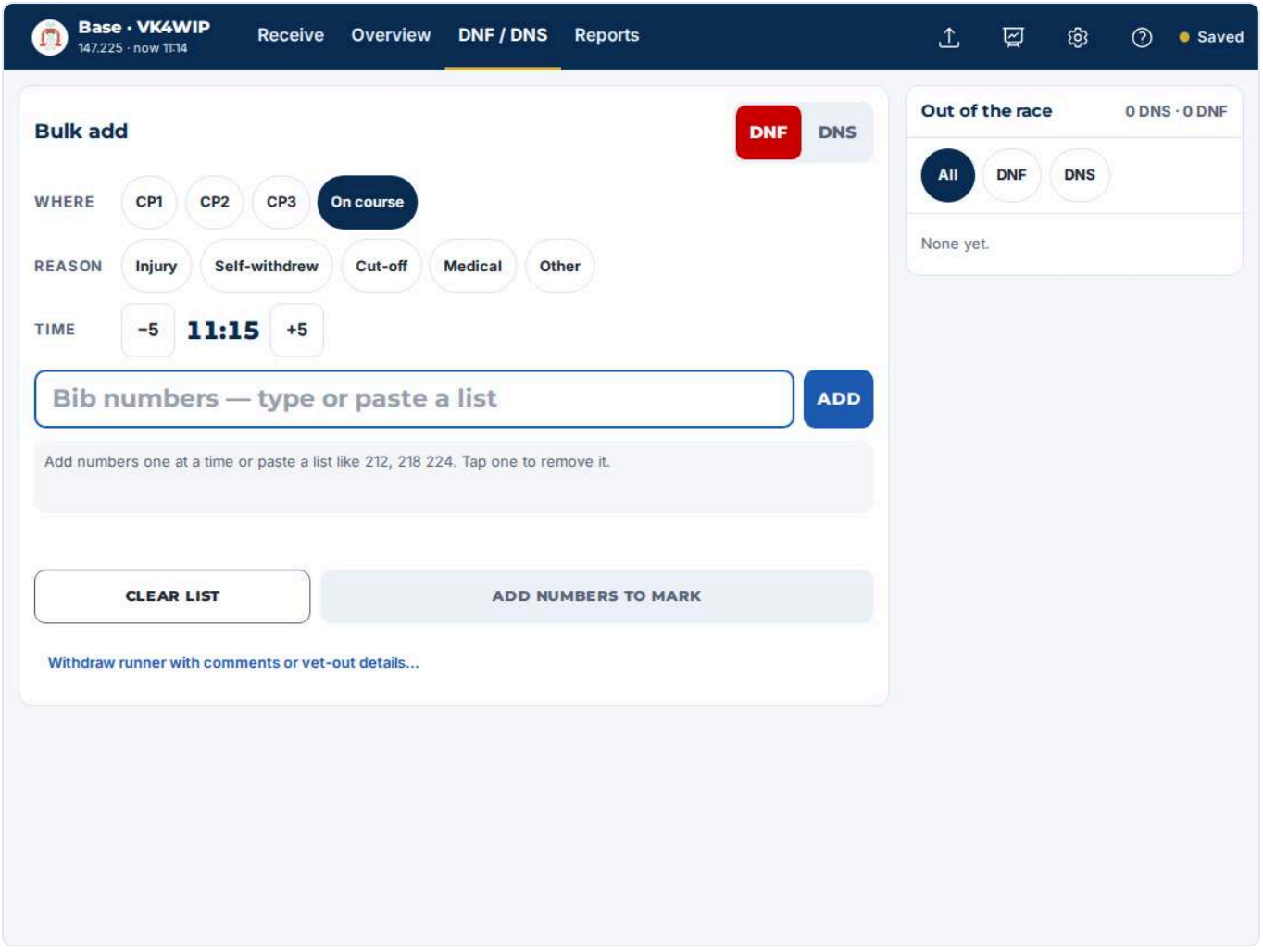
Plan	§4.9
Event	**Phone: Cancel call (two taps)**
Plan says	*Back to idle.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



Base — DNF / DNS

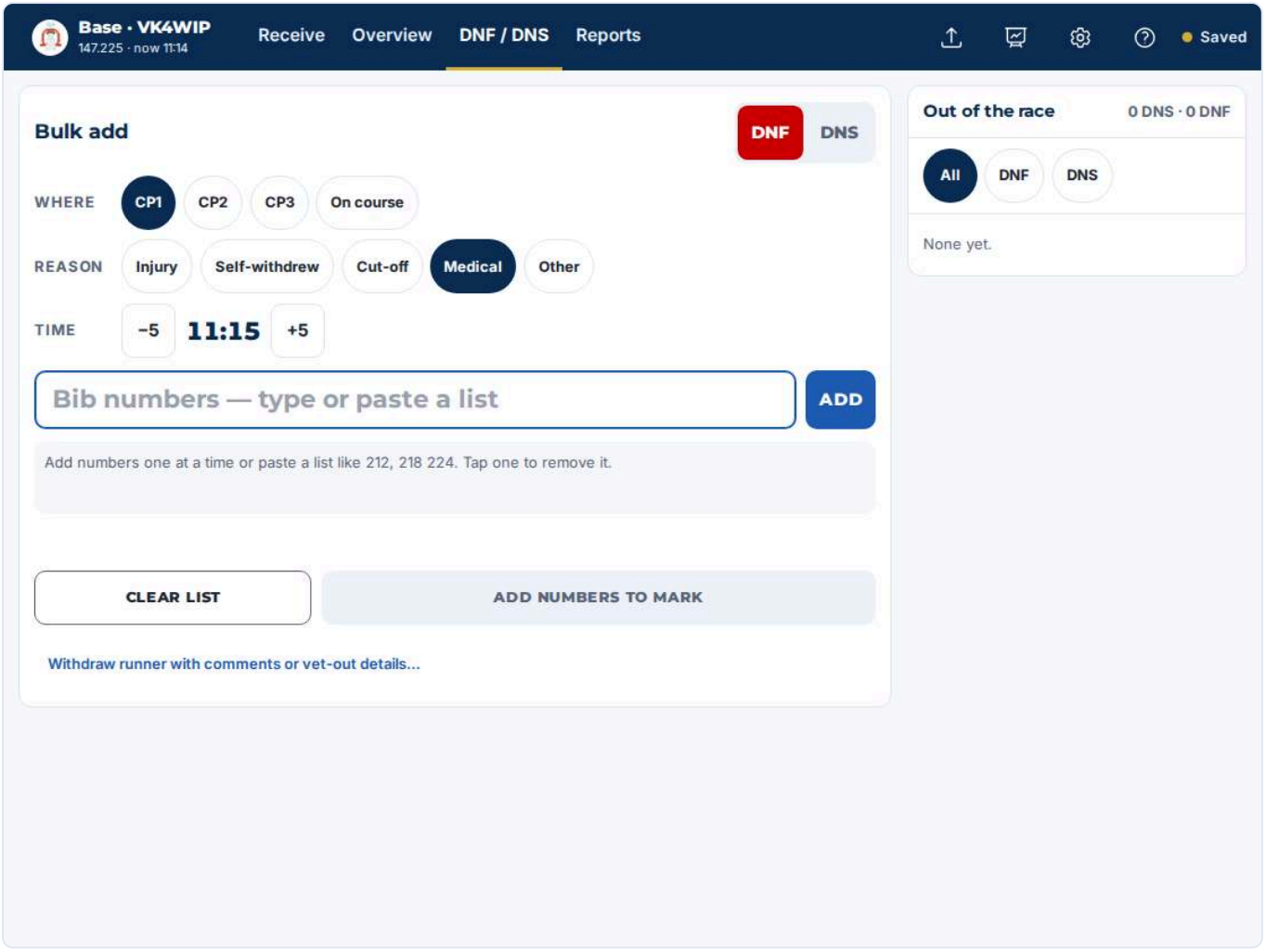
ST-01 1112×834

Plan	§4.11
Event	**Tap the "DNF / DNS" tab**
Plan says	*Bulk add panel: DNF/DNS segmented toggle (DNF active = red). DNF shows Where pills (CP1–CP4, On course), Reason pills (Injury, Self-withdrew, Cut-off, Medical, Other — optional, tap again to unset), Time (−5/+5, clamped). Out of the race panel: totals "3 DNS · 3 DNF", filter pills, rows.*
App showed	• 11:15
Result	☐ Matches plan ☐ Issue: _____



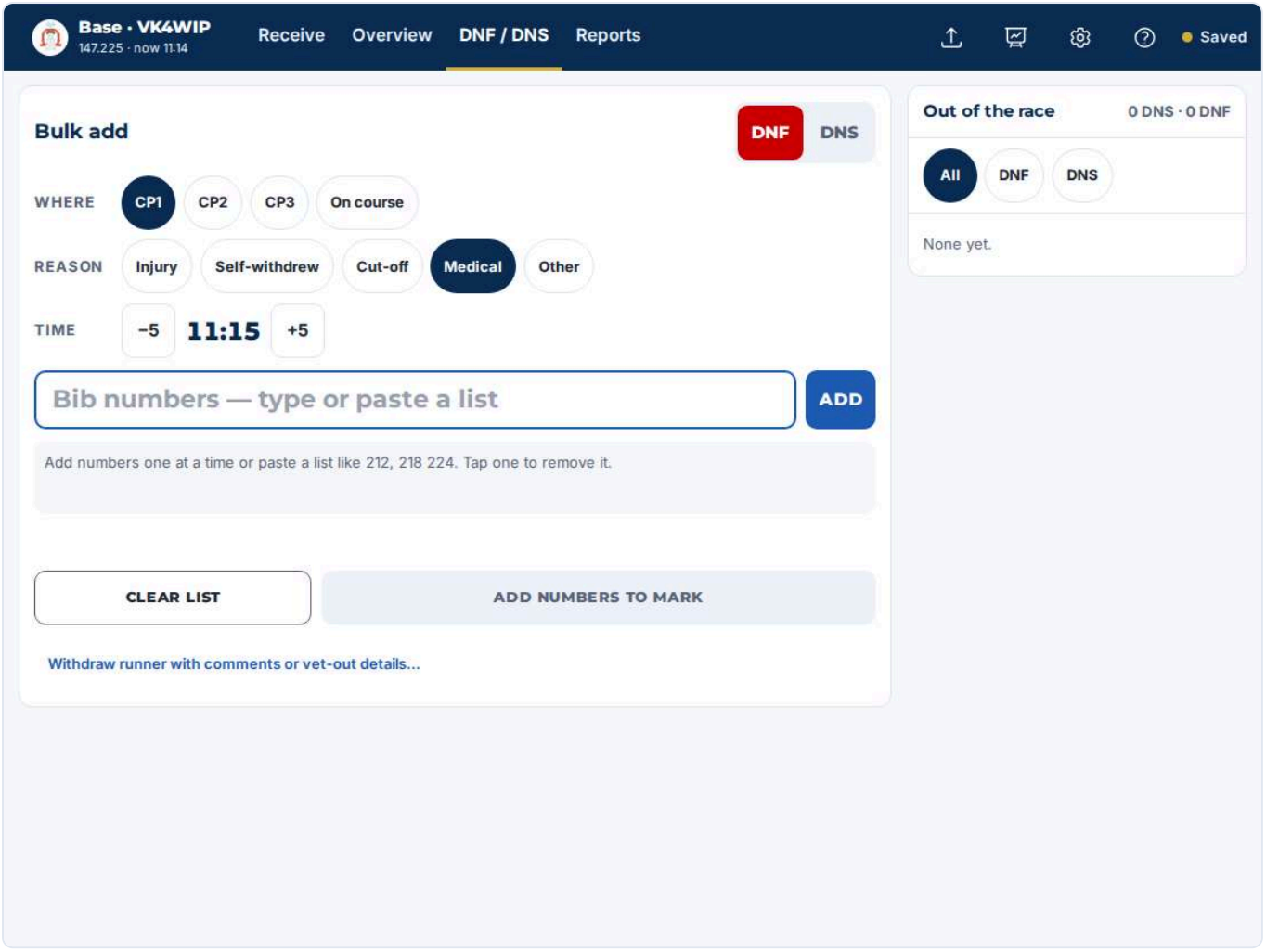
ST-02 1112×834

Plan	§4.11
Event	**Tap Where "CP1", Reason "Injury", tap "Injury" again, tap "Medical"**
Plan says	*Reason pills — optional, tap again to unset.*
App showed	• 11:15
Result	☐ Matches plan ☐ Issue: _____



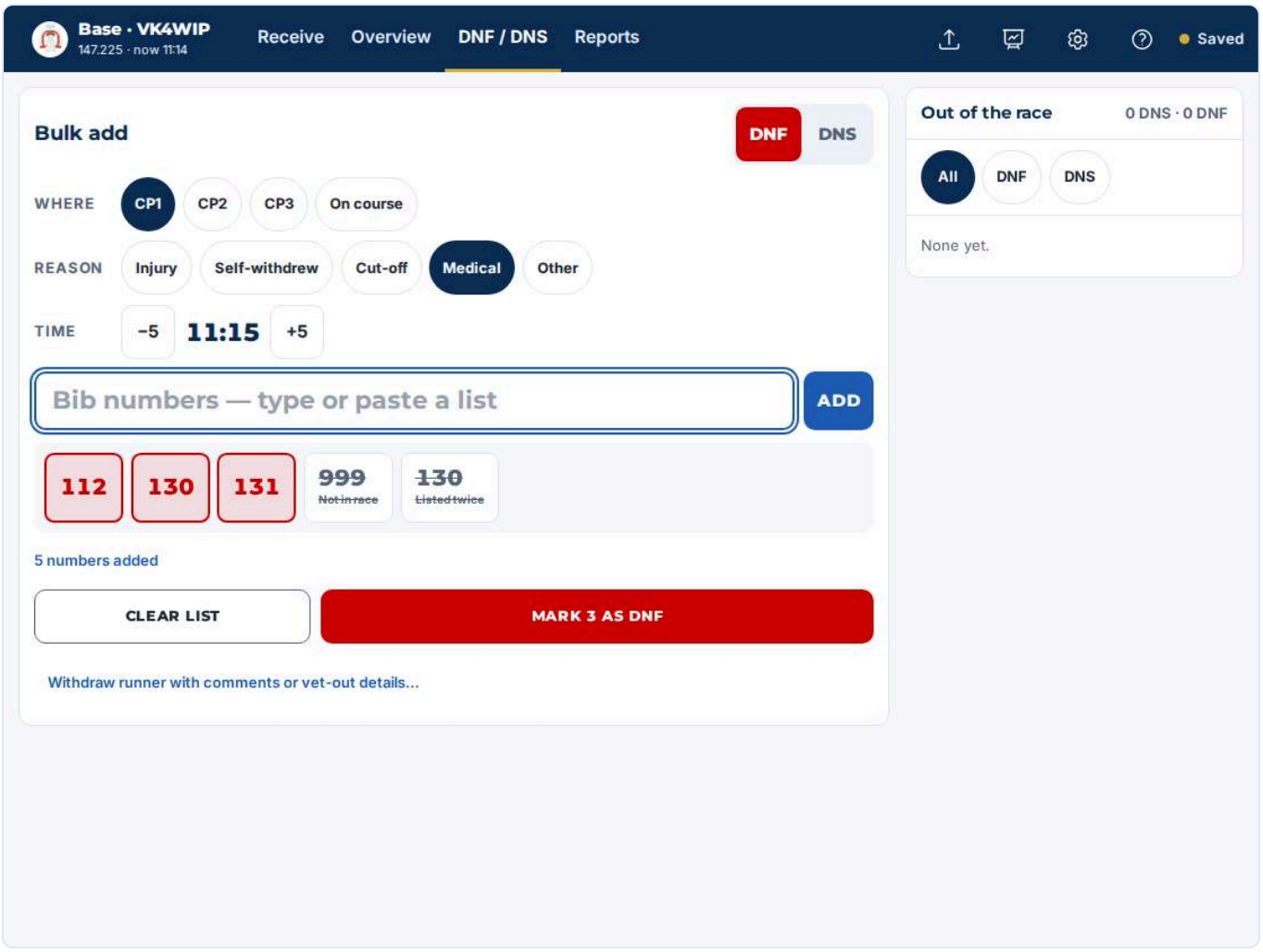
ST-03 1112×834

Plan	§4.11
Event	**Tap Time "+5"**
Plan says	*Time (−5/+5, clamped).*
App showed	• 11:15
Result	☐ Matches plan ☐ Issue: _____



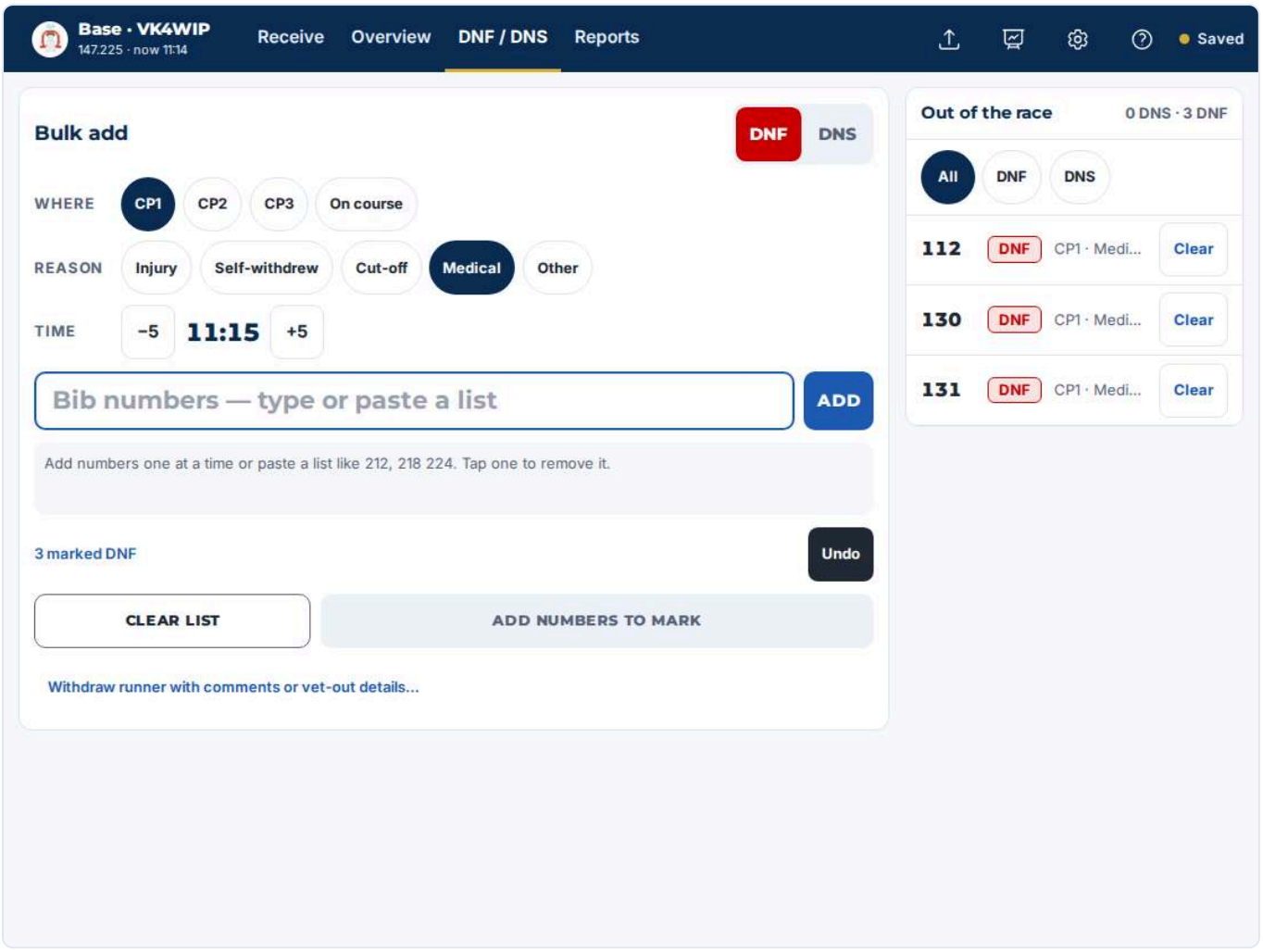
ST-04 1112×834

Plan	§4.11 validation table
Event	**Paste "112, 130 131; 999 130" and press Enter (112 was seen at CP2)**
Plan says	*Input accepts one number or a pasted list (commas, spaces, semicolons, newlines). Not in race / listed twice → rejected; DNF at CPx but seen at a later CP → warning "Seen later at CP3".*
App showed	• 11:15 • 5 numbers added
Result	☐ Matches plan ☐ Issue: _____



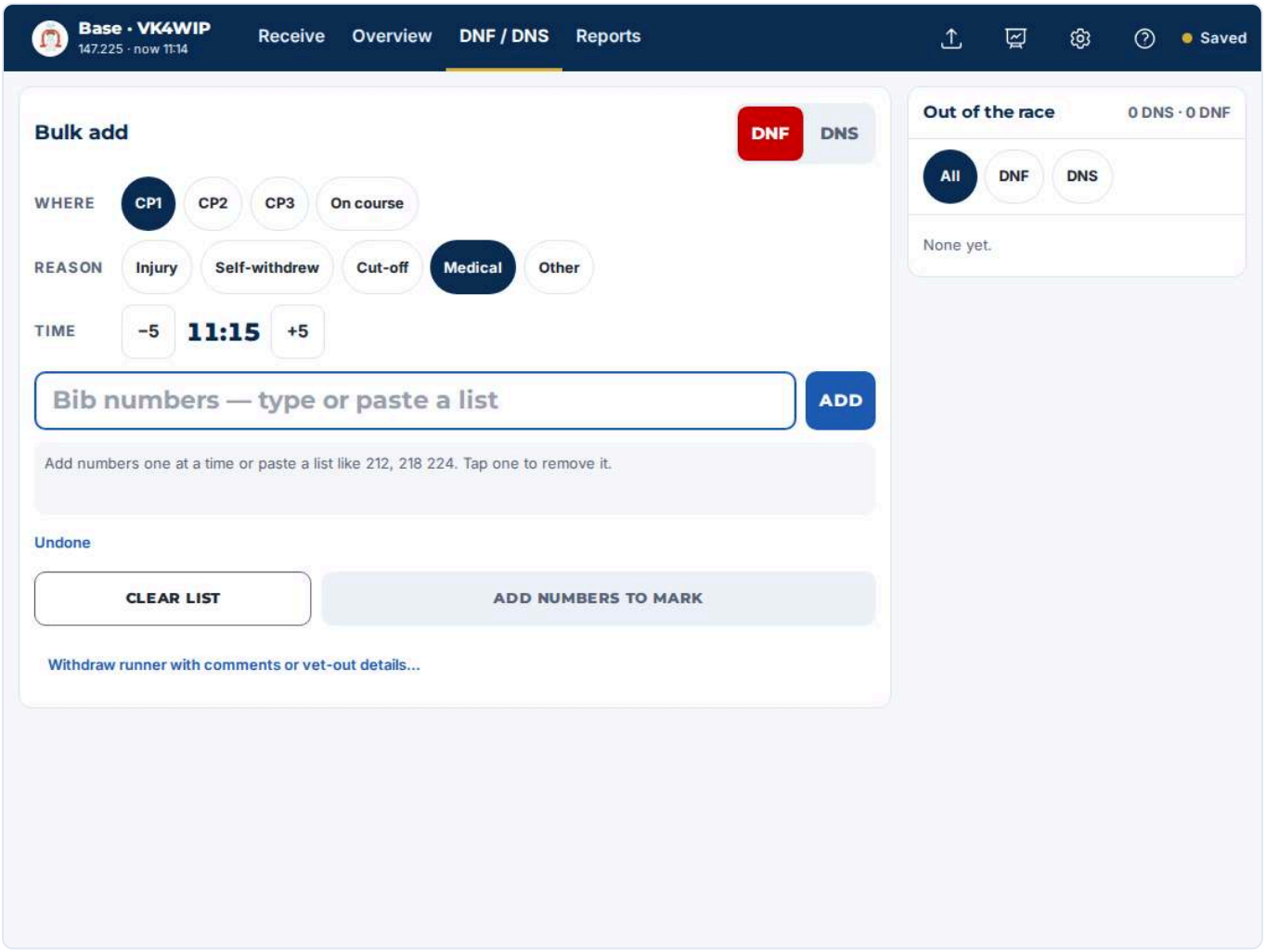
ST-05 1112×834

Plan	§4.11
Event	**Tap "Mark N as DNF"**
Plan says	*Clear list + primary Mark 6 as DNF (red for DNF, gold for DNS; disabled when 0). Undo. DNF writes withdrawal_records (checkpoint, withdrawalTime, reason) + runner status.*
App showed	• 11:15 • 3 marked DNF
Result	☐ Matches plan ☐ Issue: _____



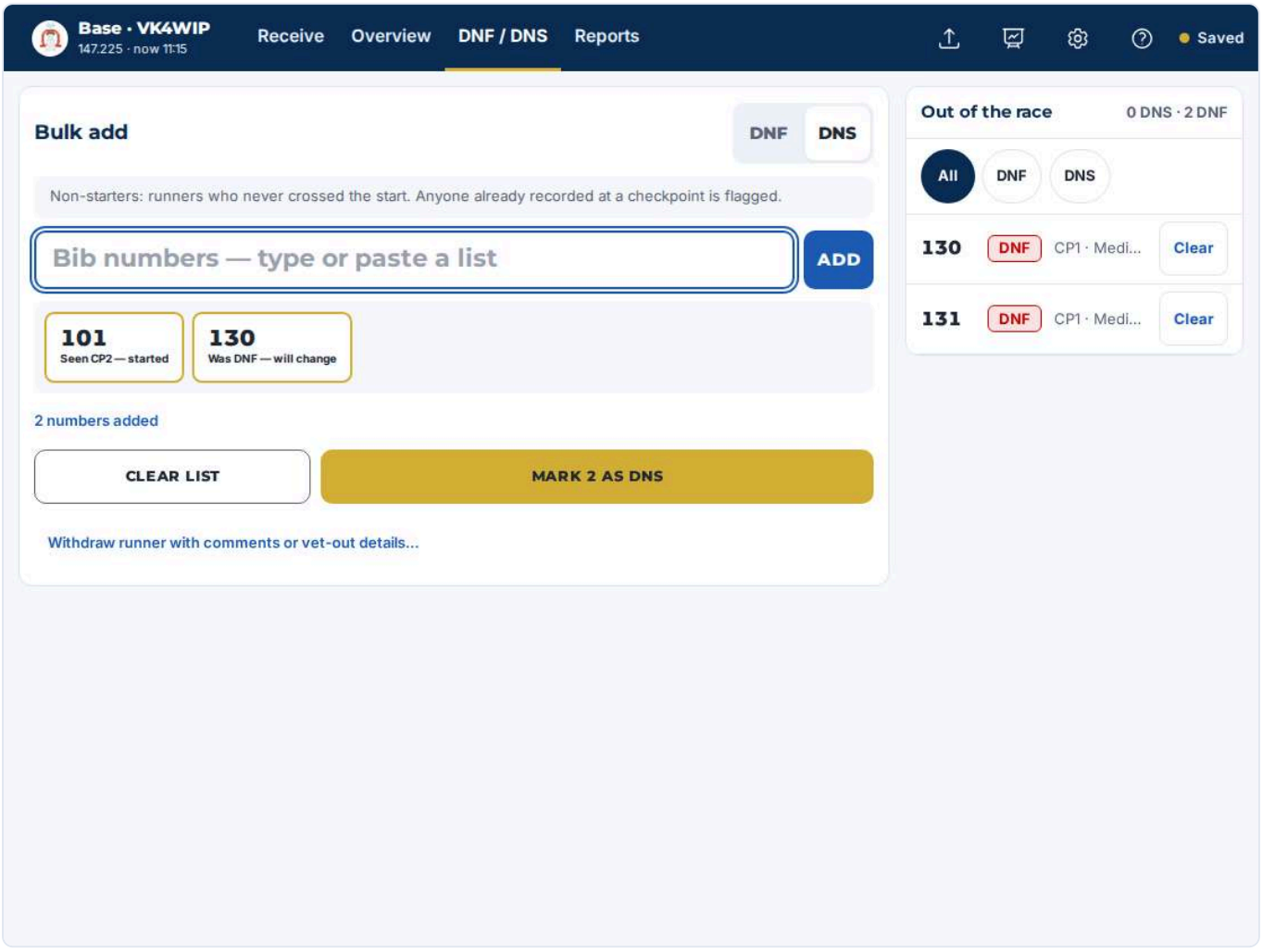
ST-06 1112×834

Plan	§4.11
Event	**Tap "Undo"**
Plan says	*Undo.*
App showed	• 11:15 • Undone
Result	☐ Matches plan ☐ Issue: _____



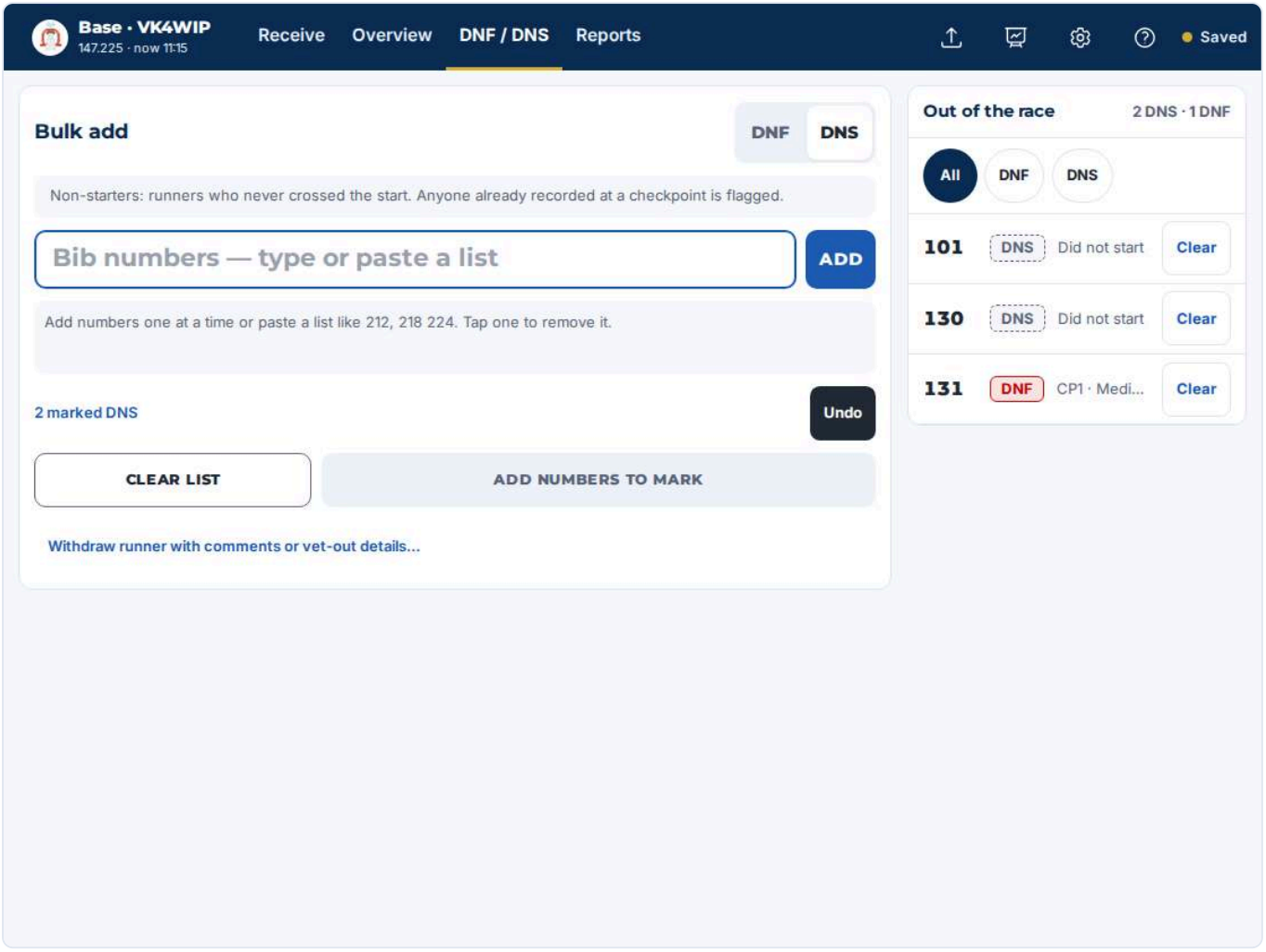
ST-07 1112×834

Plan	§4.11 validation table
Event	**Paste "130 131" again → Mark as DNF; switch to DNS, add "101 130"**
Plan says	*DNS but recorded at a checkpoint → warning "Seen CP2 — started"; currently the other status → warning "Was DNS — will change". DNS shows a one-line explainer.*
App showed	• 2 numbers added
Result	☐ Matches plan ☐ Issue: _____



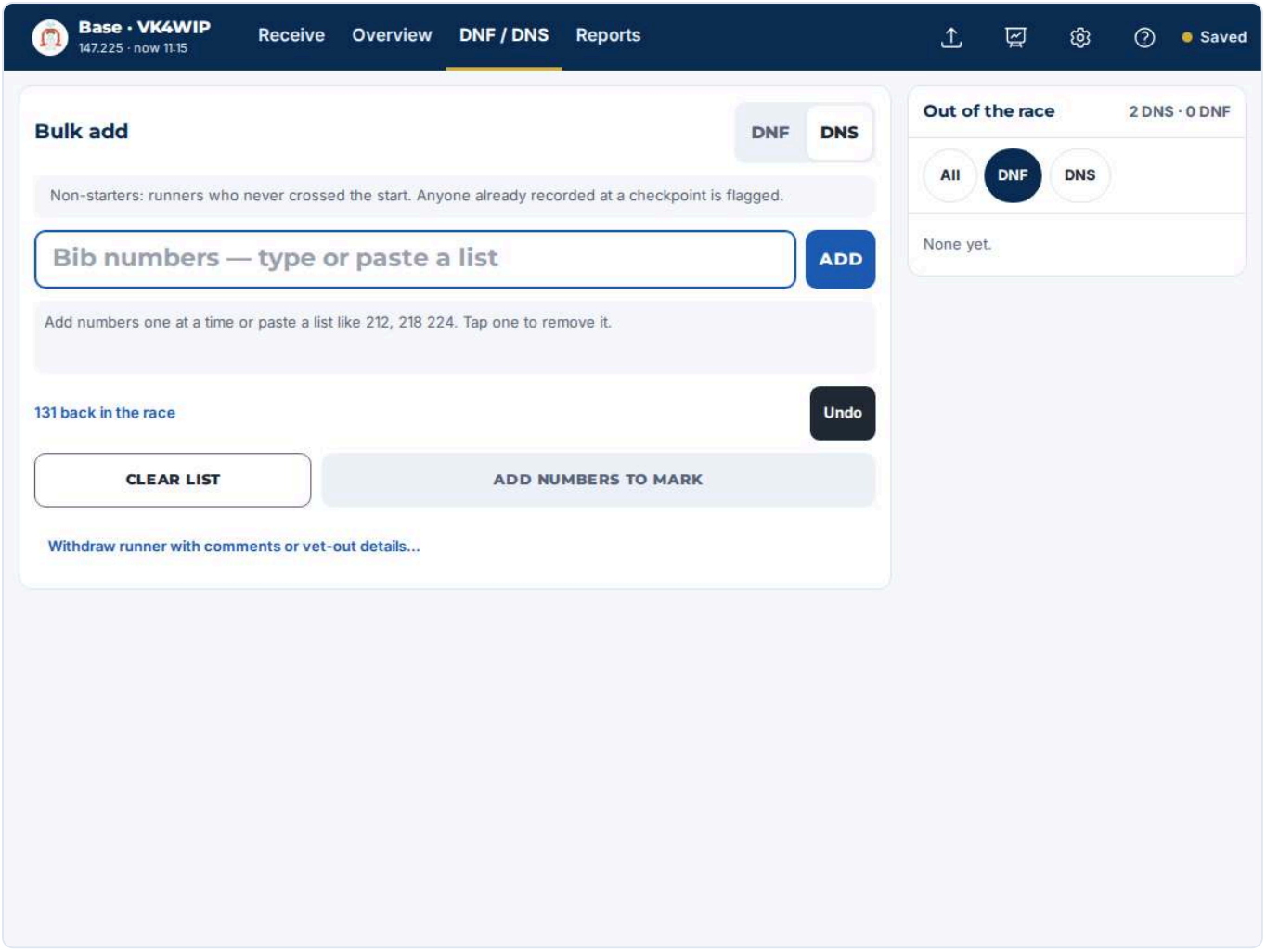
ST-08 1112×834

Plan	§4.11
Event	**Tap "Mark 2 as DNS"**
Plan says	*DNS sets runner status non-starter. Audit-log every change.*
App showed	• 2 marked DNS
Result	☐ Matches plan ☐ Issue: _____



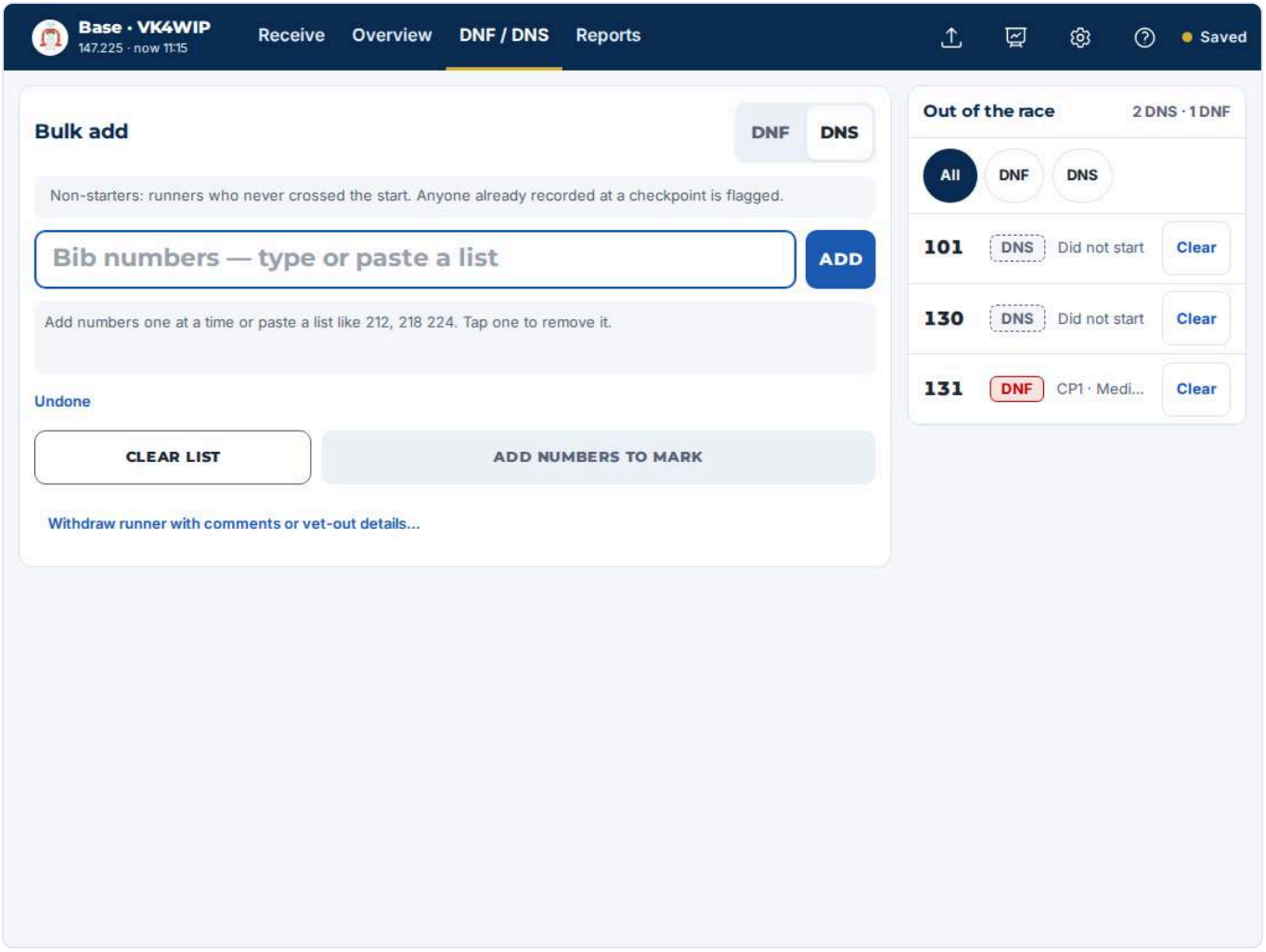
ST-09 1112×834

Plan	§4.11
Event	**Out of the race: tap filter "DNF", then "Clear" on 131**
Plan says	*Rows (bib, status chip, "CP2 · Injury · 07:12" / "Did not start", Clear → back in race, Undo).*
App showed	• 131 back in the race
Result	☐ Matches plan ☐ Issue: _____



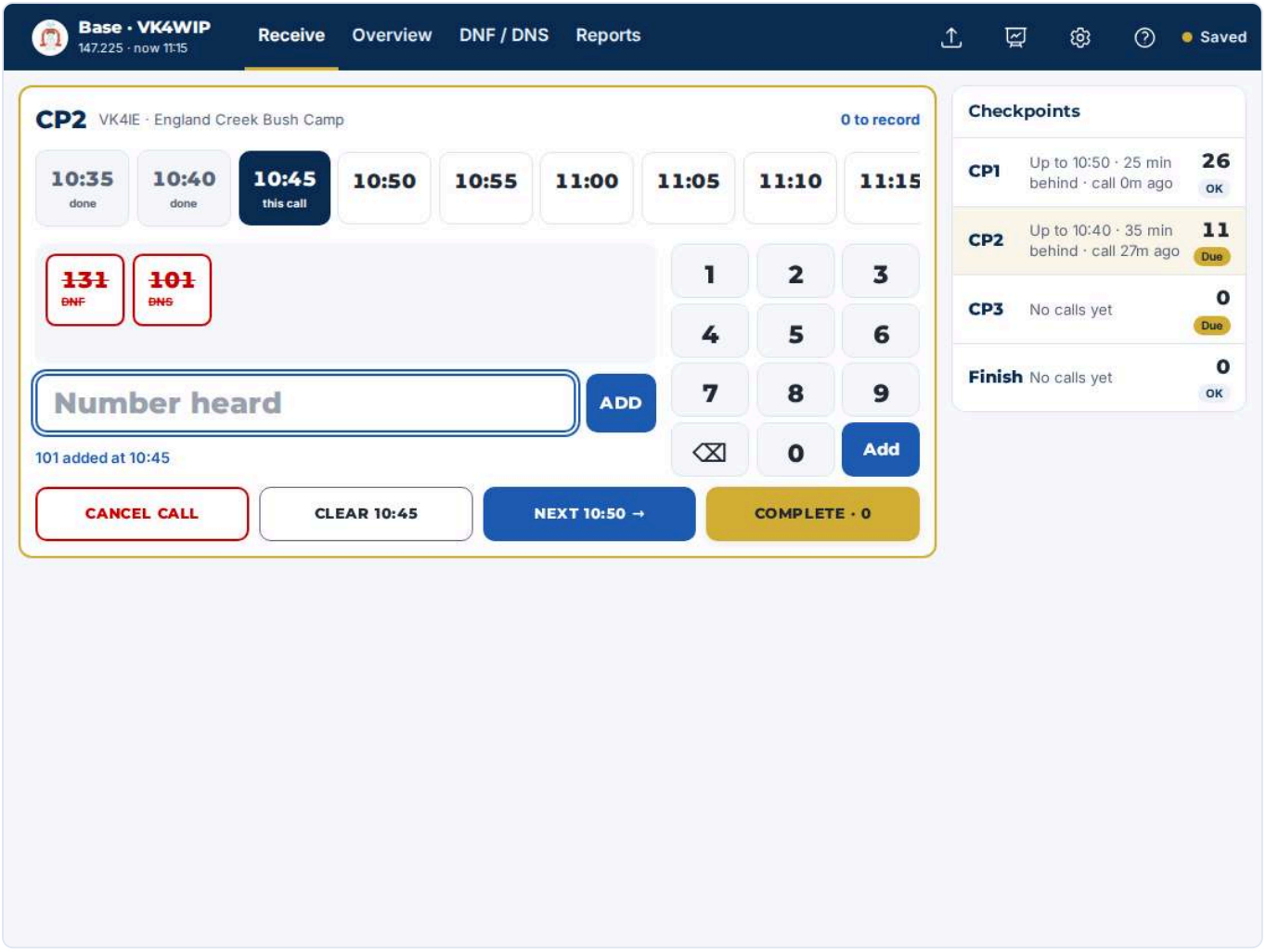
ST-10 1112×834

Plan	§4.11
Event	**Tap "Undo", then filter "All"**
Plan says	*Undo.*
App showed	• Undone
Result	☐ Matches plan ☐ Issue: _____



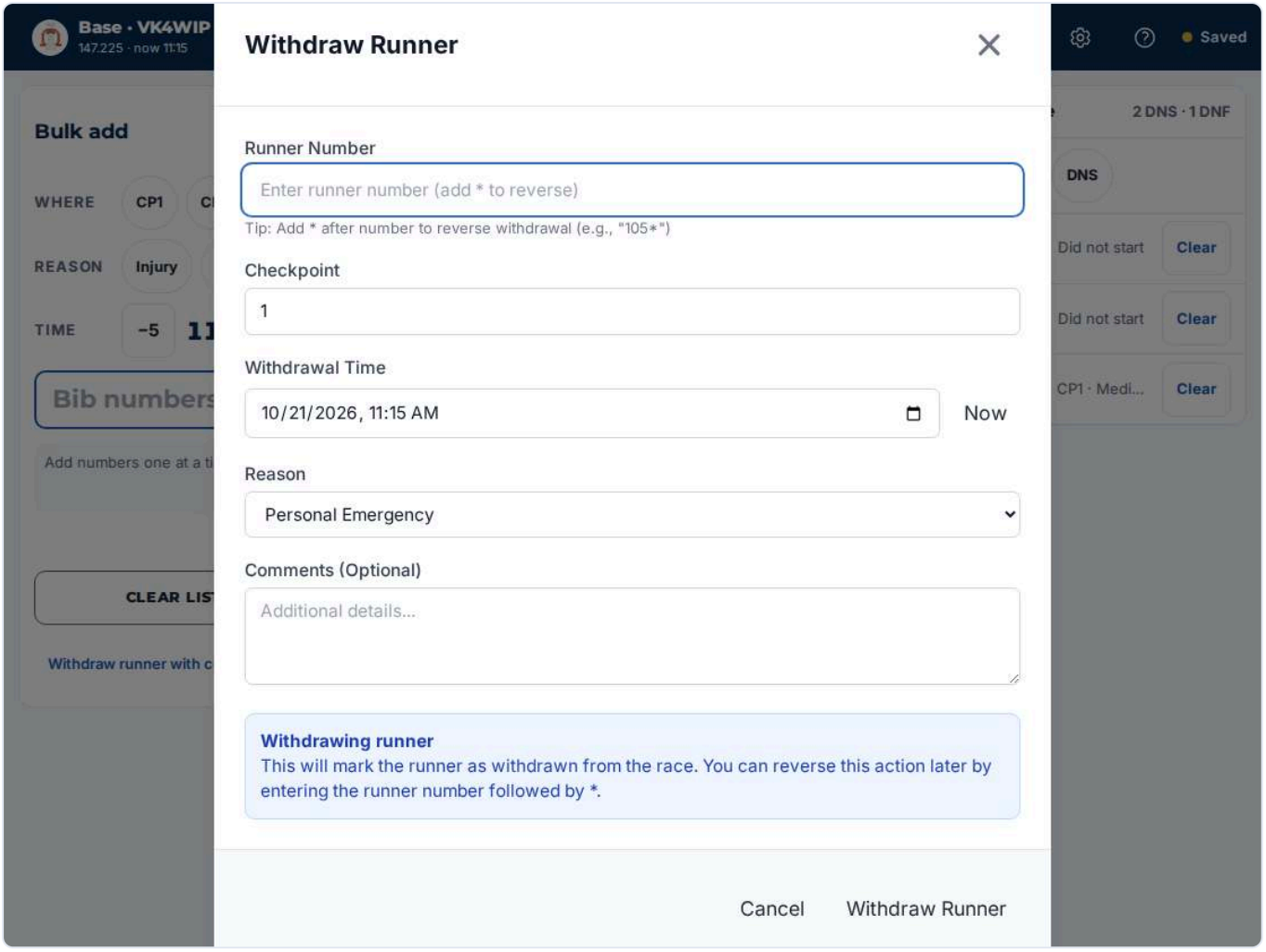
ST-11 1112×834

Plan	§4.9 validation table
Event	**Receive → CP2 → add 131 (DNF) and 101 (DNS)**
Plan says	*Marked DNF / DNS → rejected "DNF" / "DNS".*
App showed	• 101 added at 10:45
Result	☐ Matches plan ☐ Issue: _____



ST-12 1112×834

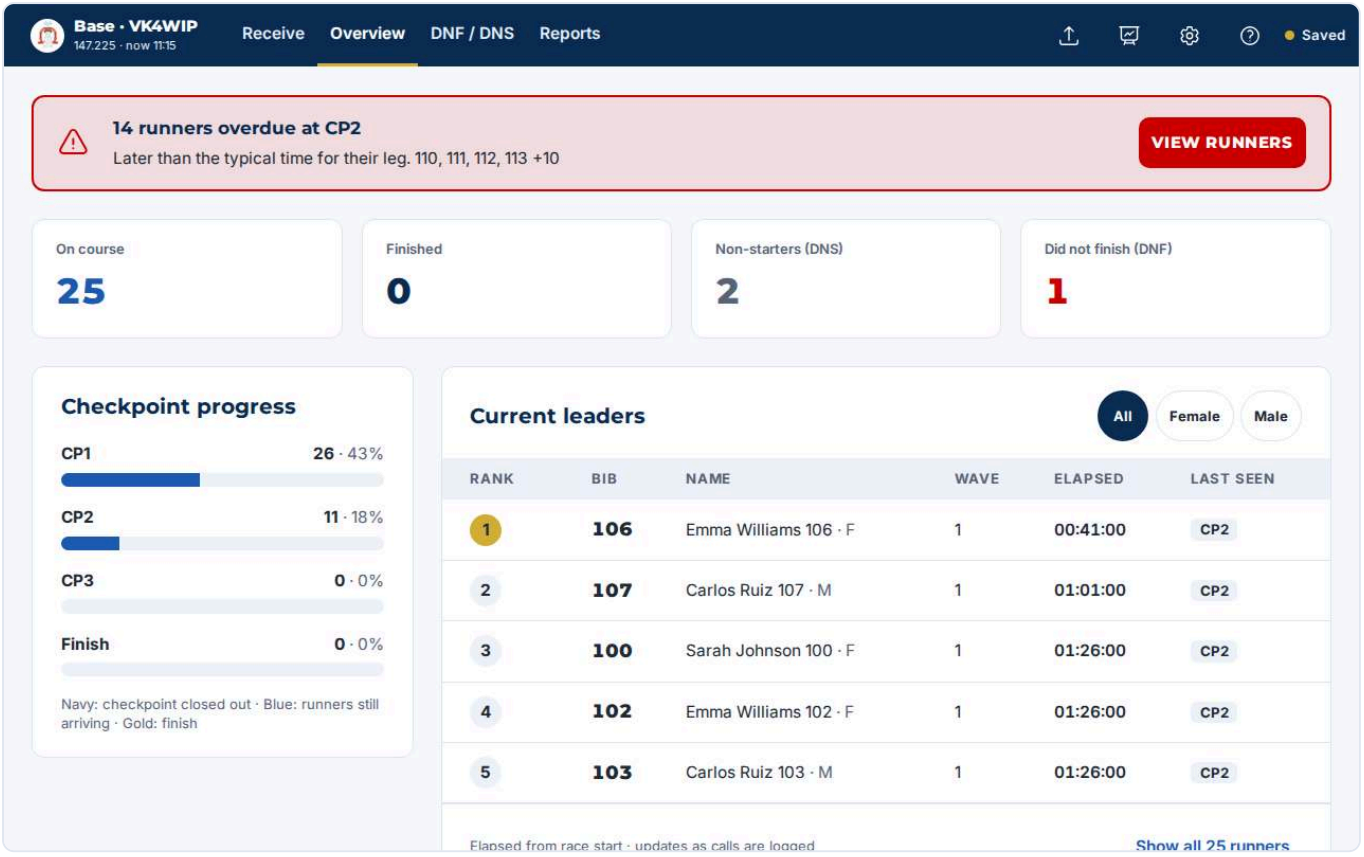
Plan	§4.11 (existing WithdrawalDialog)
Event	**Cancel call; DNF / DNS → "Withdraw runner with comments or vet-out details..."**
Plan says	*Evolves BulkDnsView.jsx + WithdrawalDialog.jsx.*
App showed	• 11:15
Result	☐ Matches plan ☐ Issue: _____



Base — Overview

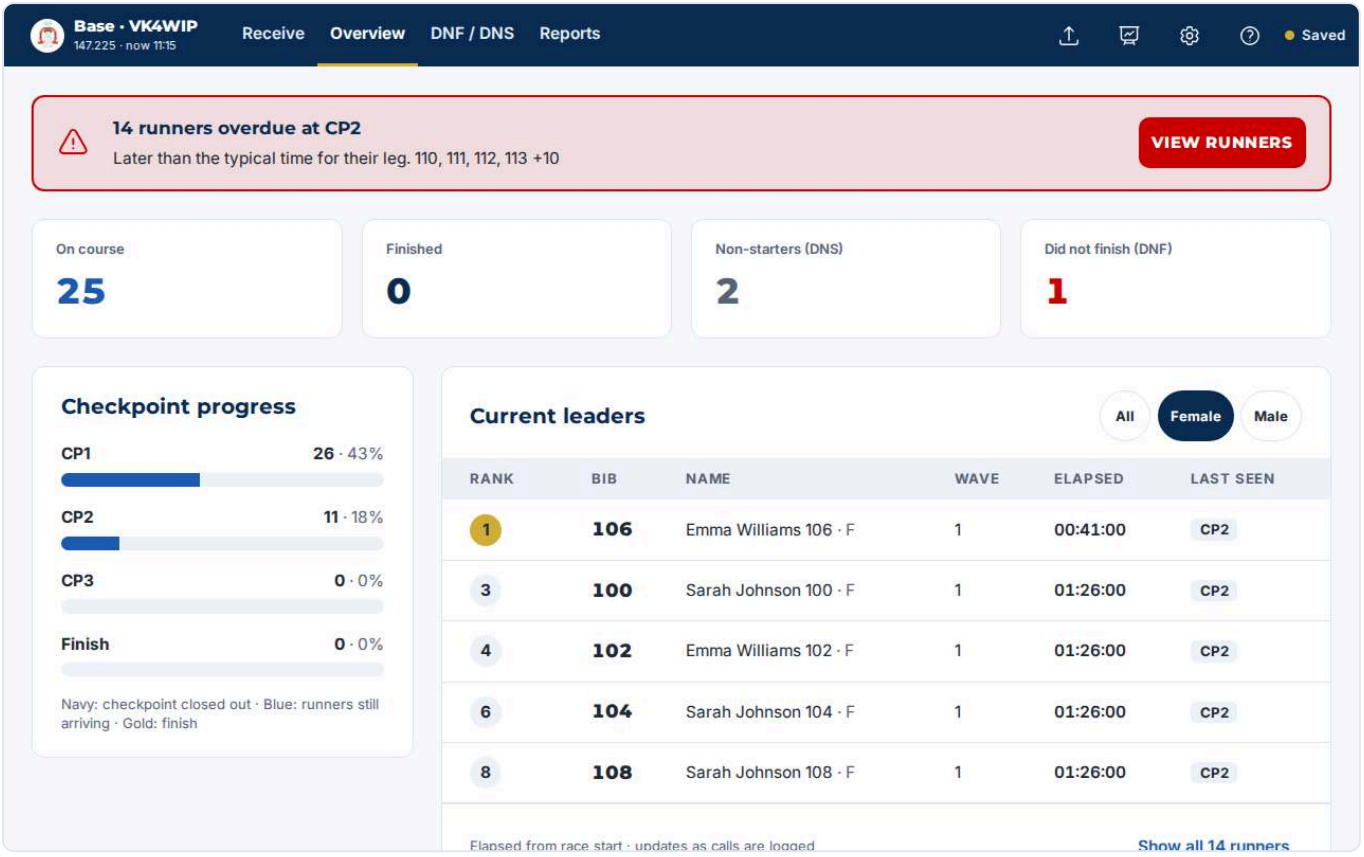
BO-01 1280×800

Plan	§4.10
Event	**Close the dialog; tap the "Overview" tab (desktop 1280)**
Plan says	*Red overdue alert banner (role="alert", "View runners" → Reports › Overdue), stat tiles (On course, Finished, DNS, DNF), checkpoint progress bars (navy = closed out, blue = arriving, gold = finish), leaderboard table with filter pills (All, Female, Male, Wave 1…), gold rank badge for #1, overflow-x box on phone.*
App showed	• 14 runners overdue at CP2 Later than the typical time for their leg. 110, 111, 112, 113 +10 VIEW RUNNERS
Result	☐ Matches plan ☐ Issue: _____



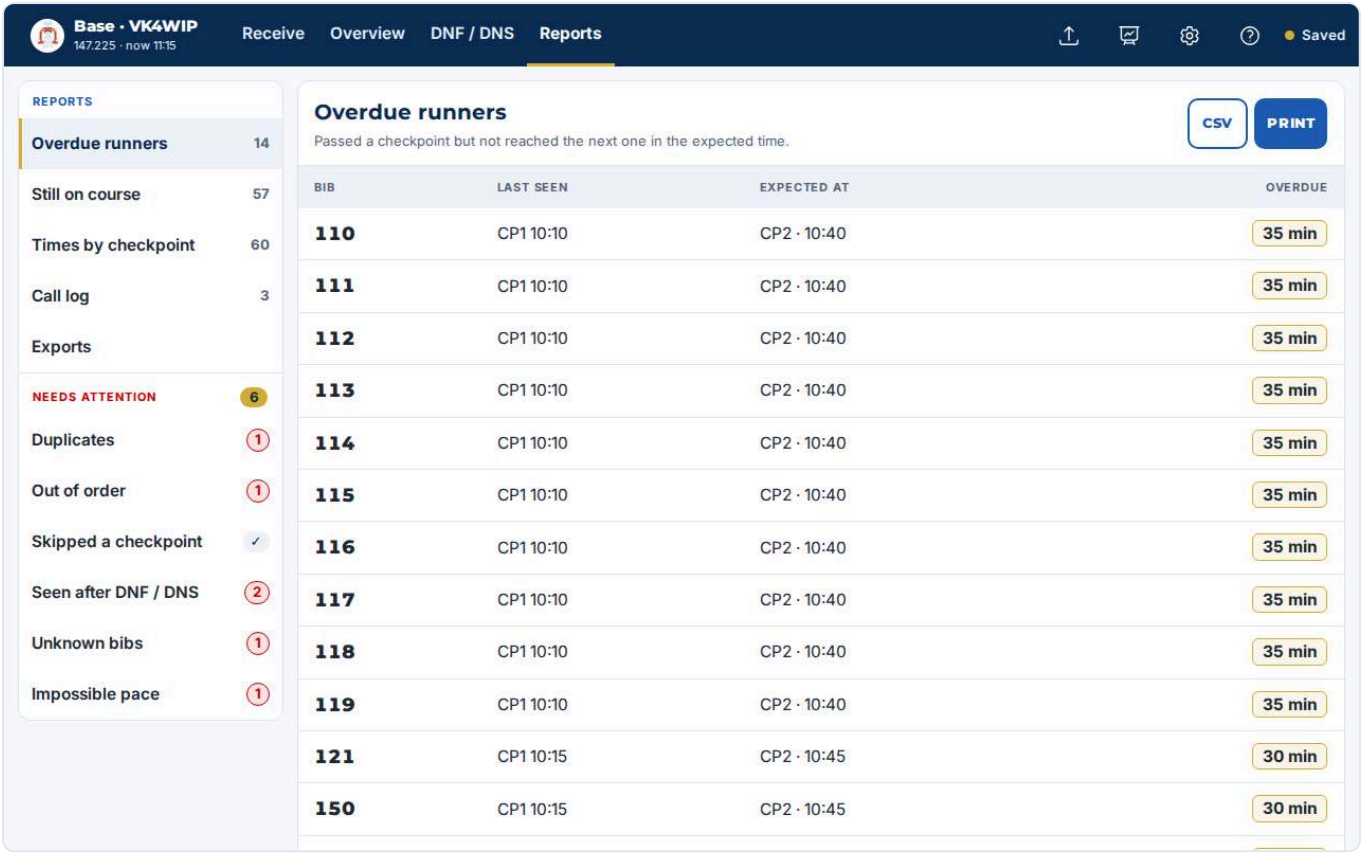
BO-02 1280×800

Plan	§4.10
Event	**Tap leader filter "Female"**
Plan says	*Leaderboard filter pills (All, Female, Male, Wave 1…).*
App showed	• 14 runners overdue at CP2 Later than the typical time for their leg. 110, 111, 112, 113 +10 VIEW RUNNERS
Result	☐ Matches plan ☐ Issue: _____



BO-03 1280×800

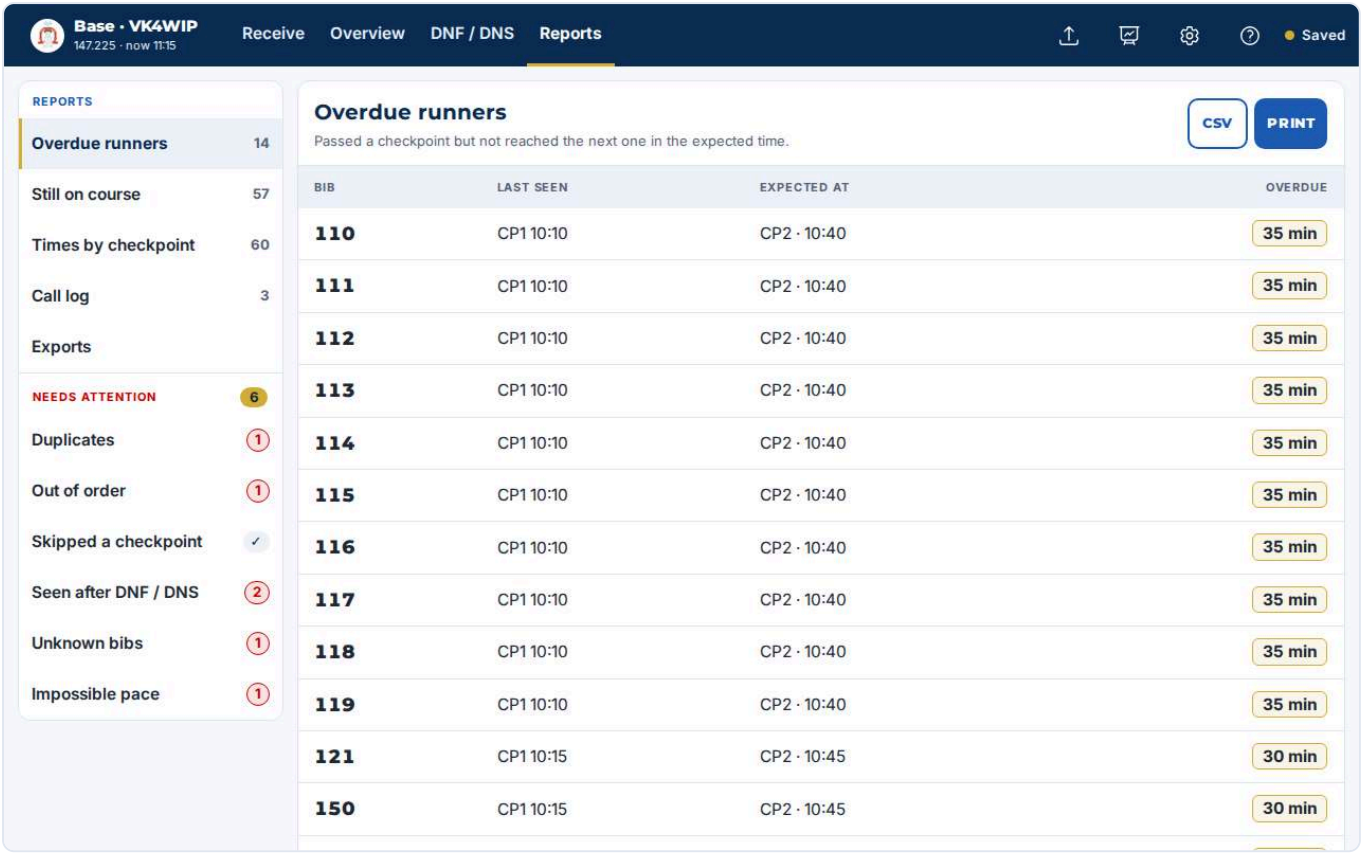
Plan	§4.10
Event	**Scroll to the top and tap "View runners" on the overdue alert**
Plan says	*"View runners" → Reports › Overdue.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



Base — Reports & data checks

RP-01 1280×800

Plan	§4.12 Reports
Event	**Reports › "Overdue runners" (opened from the alert)**
Plan says	*Bib, last seen (CP + time), expected at, overdue minutes (gold badge; red ≥ 60). Expected time = last seen + typical split for that leg (median of recorded splits). CSV + Print buttons. Sidebar (flex 1 1 250px) + report panel; "Needs attention" group with total gold badge; each item red count badge or ✓.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



RP-02 1280×800

Plan §4.12 Reports

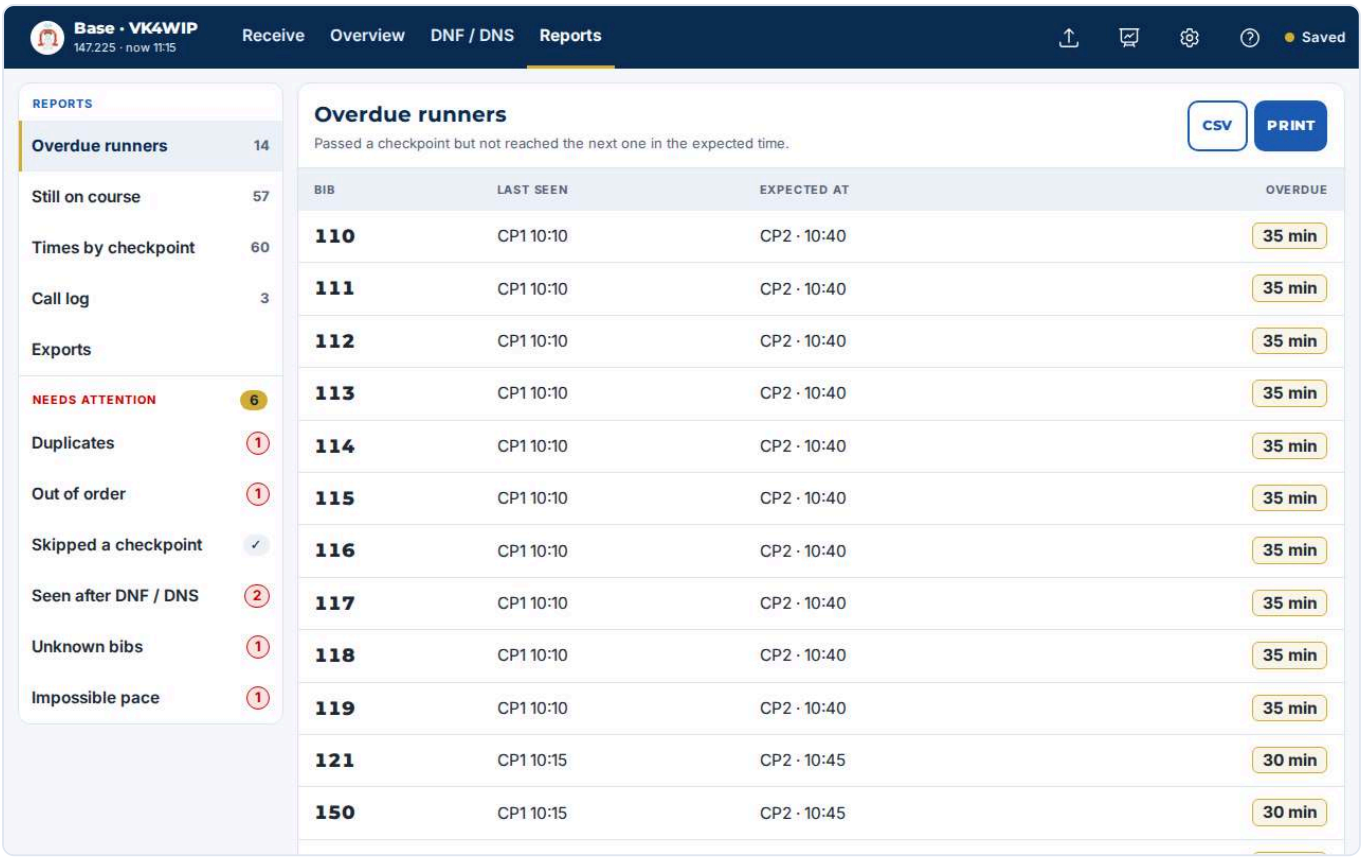
Event **Click "CSV"**

Plan says *CSV + Print buttons; reuse existing export utilities.*

App showed

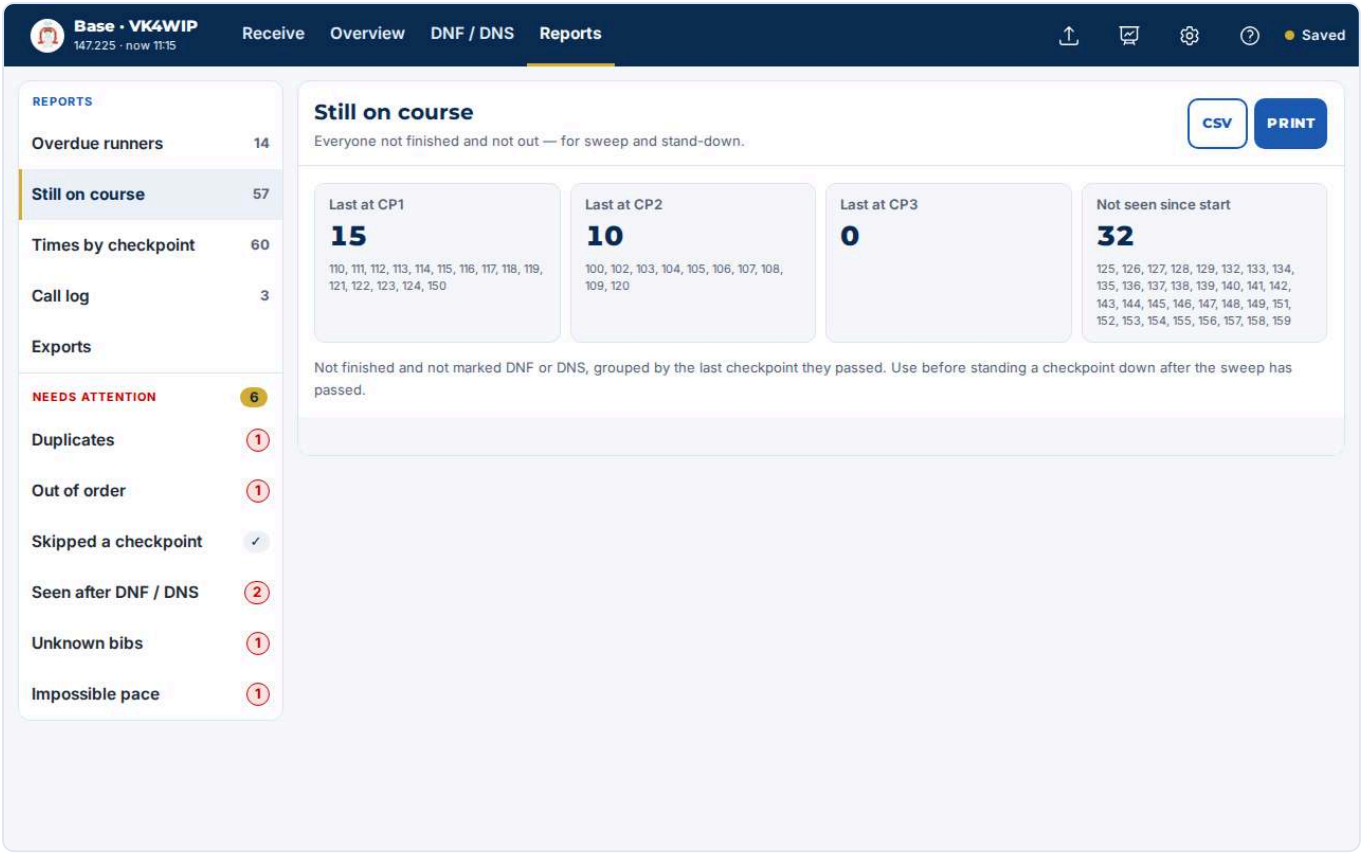
- downloaded: Mt-Glorious-Mountain-Trails-overdue-runners.csv
- first lines: Bib,Last seen,Last time,Expected at,Expected time,Overdue (min) | 110,CP1,10:10,CP2,10:40,35 | 111,CP1,10:10,CP2,10:40,35 | 112,CP1,10:10,CP2,10:40,35

Result ☐ Matches plan ☐ Issue: _____



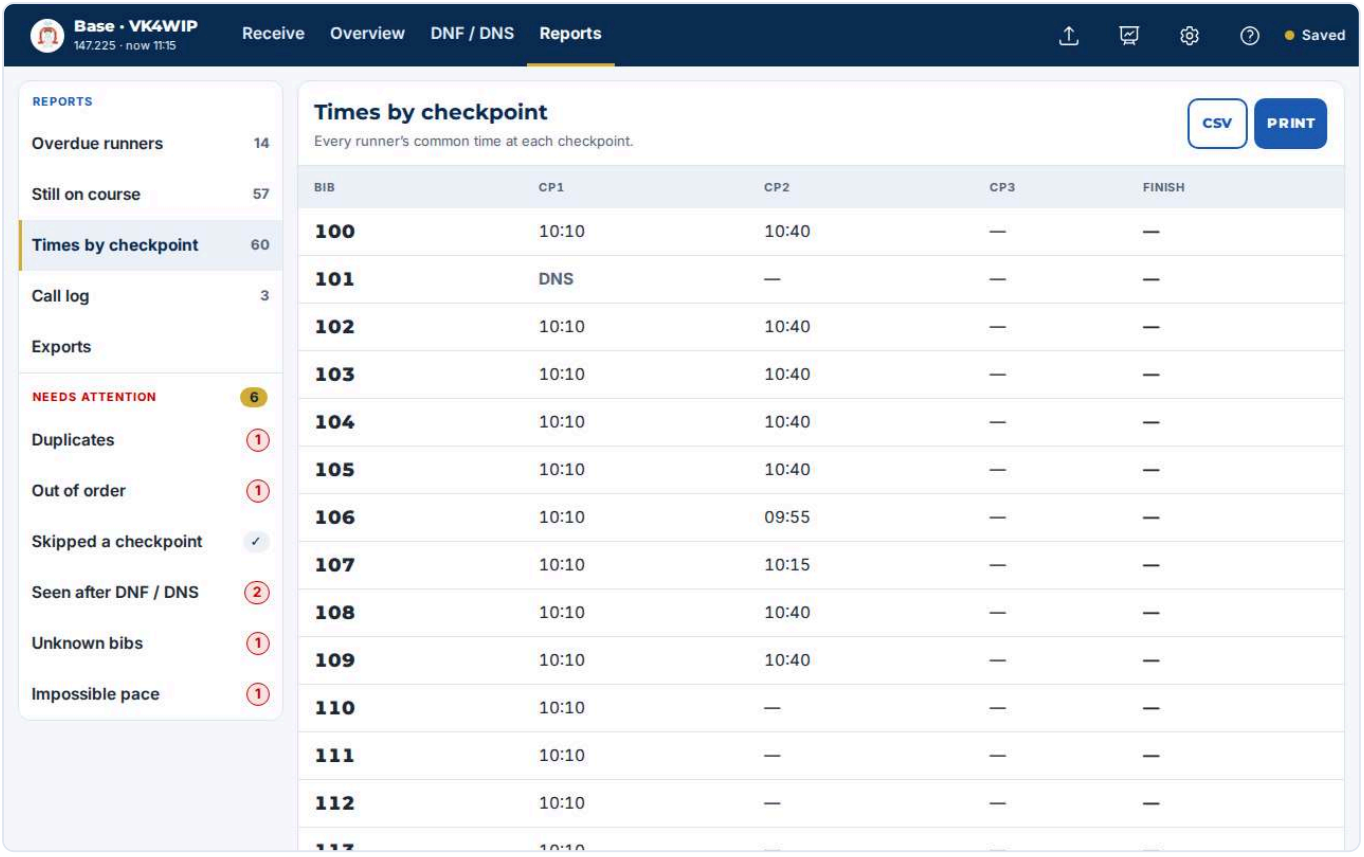
RP-03 1280×800

Plan	§4.12 Reports
Event	**Click "Still on course"**
Plan says	*Not finished and not out, grouped by last checkpoint passed (tiles). For sweep / stand-down.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



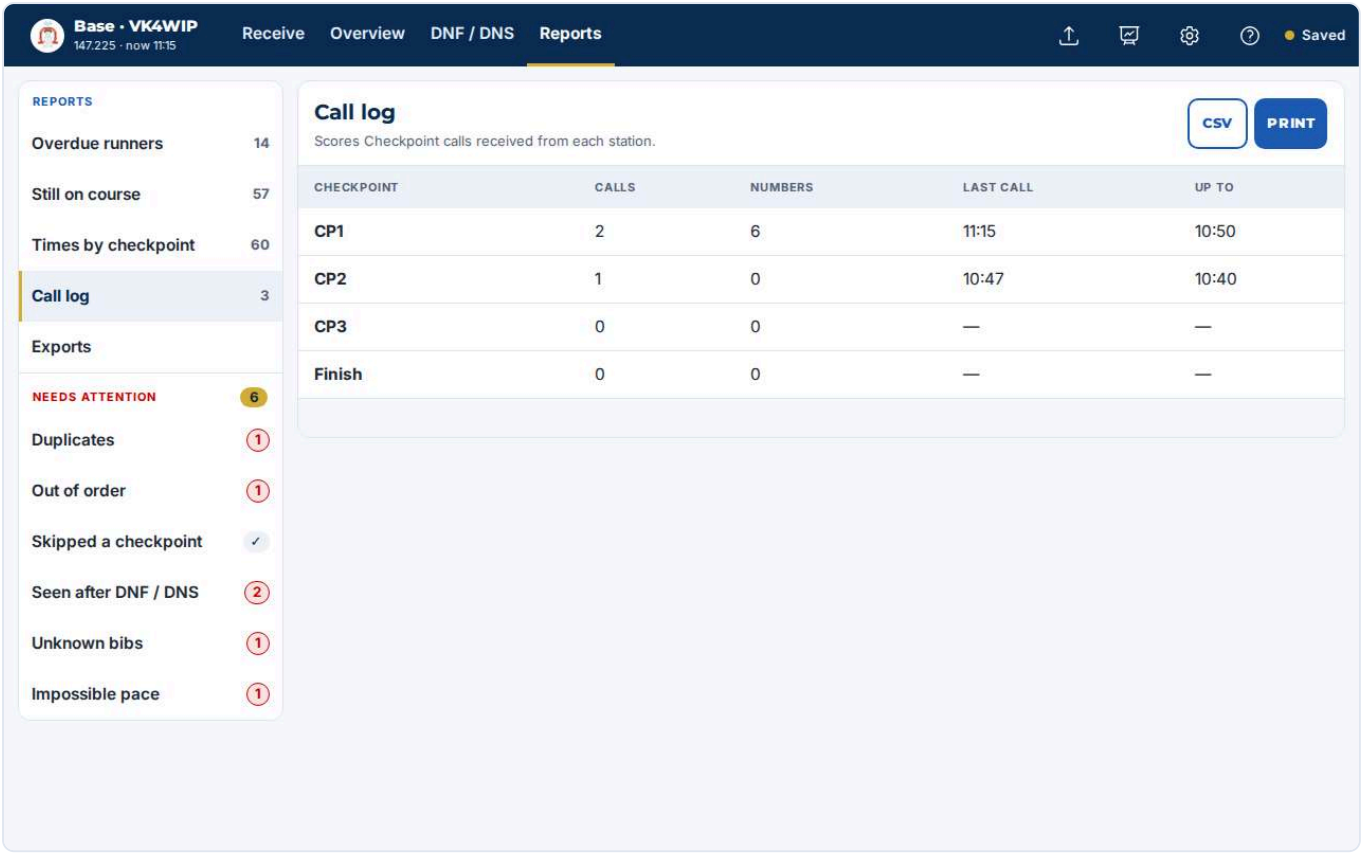
RP-04 1280×800

Plan	§4.12 Reports
Event	**Click "Times by checkpoint"**
Plan says	*Matrix bib × CP1…Finish (common times; DNF/DNS shown in place).*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



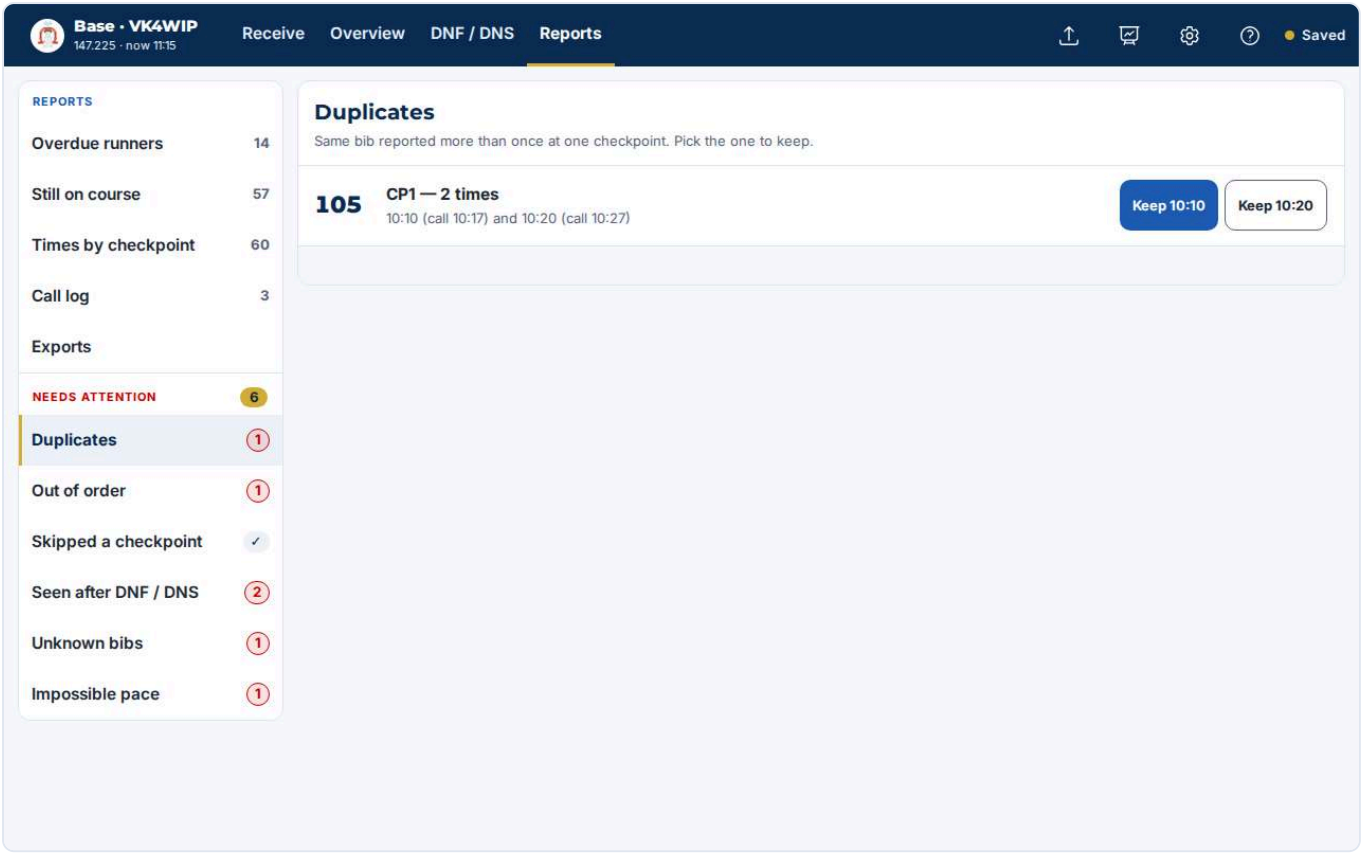
RP-05 1280×800

Plan	§4.12 Reports
Event	**Click "Call log"**
Plan says	*Per checkpoint: calls received, numbers, last call time, up to (lastBlock). From radio_calls.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



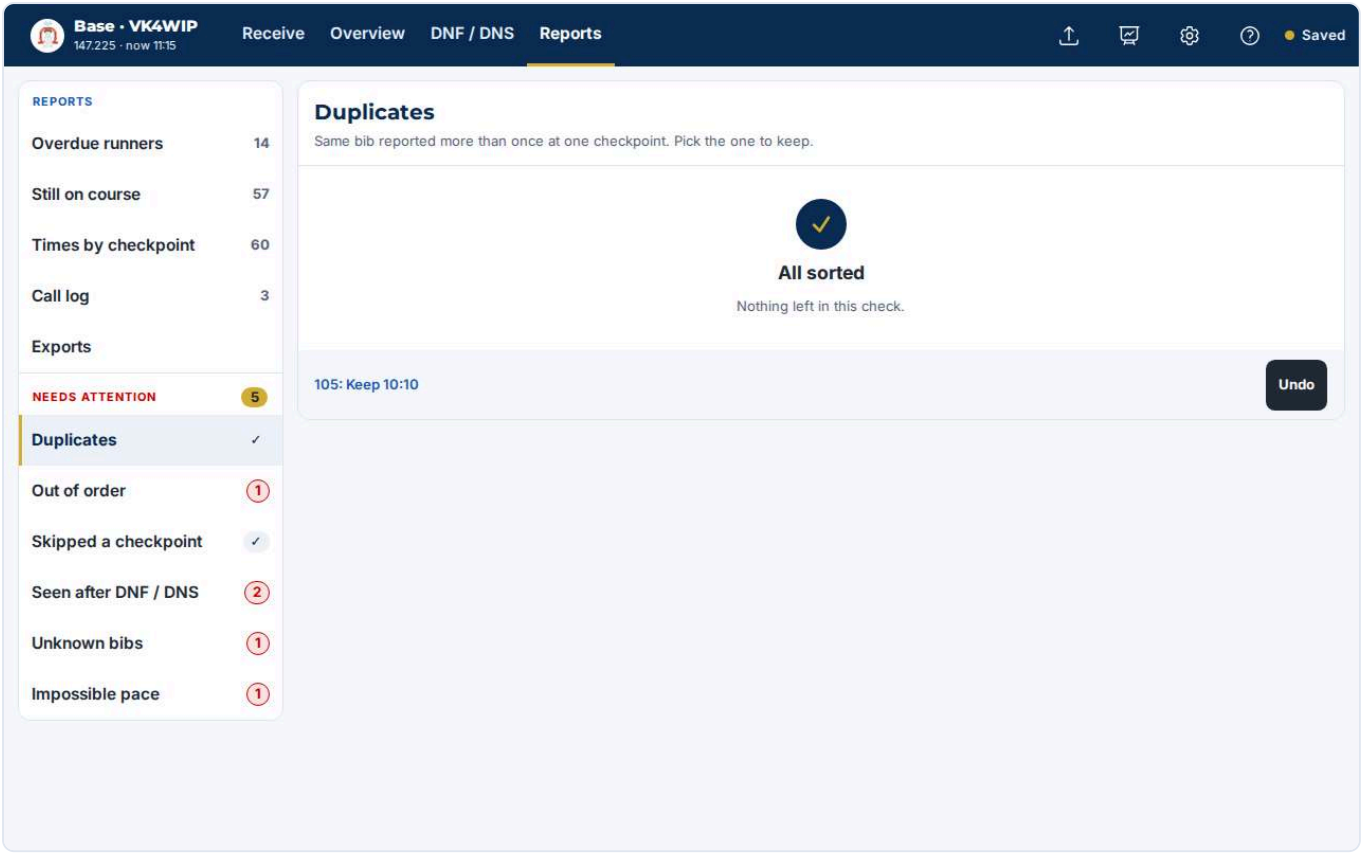
RP-06 1280×800

Plan	§4.12 Needs attention
Event	**Click "Duplicates"**
Plan says	*Same bib, same checkpoint, >1 base record (different or same time) → Keep A / Keep B (or Merge if identical).*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



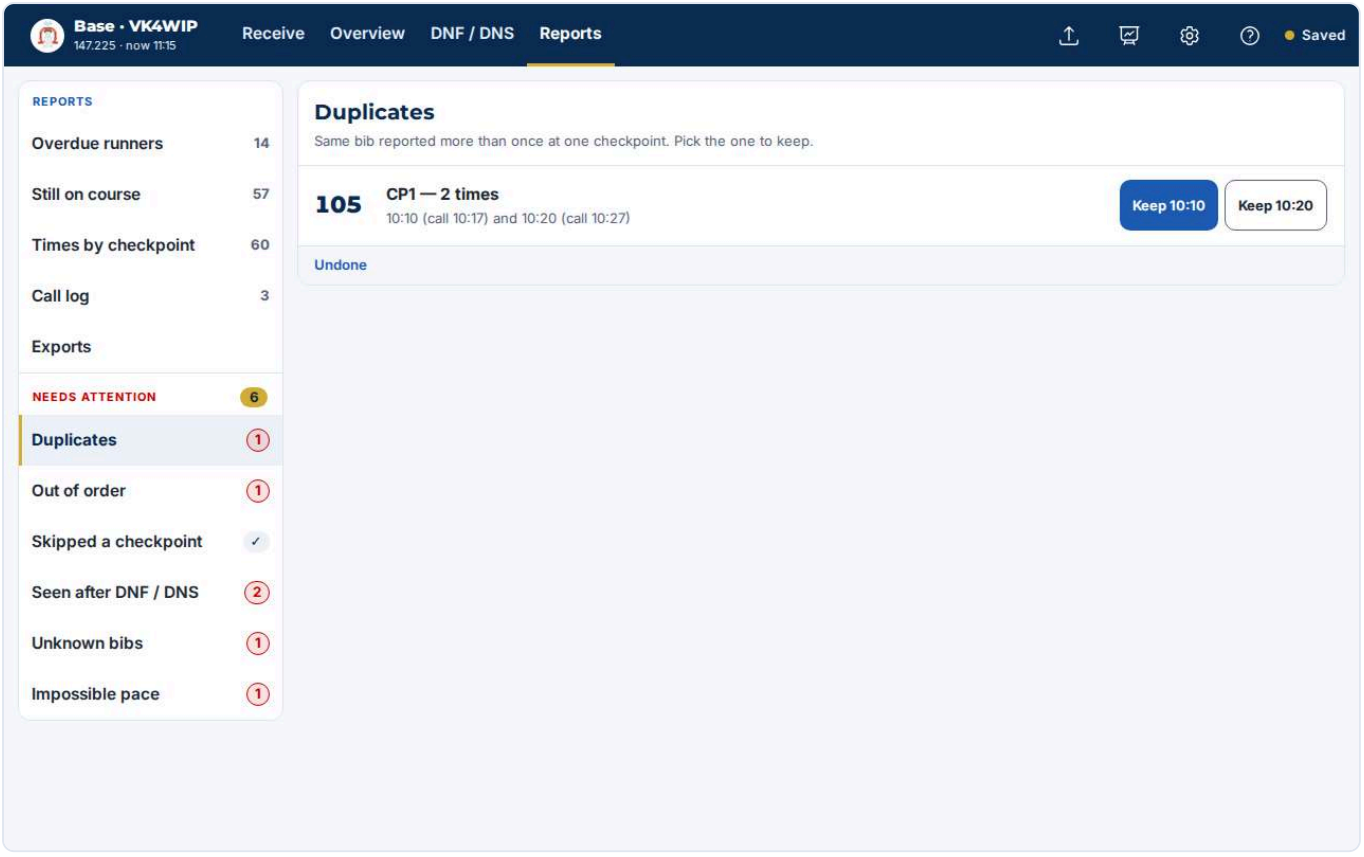
RP-07 1280×800

Plan	§4.12
Event	**Click the first "Keep HH:MM"**
Plan says	*Each issue row: bib, headline, detail, 1–2 action buttons; resolving removes it (Undo).*
App showed	• 105: Keep 10:10
Result	☐ Matches plan ☐ Issue: _____



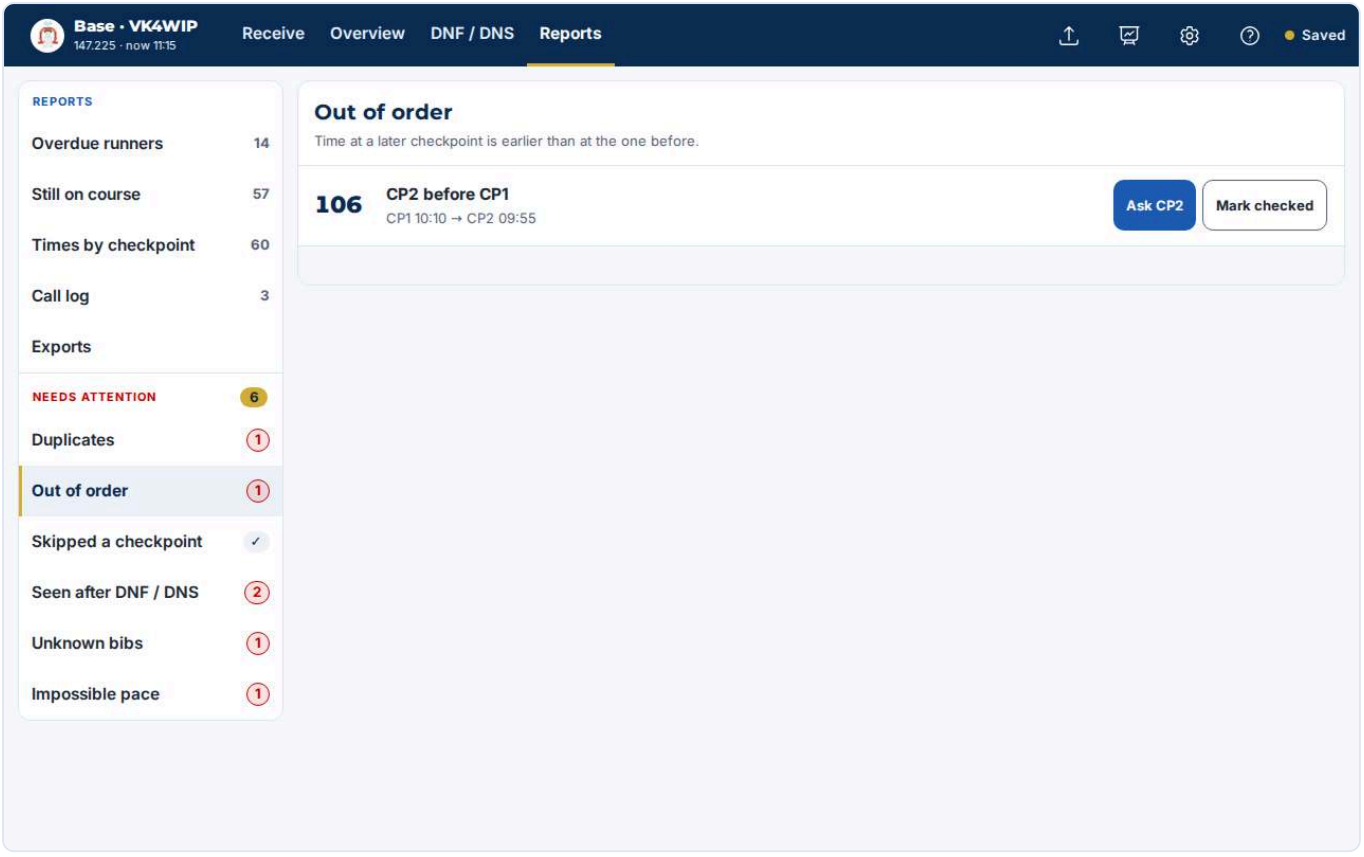
RP-08 1280×800

Plan	§4.12
Event	**Click "Undo"**
Plan says	*Resolving removes it (Undo).*
App showed	• Undone
Result	☐ Matches plan ☐ Issue: _____



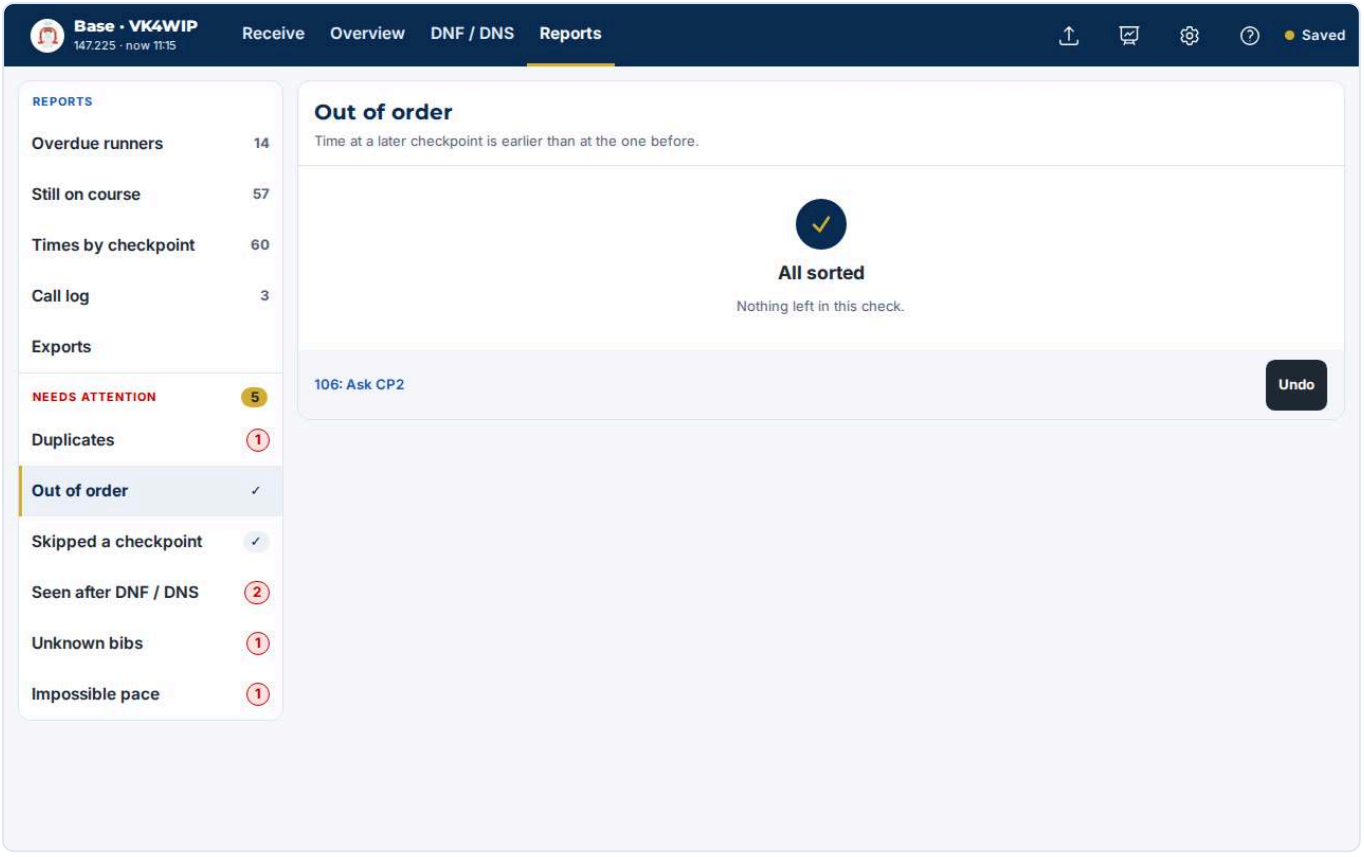
RP-09 1280×800

Plan	§4.12 Needs attention
Event	**Click "Out of order"**
Plan says	*Time at CP(n+1) < time at CP(n) → Ask CPx (flag for radio check) / Mark checked.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



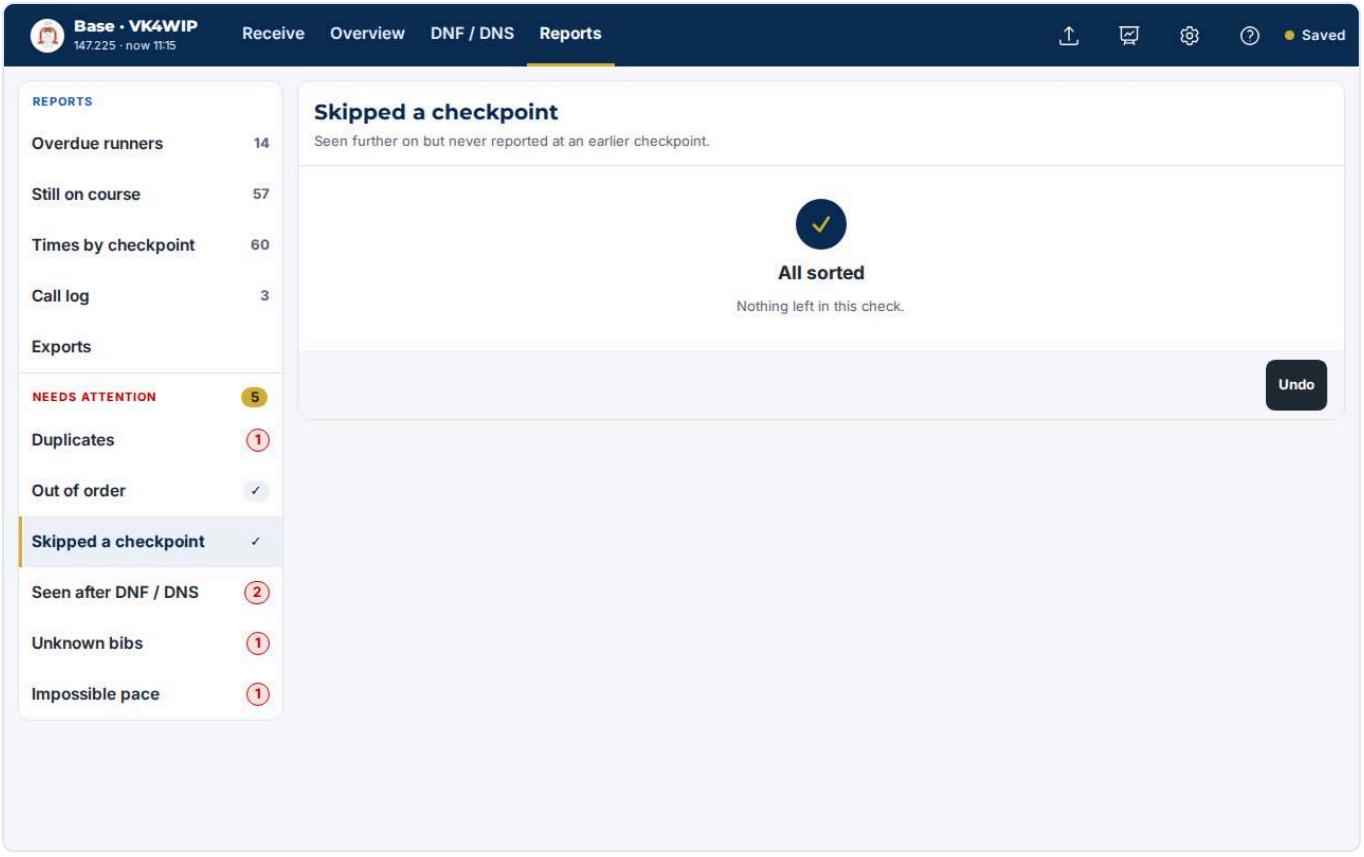
RP-10 1280×800

Plan	§4.12
Event	**Click "Ask CP2"**
Plan says	*Resolutions persist in a new data_issue_resolutions table so resolved items don't reappear.*
App showed	• 106: Ask CP2
Result	☐ Matches plan ☐ Issue: _____



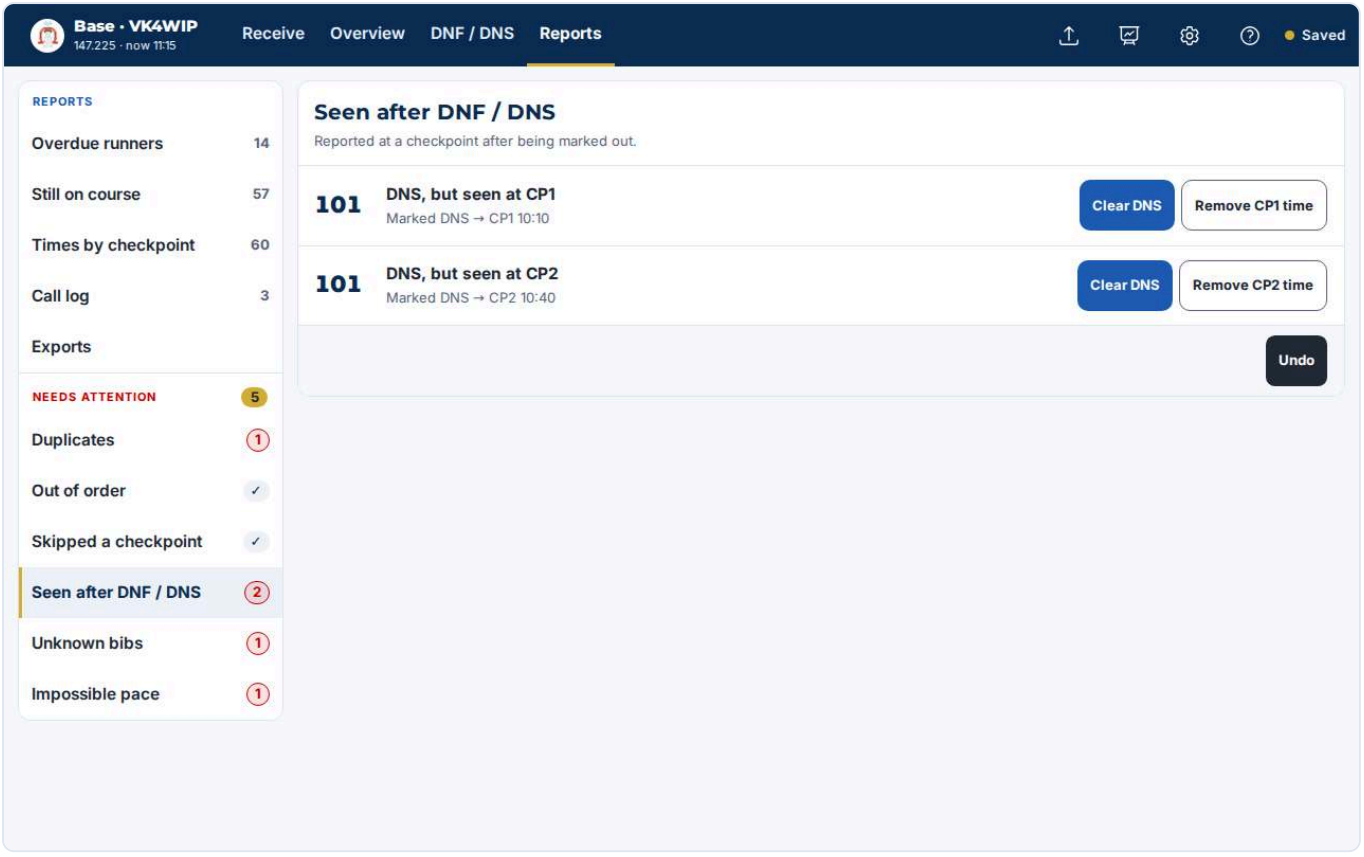
RP-11 1280×800

Plan	§4.12 Needs attention
Event	**Click "Skipped a checkpoint"**
Plan says	*Has later CP time, missing an earlier CP → Ask CPx / Mark checked.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



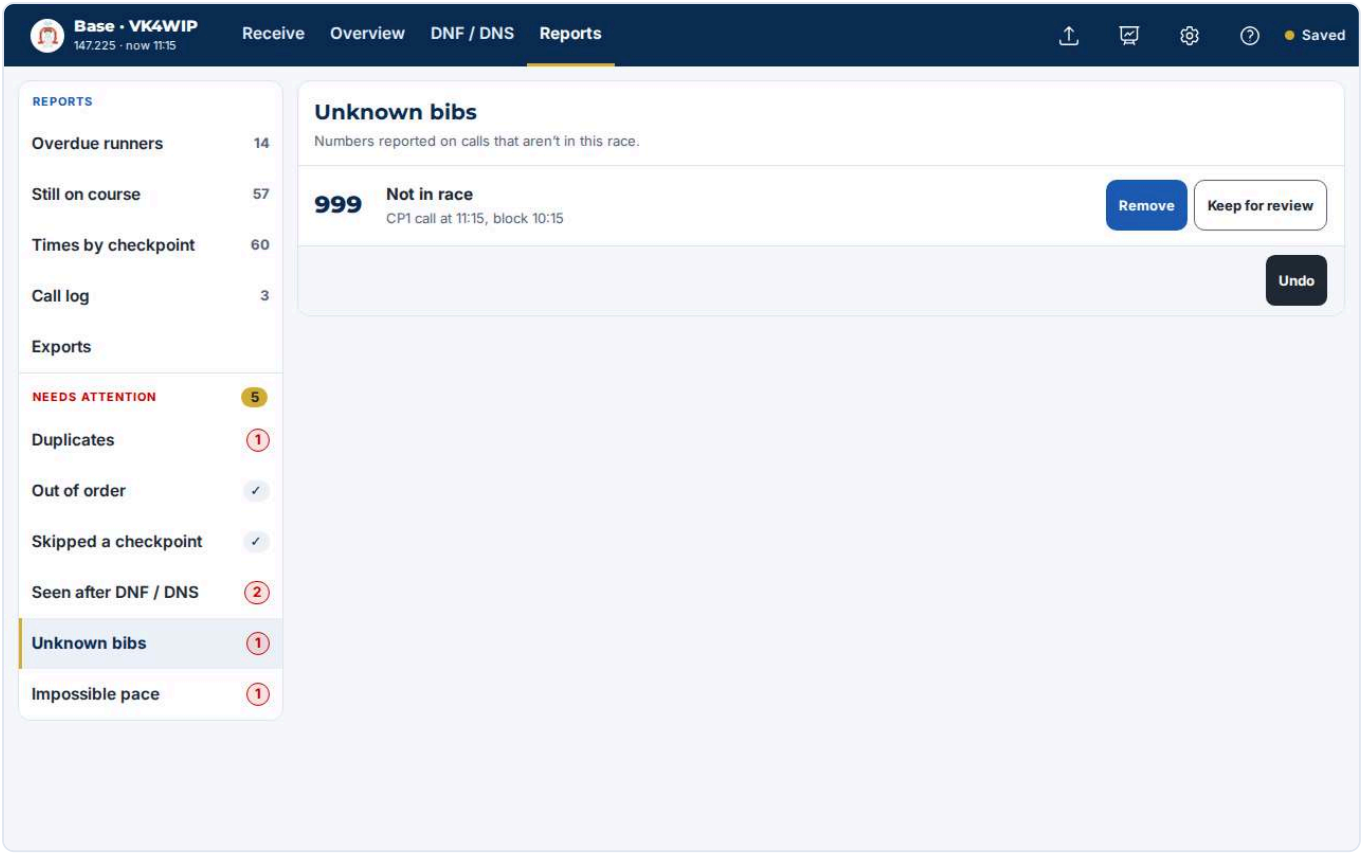
RP-12 1280×800

Plan	§4.12 Needs attention
Event	**Click "Seen after DNF / DNS"**
Plan says	*Checkpoint time after DNF time, or any time while DNS → Clear DNF/DNS / Remove that time.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



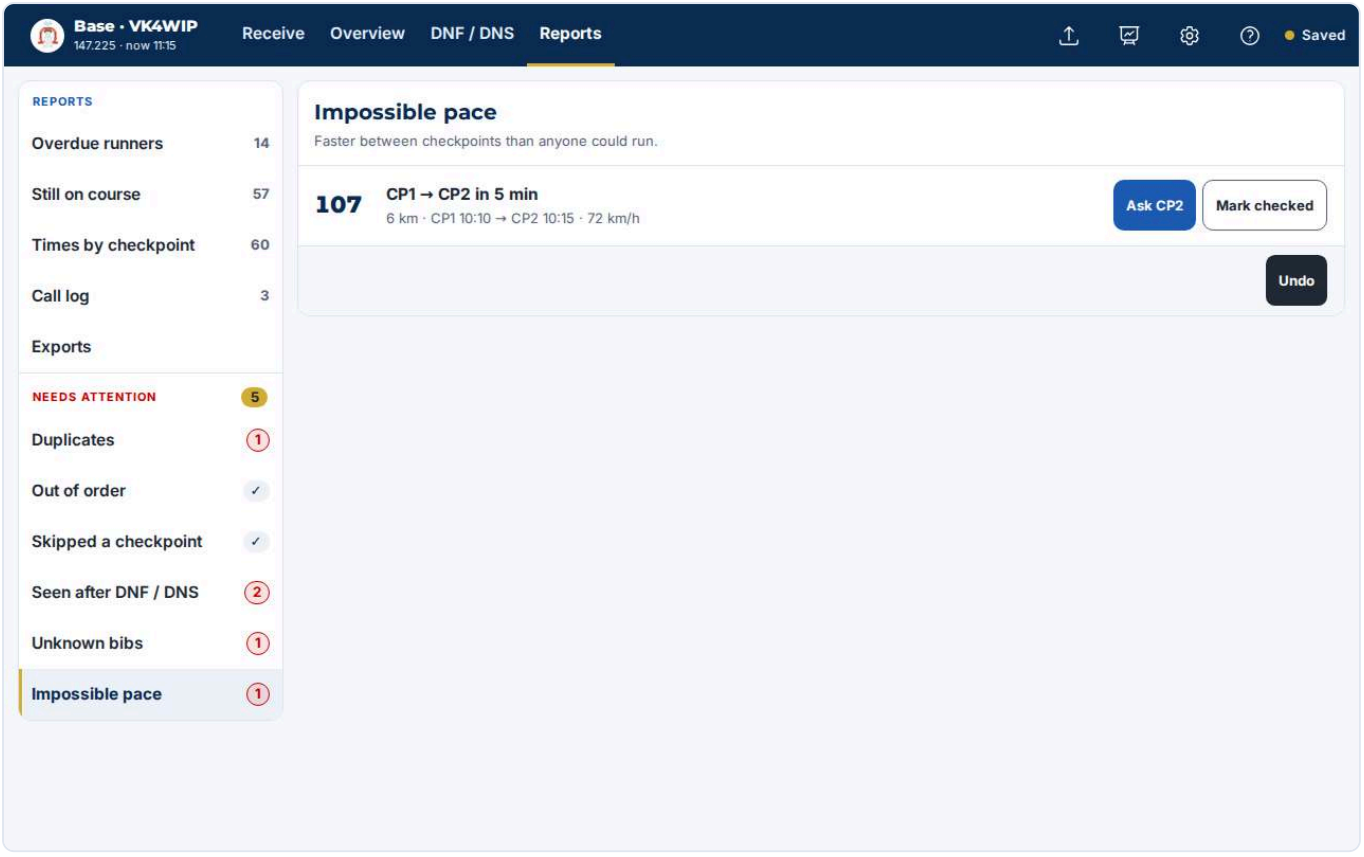
RP-13 1280×800

Plan	§4.12 Needs attention
Event	**Click "Unknown bibs"**
Plan says	*Numbers outside the race reported in calls (keep them in radio_calls.rejected) → Remove / Keep for review.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



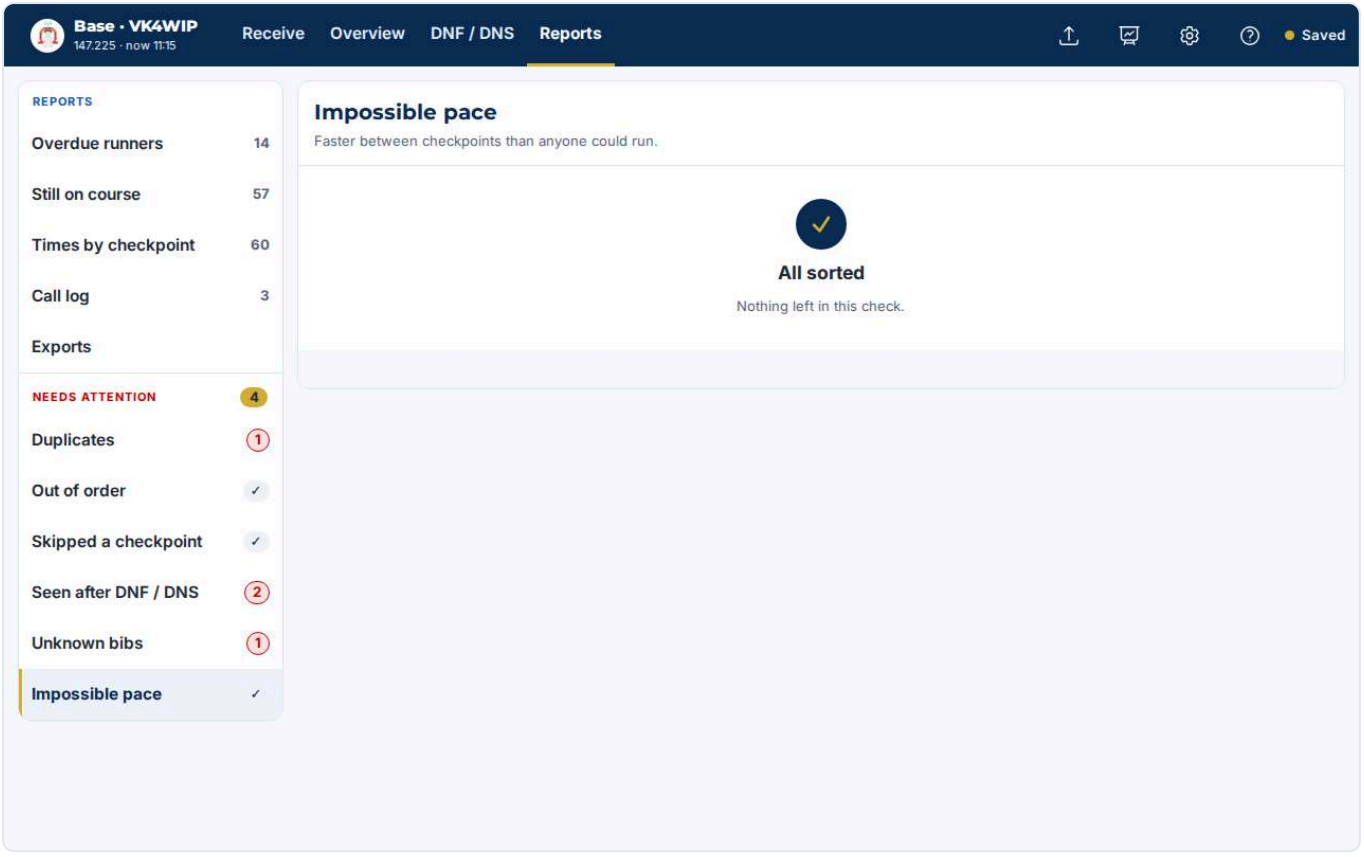
RP-14 1280×800

Plan	§4.12 Needs attention
Event	**Click "Impossible pace"**
Plan says	*Split speed above a threshold (default > 20 km/h; checkpoint distances from race config) → Ask CPx / Mark checked.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



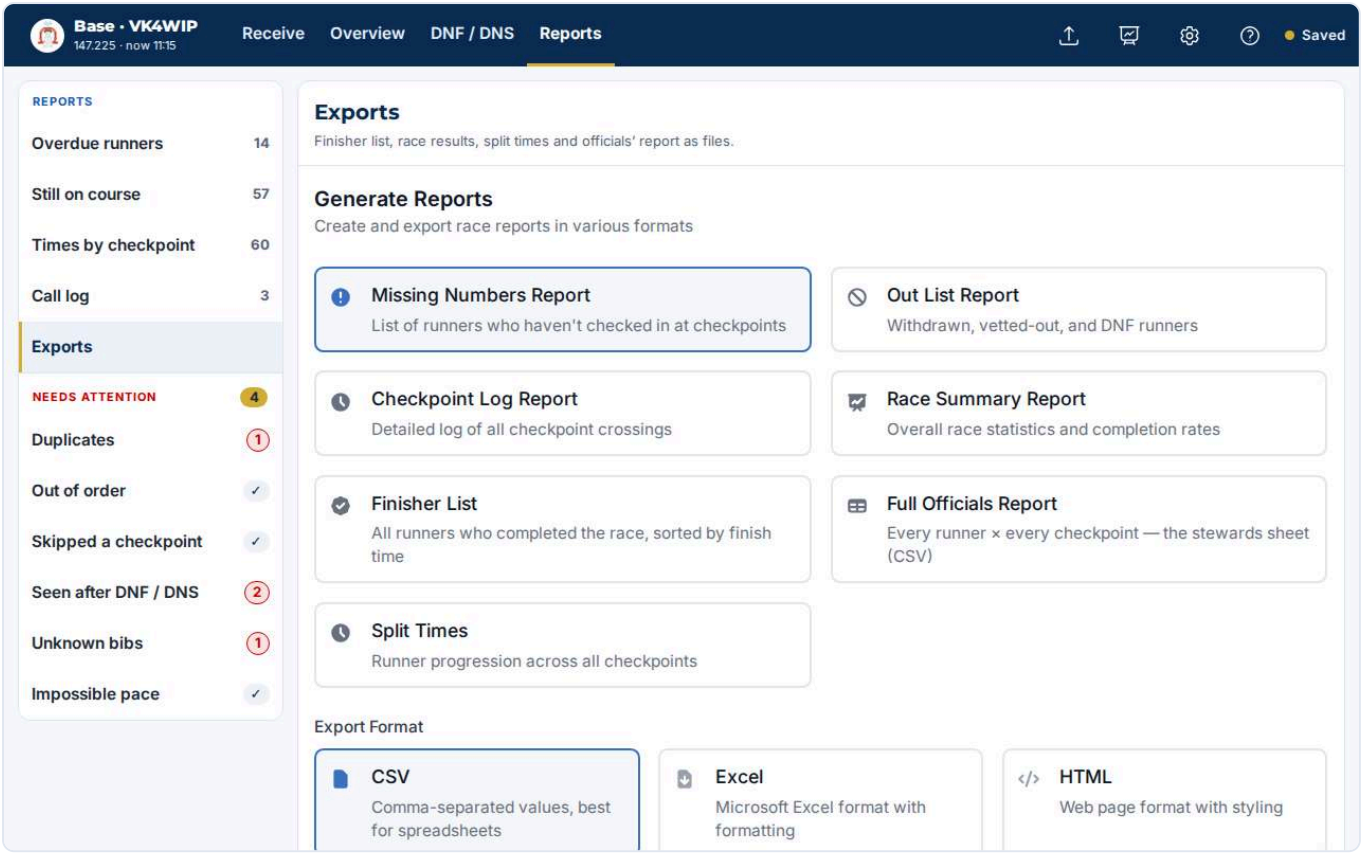
RP-15 1280×800

Plan	§4.12
Event	**Click "Mark checked", then reload the page and open Reports › Impossible pace**
Plan says	*"Mark checked" = acknowledged, data unchanged. Resolved items don't reappear.*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



RP-16 1280×800

Plan	§4.12
Event	**Click "Exports"**
Plan says	*Evolves ReportsPanel.jsx, ReportBuilder.jsx (existing exports kept).*
App showed	*(no status messages — see screenshot)*
Result	☐ Matches plan ☐ Issue: _____



Appendix — the plan (IMPLEMENTATION_SPEC.md, verbatim)

RaceTracker Pro — VK4WIP UI redesign: implementation spec

Status: approved design, ready to build. Design sources: `design/screens/*.dc.html`.

Target repo: `bwhite001/RaceTrackers` (React 18, Vite, Tailwind, Zustand, Dexie v10, PWA).

Owner: Brandon White VK4BRW. Club: Ipswich & District Radio Club, Inc. (VK4WIP).

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1. Goals

1. Restyle the whole app on the VK4WIP brand (navy / blue / gold / red, Montserrat + Inter).
2. Make checkpoint work safe and fast on a **phone**, one-handed, in sunlight.
3. Rebuild base station as a **receive** workflow — the inverse of a checkpoint read-out — that works on **tablet (primary), phone and desktop** without being tall.
4. Follow the WICEN radio procedure (see `reference/` PDF): Scores Checkpoint calls, common times in 5-minute blocks, every number read individually.
5. Let anyone fix mistakes: undo everywhere, edit times, restore accidental unmarks, base-controlled DNF/DNS, and data-check reports.

Non-goals: real-time device sync, RFID, new auth. Radio phrases are **not** shown in the UI (they live in `docs/radio-procedure.md` for the user guide).

---

2. Global rules (apply everywhere)

#	Rule
G1	**Every bib is handled individually.** Never collapse numbers into ranges ("144 to 148") in call-out, read-out, base receive, or previews. Ranges are only accepted in *Race setup* to define the field.
G2	**Common times are 5-minute blocks** (07:40, 07:45 …) unless an operator types an exact time. Block = `floor(minutes / 5) * 5`.
G3	**Times are never in the future** and never before race start. Base is often 45–60 min behind, so past blocks must be one tap away.
G4	**24-hour times** (`07:45`), Australian English, en-AU dates.
G5	**Undo** after every destructive or bulk action (snackbar/inline Undo). Two-tap confirm for "Cancel call" when it would discard data.
G6	Touch targets ≥ 44px (primary actions 48–56px). Real `<button>` / `<input>` / `<label>`; icon-only buttons need `aria-label`. Text contrast ≥ 4.5:1.
G7	**Status colours stay inside the club palette and differ by lightness/border, not hue alone** (see §3.3). Users may still recolour in Settings.
G8	One gold primary action per screen. Red only for DNF / destructive / safety alerts.
G9	No emoji in UI (replace existing ones, e.g. 📝 📋 🏃 in base views). Use line icons (Heroicons outline is already a dependency).
G10	Use `src/services/timeUtils.js` for all time maths; store ISO strings. The designs use a fixed demo clock of 07:47 — replace with real now.
G11	Respect module isolation (CLAUDE.md): checkpoint code only touches `checkpoint_runners`; base only `base_station_runners` + base tables.

---

3. Design system

3.1 Tokens (`design-tokens/`)

Replace the Tailwind-default navy/red/gold in `tailwind.config.js` and the CSS variables in `src/index.css` with the values in `design-tokens/tailwind.vk4wip.js` and `design-tokens/tokens.css`. Keep existing class names working by aliasing (`primary-*`, `navy-*`, `accent-red-*`, `accent-gold-*`, `gold-*`) — see the comments in that file.

Token	Hex	Use
navy	`#0B2D53`	App bars, headings, "called in" state, selected chips
blue	`#1E5CB3`	"Passed" state, links, focus rings, secondary buttons, info
gold	`#D4AF37`	Primary button (ink text), "call due", pending/needs-attention, active-tab underline
red	`#CC0000`	DNF, destructive actions, overdue/safety alerts only
ink	`#222A35`	Body text; text on gold
muted	`#5A6579`	Secondary text (do **not** lighten — contrast)
paper	`#F7F8FB`	Page background
band	`#EEF3FA`	Panels, selected rows, segmented-control track
border	`#E0E6F0`	Card / divider borders

surface-alt	`#E9F2FF`	Light text on navy
Tints	`rgba(30,92,179,.1)` blue · `rgba(212,175,55,.1)` gold · `rgba(204,0,0,.1)` red	Backgrounds for states

Radii: 6 (chips), 10 (buttons, inputs, tiles), 12 (cards), 14 (sheets), 9999 (pills/badges). Shadows: `0 2px 10px rgba(0,0,0,.08)` (buttons/cards), `0 10px 30px rgba(0,0,0,.18)` (sheets/toasts).

3.2 Typography

- **Montserrat** 600/700/800 — headings, bib numbers, button labels, big times. Bib numbers use `font-variant-numeric: tabular-nums`.
- **Inter** 400–700 — body, labels, tables.
- Load both via the PWA (self-host or Google Fonts with `display=swap`; precache in the service worker so they work offline — bump the PWA version per CLAUDE.md).
- Keep the existing `--font-scale` setting; all sizes scale with it.

3.3 Runner state styles (single source of truth — put in a `statusStyles.js` helper)

State	Tile / chip	Label
Not passed	white, 1px border `#E0E6F0`	—
Passed (not called in)	solid blue, white text	time `07:44`
Passed, edited after call-in	solid blue + `✎` prefix on time	"Correction" chip in lists
Called in	solid navy, white text	`✓ 07:31` (gold tick)
Unmarked earlier (restorable)	white, **2px dashed blue**	`was 07:37`
DNS	paper bg, **1px dashed muted**, number struck through	`DNS`
DNF	red tint, **2px red border**, red text	`DNF`
Base check: OK	solid blue	—
Base check: info (e.g. not seen at previous CP)	solid blue + 2px navy border	note
Base check: warning (duplicate / already logged)	white, 2px gold border	note
Base check: rejected (not in race / out)	white, 2px red border, struck through	note

3.4 Buttons

Montserrat 800, uppercase, letter-spacing .6px, radius 10, min-height 48.

- **Primary**: gold bg, ink text, shadow. One per screen.
- **Secondary**: transparent, 2px blue border, blue text.
- **Neutral solid**: blue bg, white text (Add, Next block).
- **Plain**: white, 1px muted border, ink text.
- **Destructive**: red bg white text (Mark DNF) or red outline (Cancel call, first tap).
- **Disabled/locked**: white or band bg, dashed border, muted text, `cursor: not-allowed`, `disabled` attribute.

3.5 Shared shell components (build first, Phase 0)

Component	Spec	Design ref
`AppBar`	Navy bar. Phone: back button (44px), title (Montserrat 700 18px), subtitle (12px `#E9F2FF`), right "Saved" status (8px gold dot). Base: badge (32px white circle with `public/pwa-512x512.png`), "Base · VK4WIP", "147.225 · now HH:MM", inline tabs, single row on tablet.	all
`CheckpointTabBar`	Bottom nav, 72px, 3 tabs (Mark off, Callout, Overview), line icons, active = navy text + 3px gold top border, `aria-current="page"`. Callout tab shows a gold count badge = runners passed but not called in.	`CpMarkOff`
`BaseTabs`	Inline in AppBar: Receive · Overview · DNF / DNS · Reports. Active = white bold + 3px gold underline. Horizontally scrollable on phone.	`BaseEntry`
`StatTile`	Small tiles with Montserrat 800 number + 11px label. Gold variant for "To call in".	`CpMarkOff`
`UndoBar`	Dark ink pill above tab bar or inline status line with an Undo button (gold or ink). Auto-dismiss ~6s, cleared by the next action.	`CpMarkOff`, `BaseEntry`
`BottomSheet`	Navy 55% scrim, white sheet radius 14 top, `role="dialog"`, closes on scrim tap / Cancel.	`CpMarkOff`
`TimeStepper`	−5 / −1 / big time / +1 / +5. Clamp to [race start, now].	`CpMarkOff`, `CpOverview`
`BlockPicker`	Horizontally scrolling row of 5-minute block buttons (64×48), labels `done` / `next` / `this call`, selected = navy; plus an "Exact" time input at the end (dashed when unused, navy when set).	`BaseEntry`
`NumberPad`	3×4 grid: 1–9, ⌫, 0, Add (blue). 46–52px keys. Always paired with a real numeric `<input>` that also accepts keyboard + Enter.	`BaseEntry`
`BibChip`	Number (Montserrat 800 18–20px) + optional 10–11px note; style per §3.3; tap = remove (with Undo).	`BaseEntry`

4. Screens

Each section: route → component(s) → layout → behaviour → data. Design file in brackets.

4.1 Home / role picker [`Home.dc.html`]

Route `/` → `components/Home/Homepage.jsx`.

- Navy header: badge, "RaceTracker", "Ipswich & District Radio Club · VK4WIP", "Offline ready" status. Current race card (name, date, checkpoint count, start time).
- Three role cards (≥76px): **Checkpoint**, **Base station**, **Race setup** — icon tile, title, one-line description, chevron. Navigation protection unchanged.
- Primary button at bottom: **Scan race QR code** (existing import flow).
- Footer lockup: "Ipswich & District Radio Club · VK4WIP".

4.2 Race setup [`RaceSetup.dc.html`]

Route `/race-maintenance/setup` → `components/Setup/RaceSetup.jsx`. Restyle only; keep the 4-step wizard.

- Navy header with step "Step 3 of 4 · Runners" and a 4-segment gold progress bar.
- Runner ranges input (ranges allowed **here only**), live count ("250 runners will be created") and first/last preview chips.
- Link to CSV / WebScorer import. Back (secondary) + Next (primary).

4.3 Share config [`ShareConfig.dc.html`]

Existing QR export: large QR in a white card with shadow, runner/checkpoint count; Download / Email / Copy tiles; primary "Show full screen" (`QRDisplayFullscreen`). Note "Works without internet — device to device".

4.4 Settings [`Settings.dc.html`]

`components/Settings/SettingsModal.jsx` (may become a page on phone). Sections: Display (Theme segmented: Light / Dark / Sunlight; text-size slider), Status colours (rows with swatch, name, description; "Reset to club colours"), Race data (Share or import race config).

- **Sunlight** = new high-contrast theme option (pure white bg, ink text, thicker borders). Optional for v1 — if skipped, show Light/Dark only.

4.5 Checkpoint — Mark off [`CpMarkOff.dc.html`] (interactive)

Route `/checkpoint/:checkpointId` → `views/CheckpointView.jsx` + `components/Checkpoint/RunnerGrid.jsx`, `QuickEntryBar.jsx`.

Layout (390px phone): AppBar (checkpoint number, location + operator callsign) with 3 StatTiles (Passed, Still to come, **To call in** gold) → quick-entry input + gold **Mark** → status line (aria-live) → group chips (20 per group: 100–119 …, horizontally scrolling) → 4-column tile grid (64px rows) → CheckpointTabBar.

Behaviour:

1. **Tap a not-passed tile** → mark passed at now (actualTime = now, commonTime = block).
2. **Tap a passed or called-in tile** → opens the **Edit sheet** (never unmarks directly — this replaces the current tap-again-to-unmark / double-click behaviour).
3. **Edit sheet**: bib, status line ("Passed · not called in yet" / "Called in to base"), "Marked at HH:MM". TimeStepper (clamped). Primary **Save HH:MM** (or "Keep time" if unchanged). Secondary **Unmark runner**. Cancel.
4. If the runner was **called in** and the time changes → warning box "Base already has 07:31. A changed time goes back on the callout sheet as a correction." Saving sets status back to passed with `correction = true` so it reappears on the callout sheet; tile shows `✎`.
5. **Unmark** → tile becomes "unmarked earlier" (dashed blue, `was HH:MM`) and an UndoBar "142 unmarked · Undo" appears. Undo restores the original record exactly.
6. **Tap an unmarked-earlier tile** → sheet in restore mode: "Unmarked earlier · It had been marked at 07:37", TimeStepper preset to the original, primary **Mark at 07:37**, secondary **Use now (HH:MM)**.
7. Quick entry accepts multiple numbers separated by commas/spaces (individual numbers only), marks each at now, jumps the grid to the group of the first number, message "3 runners marked at 07:47".
8. DNS/DNF tiles are not editable here (use Overview).

Data (`checkpoint_runners`): add `originalTime` (first mark), `editedAt`, `correction` (bool), `unmarkedAt` + `unmarkedTime` (for restore). See §5.

4.6 Checkpoint — Callout sheet [`CpCallout.dc.html`]

`components/Checkpoint/CalloutSheet.jsx`.

- AppBar subtitle: `CP3 · VK4PMC → VK4WIP · 147.225` (checkpoint, operator callsign, base callsign, frequency — from race config; hide pieces that aren't configured).
- **Call-due banner** (gold) when either: ≥ 5 runners waiting, or the oldest uncalled runner is ≥ 5 min old (configurable; WICEN brief: every 5–10 competitors, ≤ ~5 min delay). Text: "Call due — 9 waiting, oldest 6 min".
- Called-in blocks collapse to one compact row (navy tick, `07:30 – 07:35 · 4`, numbers, "Sent 07:36").
- **Next call card** (gold border, gold tint): heading "Next call · 2 blocks", badge "9 competitors", then one white card per pending block: `Common time 07:40 · 5` + the numbers in Montserrat 800 22px separated by `·`.
- Buttons: **Read out** (secondary, opens 4.7) + **Base copied — mark sent** (primary) → marks every runner in the pending blocks `calledIn = true`, `callInTime = now`.
- Below: **Official message** and **Request stand down** (plain buttons; v1 can open a simple notes modal / log entry).
- Corrections (runners with `correction = true`) appear in the pending list tagged "Correction".
- **No radio phrases on screen.**

4.7 Checkpoint — Read-out mode (busy blocks) [`CpReadout.dc.html`] (interactive)

New view, e.g. `modules/checkpoint-operations/components/ReadOutView.jsx`, opened from Callout.

- Header: "Read-out", `CP3 · block 07:45–07:50 · 27 competitors`, progress "X of N copied by base", segmented progress bar (one segment per over: gold = done, white = current, faint = to come).
- Numbers are split into **overs of up to 5 numbers** (`overSize` setting, default 5, range 3–10). **Each bib is its own row** (G1).
- Current over card: lead line `Common time 07:45` (first over) / `Continuing 07:45` (later overs); one row per number with a **checkbox** (28px, whole row is the label, ≥56px tall). The **next unticked** row is highlighted (band bg + 2px blue border + "Read next"); ticked rows go muted with strike-through.
- Primary **Base copied — next** is **locked until every number in the over is ticked** (shows "Tick each number as you read · 2 of 5").
- **Reset** (secondary) clears ticks on the current over. **Previous over** steps back.
- Finishing the last over → done state "All 27 copied", which marks the block called in.
- Persist read-out progress so an interrupted read resumes at the same over.

4.8 Checkpoint — Overview (fix anything) [`CpOverview.dc.html`] (interactive)

Existing overview tab.

- Header: % through, "142 of 250 through", stacked bar (called in / passed not called / to come / out), `role="img"` with aria-label.
- 4 StatTiles: Passed, To come, DNS, DNF. Filter pills: All · Passed · To come · DNS / DNF.
- Scrollable runner list; **tap a row to expand** (one open at a time, `aria-expanded`). Actions by status:

Status	Actions
Called in	TimeStepper, **Undo call-in**, **Reset — not passed**, **Mark DNF** (+ note "Base already has HH:MM. Changing the time sends a correction.")
Passed	TimeStepper, **Mark called in**, **Reset — not passed**, **Mark DNF**
To come	**Passed now (HH:MM)** (gold), **Mark DNS**, **Mark DNF**
DNS / DNF	**Clear DNS/DNF**, **Passed now** (+ note that base is flagged)

- Changing the time of a called-in runner → status passed + `correction = true`.
- Every action shows UndoBar; totals, bar and the Callout badge update live.

4.9 Base — Receive [`BaseEntry.dc.html` , tablet `BaseEntryTablet.dc.html` , phone `BaseEntryPhone.dc.html`] (interactive)

Route `/base-station/operations`, tab **Receive** → replaces `DataEntry` / `BatchEntryLayout` UI in `modules/base-operations`. **One responsive component** — no separate tablet/phone builds.

Layout: AppBar + BaseTabs (single row on tablet). Main = flex-wrap: receive panel (`flex: 999 1 480px`) + checkpoint board (`flex: 1 1 260px`). Tablet landscape (1112×834) and desktop: side by side, **must fit in 834px height without page scroll**. Phone: stacks, receive panel first.

Idle — "Who's calling?": grid (`auto-fill, minmax(140px,1fr)`) of compact buttons (≥64px) for Start, CP1–CP4, Finish: short name, status pill (Due gold / OK / Closed), `VK4PMC · up to 07:15`. "Due" = runners expected and no call for ≥ 8 min (constant `CALL_DUE_MINUTES = 8`, configurable).

Tap a checkpoint → straight into receiving (no separate acknowledge step):

- Header row: CP short name, operator · location, "N to record".
- **BlockPicker** from `lastBlock − 5` to the current block. Default selection = **that checkpoint's next unreceived block** (`lastBlock + 5` , capped at now) — base is usually behind. Chips: `done` (≤ lastBlock, muted), `next` , `this call` (has numbers in this call, blue border), selected = navy. "Exact" input accepts `HH:MM` or `HHMM` between race start and now; sets a non-block time and shows it selected.
- **Numbers box** (scrolls internally, max ~210px): shows **only the current block's numbers**. Switching block clears the box and the typed number (nothing is lost). Other blocks in the call show underneath as pills: "Also in this call: 07:40 · 5" — tap to switch back.
- Input + **Add** + NumberPad side by side (pad `flex: 0 1 220px`). Enter adds. Each added number becomes a BibChip validated live:

Check	Kind	Note
Not in race range	rejected (red)	"Not in race"
Marked DNF / DNS	rejected	"DNF" / "DNS"
Same bib earlier in this call	warning (gold)	"Said twice"
Already logged at this checkpoint	warning	"Had 07:40"
Not logged at previous checkpoint	info (navy border)	"Not seen CP2"
Otherwise	ok (blue)	—

Only **ok + info** are recorded. Tap a chip to remove (Undo).

- Status line (aria-live) + inline **Undo** when available.
- **Action row** (`repeat(auto-fit, minmax(110px,1fr))`):
- **Cancel call** — red outline; if numbers exist, first tap turns solid red "Tap again to cancel" and says how many will be dropped.
- **Clear 07:40** — clears all numbers in the current block (Undo).
- **Next 07:45 →** — move to the next 5-minute block, clear the box; disabled "At now" at the current block.
- **Complete · N** — gold primary; records the call.
- **Complete** → writes base_station_runners (`commonTime` = each number's block, `markOffEntryTime` = now, `checkpointNumber`), writes a `radio_calls` record, updates the board (count, last call = now, `lastBlock` = max block received). Done card: "N recorded at CP3", "Blocks 07:40, 07:45 · 2 skipped (flagged)", **Undo** (reopens the call, reverts writes) and **Next call**.
- **Checkpoint board**: rows with short name, "Up to 07:15 · 30 min behind · call 11m ago", count, Due/OK/Closed pill; active checkpoint row gold-tinted. "behind" shown when ≥ 15 min.

4.10 Base — Overview [`BaseOverview.dc.html`]

Existing overview tab restyled: red overdue alert banner (`role="alert"` , "View runners" → Reports › Overdue), stat tiles (On course, Finished, DNS, DNF), checkpoint progress bars (navy = closed out, blue = arriving, gold = finish), leaderboard table with filter pills (All, Female, Male, Wave 1…), gold rank badge for #1, overflow-x box on phone.

4.11 Base — DNF / DNS [`BaseStatus.dc.html`] (interactive)

Tab **DNF / DNS** → evolves `BulkDnsView.jsx` + `WithdrawalDialog.jsx` . Base is the authority for race-wide DNF/DNS.

- **Bulk add** panel: DNF/DNS segmented toggle (DNF active = red). DNF shows **Where** pills (CP1–CP4, On course), **Reason** pills (Injury, Self-withdrew, Cut-off, Medical, Other — optional, tap again to unset), **Time** (−5/+5, clamped). DNS shows a one-line explainer.
- Input accepts **one number or a pasted list** (`212, 218 224` — commas, spaces, semicolons, newlines). Chips validated:

Check	Kind	Note
Not in race / listed twice	rejected	"Not in race" / "Listed twice"
Already this status	rejected	"Already DNF"
Currently the other status	warning	"Was DNS — will change"
DNS but recorded at a checkpoint	warning	"Seen CP2 — started"
DNF at CPx but seen at a later CP	warning	"Seen later at CP3"

- **Clear list** + primary **Mark 6 as DNF** (red for DNF, gold for DNS; disabled when 0). Undo.
- **Out of the race** panel: totals "3 DNS · 3 DNF", filter pills, rows (bib, status chip, "CP2 · Injury · 07:12" / "Did not start", **Clear** → back in race, Undo).
- DNF writes `withdrawal_records` (checkpoint, withdrawalTime, reason) + runner status; DNS sets runner status `non-starter` . Audit-log every change.
- Open question for Brandon (default shown): checkpoint-marked DNFs count immediately; a future "confirm" queue is out of scope.

4.12 Base — Reports & data checks [`BaseReports.dc.html`] (interactive)

Tab **Reports** → evolves `ReportsPanel.jsx` , `ReportBuilder.jsx` , `DuplicateEntriesDialog.jsx` , `MissingNumbersList.jsx` .

Layout: sidebar (`flex: 1 1 250px`) + report panel (`flex: 999 1 520px`); stacks on phone.

Reports (CSV + Print buttons; reuse existing export utilities):

Report	Content
Overdue runners	Bib, last seen (CP + time), expected at, overdue minutes (gold badge; red ≥ 60). Expected time = last seen + typical split for that leg (median of recorded splits).
Still on course	Not finished and not out, grouped by last checkpoint passed (tiles). For sweep / stand-down.
Times by checkpoint	Matrix bib × CP1…Finish (common times; DNF/DNS shown in place).
Call log	Per checkpoint: calls received, numbers, last call time, up to (lastBlock). From `radio_calls` .

Needs attention (sidebar group with total gold badge; each item red count badge or ✓). Each issue row: bib, headline, detail, 1–2 action buttons; resolving removes it (Undo).

Check	Detection	Actions
Duplicates	same bib, same checkpoint, >1 base record (different or same time)	Keep A / Keep B (or Merge if identical)
Out of order	time at CP(n+1) < time at CP(n)	Ask CPx (flag for radio check) / Mark checked
Skipped a checkpoint	has later CP time, missing an earlier CP	Ask CPx / Mark checked
Seen after DNF / DNS	checkpoint time after DNF time, or any time while DNS	Clear DNF/DNS / Remove that time
Unknown bibs	numbers outside the race reported in calls (keep them in `radio_calls.rejected`)	Remove / Keep for review
Impossible pace	split speed above a threshold (default > 20 km/h; checkpoint distances from race config, e.g. CP1 5 km, CP2 11 km, CP3 14.5 km, CP4 17.5 km)	Ask CPx / Mark checked

Resolutions persist in a new `data_issue_resolutions` table so resolved items don't reappear; "Mark checked" = acknowledged, data unchanged.

---

5. Data model changes (Dexie v11 — additive)

Add `this.version(11).stores({...})` , keeping v10 tables. New indexed fields only where queried; others are data columns.

```
this.version(11).stores({
  checkpoint_runners: "++id, [raceId+checkpointNumber+number], raceId, checkpointNumber, number, actualTime, commonTime, calledIn,
markOffTime, callInTime, status, notes, correction",
```

```
  base_station_runners: "++id, [raceId+checkpointNumber+number], raceId, checkpointNumber, number, commonTime, markOffEntryTime,
status, notes, callId",
  radio_calls: "++id, [raceId+checkpointNumber], raceId, checkpointNumber, receivedAt, undoneAt",
  data_issue_resolutions: "++id, [raceId+issueKey], raceId, issueKey, resolvedAt",
  readout_progress: "++id, [raceId+checkpointNumber+blockTime], raceId, checkpointNumber, blockTime",
});
```

Table / field	Purpose
`checkpoint_runners.originalTime`	first marked time (restore)
`.editedAt`, `.correction`	time changed after call-in → resend
`.unmarkedAt`, `.unmarkedTime`	"was HH:MM" restore after accidental unmark
`base_station_runners.callId`	link to `radio_calls`; Undo call deletes by callId
`.manualTime` (bool)	exact time, not a 5-min block
`radio_calls`	`{raceId, checkpointNumber, receivedAt, blocks:[{time, numbers[]}], recorded, skipped: [{number, reason}], rejected:[…], undoneAt}`
`checkpoints` data columns	`operatorCallsigns[]`, `distanceKm`, `locationName` (Receive board, pace check)
`races` data columns	`baseCallsign` (VK4WIP), `frequency` (147.225), `callDueMinutes`, `overSize`
`withdrawal_records.reason`	DNF reason (Injury, Self-withdrew, Cut-off, Medical, Other)
`data_issue_resolutions`	`{issueKey: "dup:142:3", resolution, resolvedAt}`
`readout_progress`	`{tickedNumbers[], overIndex}` for resuming a read-out

Never cross `checkpoint_runners` ↔ `base_station_runners` (G11). Audit-log all base edits (`audit_log`).

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6. State & logic notes

- Put pure logic in testable services:
- `services/blocks.js`: `toBlock(t)`, `blocksBetween(from, to)`, `nextBlock(t, now)`, `clampTime(t, raceStart, now)`.
- `services/receiveValidation.js`: `checkBaseNumber(num, ctx)` → `{kind: ok|info|warn|bad, note}` per §4.9 table.
- `services/statusValidation.js`: DNF/DNS checks per §4.11.
- `services/dataChecks.js`: one function per check in §4.12, each returning issues with stable `issueKey`s.
- `services/readout.js`: `splitIntoOvers(numbers, overSize)` — **no range collapsing**.
- `services/callout.js`: `isCallDue({waiting, oldestMinutes}, settings)`.
- Base receive call state can live in `baseOperationsStore` as a draft (survives tab switches, cleared on Complete/Cancel).
- Undo: keep a single-level snapshot per screen (`undoSnapshot`), cleared by the next action.

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7. Accessibility checklist

- All interactive elements are native buttons/inputs; tab bar uses `nav` + `aria-current`.
- Sheets: `role="dialog"`, labelled, focus trapped, Escape closes.
- Live regions for status lines (`aria-live="polite"`) and alerts (`role="alert"`).
- Checkbox rows in Read-out are `<label>` wrapping a real checkbox.
- Progress bars expose an aria-label with the counts.
- States never rely on colour alone (borders, strike-through, text labels).

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8. Testing (required by CLAUDE.md)

Unit (Vitest): blocks, receive validation, status validation, each data check, `splitIntoOvers` (assert never produces ranges), call-due logic, restore-after-unmark, correction flag.

Playwright journeys (add to `test/e2e/playwright/`, register in `generate-guide.js` `SECTION_MAP` / `CHAPTERS`, regenerate the guide):

1. `checkpoint-edit-time` — mark, open sheet, change time, save; unmark + Undo; unmark, later restore original time.
2. `checkpoint-correction` — called-in runner time edited → appears on callout as correction.
3. `checkpoint-readout` — 12 waiting runners → read-out with 3 overs; next locked until all ticked; Reset clears ticks.
4. `checkpoint-overview-fix` — undo call-in, reset to not passed, mark DNS, clear DNS, each with Undo.
5. `base-receive-blocks` — pick behind checkpoint, default block = next unreceived; add 5, Next block, add 5; switch back via "Also in this call"; Complete; board updates; Undo call.
6. `base-receive-validation` — unknown bib, duplicate, already logged, not seen at previous CP → correct chip states and only valid numbers recorded.
7. `base-receive-actions` — Clear block + Undo; Cancel call two-tap.
8. `base-bulk-dnf` — paste list, warnings shown, apply, clear one, Undo.
9. `base-reports-checks` — seed a duplicate and an out-of-order time; resolve both; counts reach ✓.
10. Viewports: run 5 at 1112×834 (tablet landscape — assert no vertical page scroll on Receive) and 390×844.

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9. Phases

Phase	Scope	Done when

0	Tokens (`design-tokens/`), fonts, `statusStyles.js`, AppBar, CheckpointTabBar, BaseTabs, UndoBar, BottomSheet, TimeStepper, BlockPicker, NumberPad, BibChip; remove emoji	Existing screens render in new palette; tests green
1	Home, Race setup, Share config, Settings restyle	Matches designs
2	Mark off (edit sheet, undo, restore) + Overview (fix anything); v11 checkpoint fields	Journeys 1, 2, 4
3	Callout sheet + Read-out	Journey 3
4	Base Receive + `radio_calls` + board	Journeys 5–7, tablet height check
5	DNF / DNS bulk + Base overview restyle	Journey 8
6	Reports + data checks + resolutions	Journey 9; user guide regenerated

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10. Sample data (for seeds / demos only)

From the WICEN brief (`reference/`): Mt Glorious Mountain Trails, Sun 8 Nov 2026, start 06:30, base VK4WIP on 147.225 MHz (+0.6, CTCSS 91.5), simplex fallback 147.500.

CP	Location	Operators	Distance
Start	White Cedar Picnic Area	VK4BRW	0 km
CP1	Northbrook Bush Camp	VK4MUD, VK4MKD, VK4EAR	5 km
CP2	England Creek Bush Camp	VK4IE, VK4TBA	11 km
CP3	England Creek Rd & Joyners Ridge Rd	VK4PMC, VK4JBX	14.5 km
CP4	Near Apiary Site, Joyners Ridge Rd	VK4AJG, VK4DTB	17.5 km
Finish	Maiala Picnic Area	VK4GJW, VK4ENE, VK4NNA	22 km

Runner counts and bib numbers in the designs are illustrative.